

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
```

```

nan_ratio = df[cols_to_check].isna().mean()
valid_features = nan_ratio[nan_ratio <= 0.15].index
valid_features_list = list(valid_features)

#df = df[valid_features]
# Removing of last np.nan
mask = np.isfinite(df[valid_features]).all(axis=1)
df = df[mask]

cols = list(dict.fromkeys(valid_features_list + #target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

new_feats_X = [x for x in features_X if x in valid_features_list]
new_feats_add = [x for x in feats_eng if x in valid_features_list]

data["df"]=df
data["features_X"] = new_feats_X
data["feats_added"] = new_feats_add

column_removed = [x for x in cols_to_check if x not in valid_features_list]
print(f'Column removed because np.isnan > 15%: {column_removed}')

return data

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":

```

```

df["campaign"] = df["datetime"].apply(assign_campaign)
df = df[df["campaign"] != "other"].copy()
df["innova_point"] = df.apply(assign_innova_point, axis=1)
df = df[df["innova_point"] != "no"].copy()
with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
    #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
        dict_features = json.load(f)
elif target_col[0] == "i_CO2":
    df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
    df = df[df["campaign"] != "other"].copy()
    df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
    df = df[df["innova_point"] != "no"].copy()
    with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
        dict_features = json.load(f)
else:
    raise ValueError("The target col do not match any existing function")

df["hot"] = df.apply(assign_hot_season,axis=1)
df["umidity_range"] = df.apply(assign_humidity, axis = 1)
#Removing of rabbit 4
df["columnn"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}

```

```

        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")

```

Column removed because np.isnan > 15%: []

1.2 Metadata

Nombre de lignes df	7224
Nombre de lignes df clean	7224
Nombre de features	86
% de NaN	0.0
Taille dataset entraînement	5056
Taille dataset test	2167
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_8h_D1_energy, r_NH3_wl_8h_D2_energy, r_NH3_wl_8h_D3_energy, r_NH3_wl_8h_A_energy, r_NH3_wl_8h_A_rms, r_NH3_wl_8h_D1_energy_rel, r_NH3_wl_8h_D2_energy_rel, r_NH3_wl_8h_D3_energy_rel, r_NH3_wl_8h_A_energy_rel, r_NH3_wl_12h_D1_energy, r_NH3_wl_12h_D2_energy, r_NH3_wl_12h_D3_energy, r_NH3_wl_12h_A_energy, r_NH3_wl_12h_A_rms, r_NH3_wl_12h_D1_energy_rel, r_NH3_wl_12h_D2_energy_rel, r_NH3_wl_12h_D3_energy_rel, r_NH3_wl_12h_A_energy_rel, r_NH3_wl_24h_D1_energy, r_NH3_wl_24h_D2_energy, r_NH3_wl_24h_D3_energy, r_NH3_wl_24h_A_energy, r_NH3_wl_24h_A_rms, r_NH3_wl_24h_D1_energy_rel, r_NH3_wl_24h_D2_energy_rel, r_NH3_wl_24h_D3_energy_rel, r_NH3_wl_24h_A_energy_rel, r_temperature_wl_8h_D1_energy, r_temperature_wl_8h_D2_energy, r_temperature_wl_8h_D3_energy, r_temperature_wl_8h_A_energy, r_temperature_wl_8h_A_rms, r_temperature_wl_8h_D1_energy_rel, r_temperature_wl_8h_D2_energy_rel, r_temper-

ature_wl_8h_D3_energy_rel, r_temperature_wl_8h_A_energy_rel, r_temperature_wl_12h_D1_energy, r_temperature_wl_12h_D2_energy, r_temperature_wl_12h_D3_energy, r_temperature_wl_12h_A_energy, r_temperature_wl_12h_A_rms, r_temperature_wl_12h_D1_energy_rel, r_temperature_wl_12h_D2_energy_rel, r_temperature_wl_12h_D3_energy_rel, r_temperature_wl_12h_A_energy_rel, r_temperature_wl_24h_D1_energy, r_temperature_wl_24h_D2_energy, r_temperature_wl_24h_D3_energy, r_temperature_wl_24h_A_energy, r_temperature_wl_24h_A_rms, r_temperature_wl_24h_D1_energy_rel, r_temperature_wl_24h_D2_energy_rel, r_temperature_wl_24h_D3_energy_rel, r_temperature_wl_24h_A_energy_rel, r_umidity_wl_8h_D1_energy, r_umidity_wl_8h_D2_energy, r_umidity_wl_8h_D3_energy, r_umidity_wl_8h_A_energy, r_umidity_wl_8h_A_rms, r_umidity_wl_8h_D1_energy_rel, r_umidity_wl_8h_D2_energy_rel, r_umidity_wl_8h_D3_energy_rel, r_umidity_wl_8h_A_energy_rel, r_umidity_wl_12h_D1_energy, r_umidity_wl_12h_D2_energy, r_umidity_wl_12h_D3_energy, r_umidity_wl_12h_A_energy, r_umidity_wl_12h_A_rms, r_umidity_wl_12h_D1_energy_rel, r_umidity_wl_12h_D2_energy_rel, r_umidity_wl_12h_D3_energy_rel, r_umidity_wl_12h_A_energy_rel, r_umidity_wl_24h_D1_energy, r_umidity_wl_24h_D2_energy, r_umidity_wl_24h_D3_energy, r_umidity_wl_24h_A_energy, r_umidity_wl_24h_A_rms, r_umidity_wl_24h_D1_energy_rel, r_umidity_wl_24h_D2_energy_rel, r_umidity_wl_24h_D3_energy_rel, r_umidity_wl_24h_A_energy_rel

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermapping and cluster list

The following section has been done with a threshold chosen of : 0.8

cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, __, T, H, I ; **cluster 6:** r_NH3_wl_8h_D1_energy ; **cluster 7:** r_NH3_wl_8h_D2_energy ; **cluster 8:** r_NH3_wl_8h_D3_energy ; **cluster 9:** r_NH3_wl_8h_A_energy ; **cluster 10:** r_NH3_wl_8h_A_rms ; **cluster 11:** r_NH3_wl_8h_A_energy_rel, r_NH3_wl_8h_D2_energy_rel, r_NH3_wl_8h_D1_energy_rel ; **cluster**

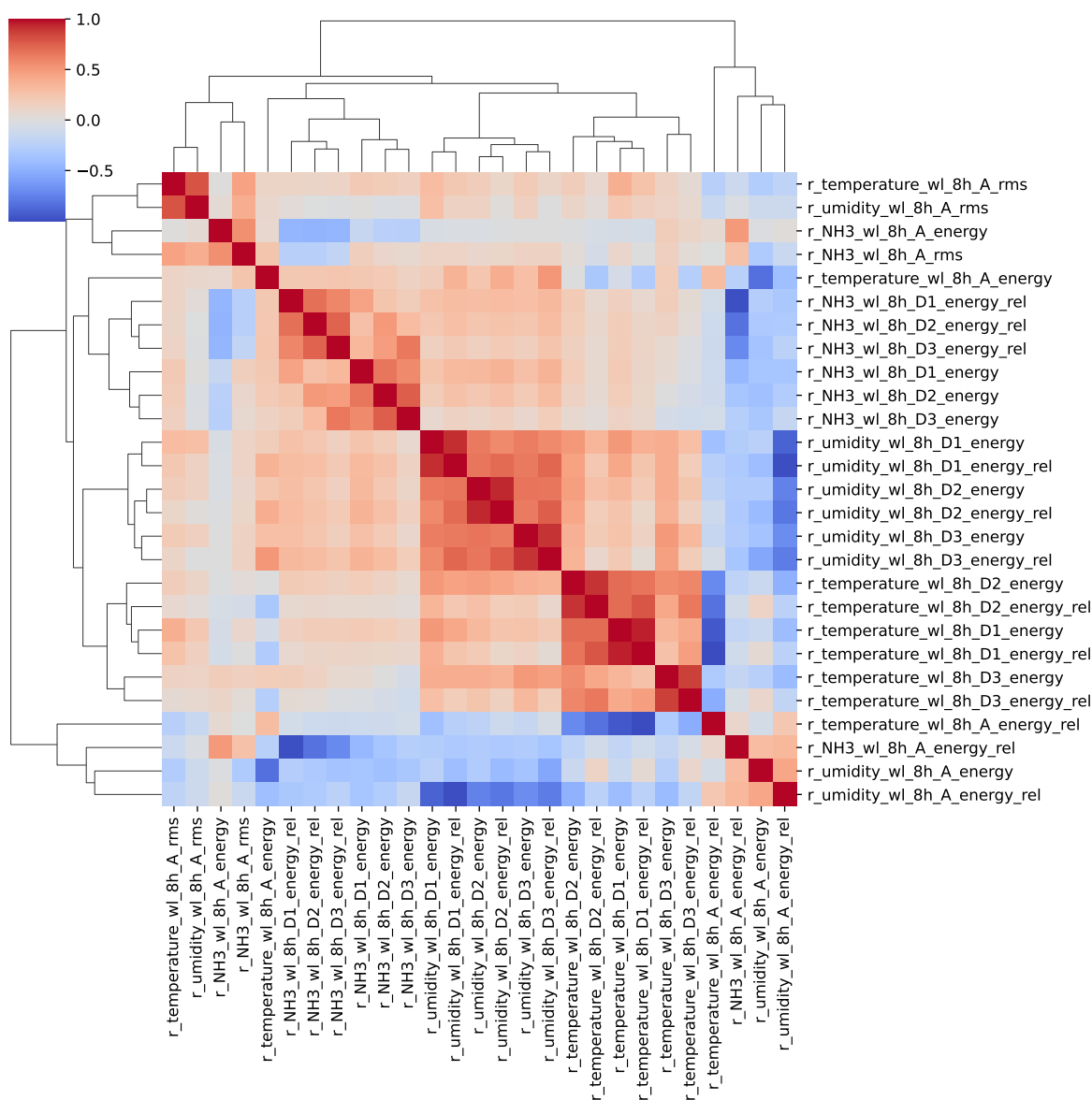
12: r_NH3_wl_8h_D3_energy_rel ; **cluster 13:** r_temperature_wl_8h_D1_energy_rel, r_temperature_wl_8h_D2_energy, r_temperature_wl_8h_D2_energy_rel, r_temperature_wl_8h_D1_energy, r_temperature_wl_8h_A_energy_rel ; **cluster 14:** r_temperature_wl_8h_D3_energy_rel, r_temperature_wl_8h_D3_energy ; **cluster 15:** r_temperature_wl_8h_A_energy, r_umidity_wl_8h_A_energy ; **cluster 16:** r_temperature_wl_8h_A_rms ; **cluster 17:** r_umidity_wl_8h_D1_energy_rel, r_umidity_wl_8h_A_energy_rel, r_umidity_wl_8h_D1_energy ; **cluster 18:** r_umidity_wl_8h_D2_energy_rel, r_umidity_wl_8h_D2_energy ; **cluster 19:** r_umidity_wl_8h_D3_energy, r_umidity_wl_8h_D3_energy_rel ; **cluster 20:** r_umidity_wl_8h_A_rms ; **cluster 21:** r_NH3_wl_12h_D1_energy ; **cluster 22:** r_NH3_wl_12h_D2_energy ; **cluster 23:** r_NH3_wl_12h_D3_energy_rel, r_NH3_wl_12h_D3_energy ; **cluster 24:** r_NH3_wl_12h_A_energy ; **cluster 25:** r_NH3_wl_12h_A_rms ; **cluster 26:** r_NH3_wl_12h_D2_energy_rel, r_NH3_wl_12h_D1_energy_rel, r_NH3_wl_12h_A_energy_rel ; **cluster 27:** r_temperature_wl_12h_A_energy_rel, r_temperature_wl_12h_D2_energy, r_temperature_wl_12h_D2_energy_rel, r_temperature_wl_12h_D1_energy_rel, r_temperature_wl_12h_D1_energy ; **cluster 28:** r_temperature_wl_12h_D3_energy_rel, r_temperature_wl_12h_D3_energy ; **cluster 29:** r_temperature_wl_12h_A_energy ; **cluster 30:** r_temperature_wl_12h_A_rms ; **cluster 31:** r_umidity_wl_12h_A_energy_rel, r_umidity_wl_12h_D1_energy_rel, r_umidity_wl_12h_D2_energy_rel, r_umidity_wl_12h_D1_energy, r_umidity_wl_12h_D2_energy ; **cluster 32:** r_umidity_wl_12h_D3_energy, r_umidity_wl_12h_D3_energy_rel ; **cluster 33:** r_umidity_wl_12h_A_energy ; **cluster 34:** r_umidity_wl_12h_A_rms ; **cluster 35:** r_NH3_wl_24h_D2_energy, r_NH3_wl_24h_D3_energy, r_NH3_wl_24h_D1_energy, r_NH3_wl_24h_D1_energy_rel, r_NH3_wl_24h_D3_energy_rel, r_NH3_wl_24h_D2_energy_rel, r_NH3_wl_24h_A_energy_rel ; **cluster 36:** r_NH3_wl_24h_A_rms, r_NH3_wl_24h_A_energy ; **cluster 37:** r_temperature_wl_24h_D1_energy, r_temperature_wl_24h_D1_energy_rel, r_temperature_wl_24h_A_energy_rel ; **cluster 38:** r_temperature_wl_24h_D2_energy, r_temperature_wl_24h_D2_energy_rel ; **cluster 39:** r_temperature_wl_24h_D3_energy, r_temperature_wl_24h_D3_energy_rel ; **cluster 40:** r_temperature_wl_24h_A_energy ; **cluster 41:** r_temperature_wl_24h_A_rms ; **cluster 42:** r_umidity_wl_24h_D3_energy_rel, r_umidity_wl_24h_D2_energy_rel, r_umidity_wl_24h_D2_energy, r_umidity_wl_24h_D1_energy, r_umidity_wl_24h_A_energy_rel, r_umidity_wl_24h_D1_energy_rel, r_umidity_wl_24h_D3_energy ; **cluster 43:** r_umidity_wl_24h_A_energy ; **cluster 44:** r_umidity_wl_24h_A_rms ;

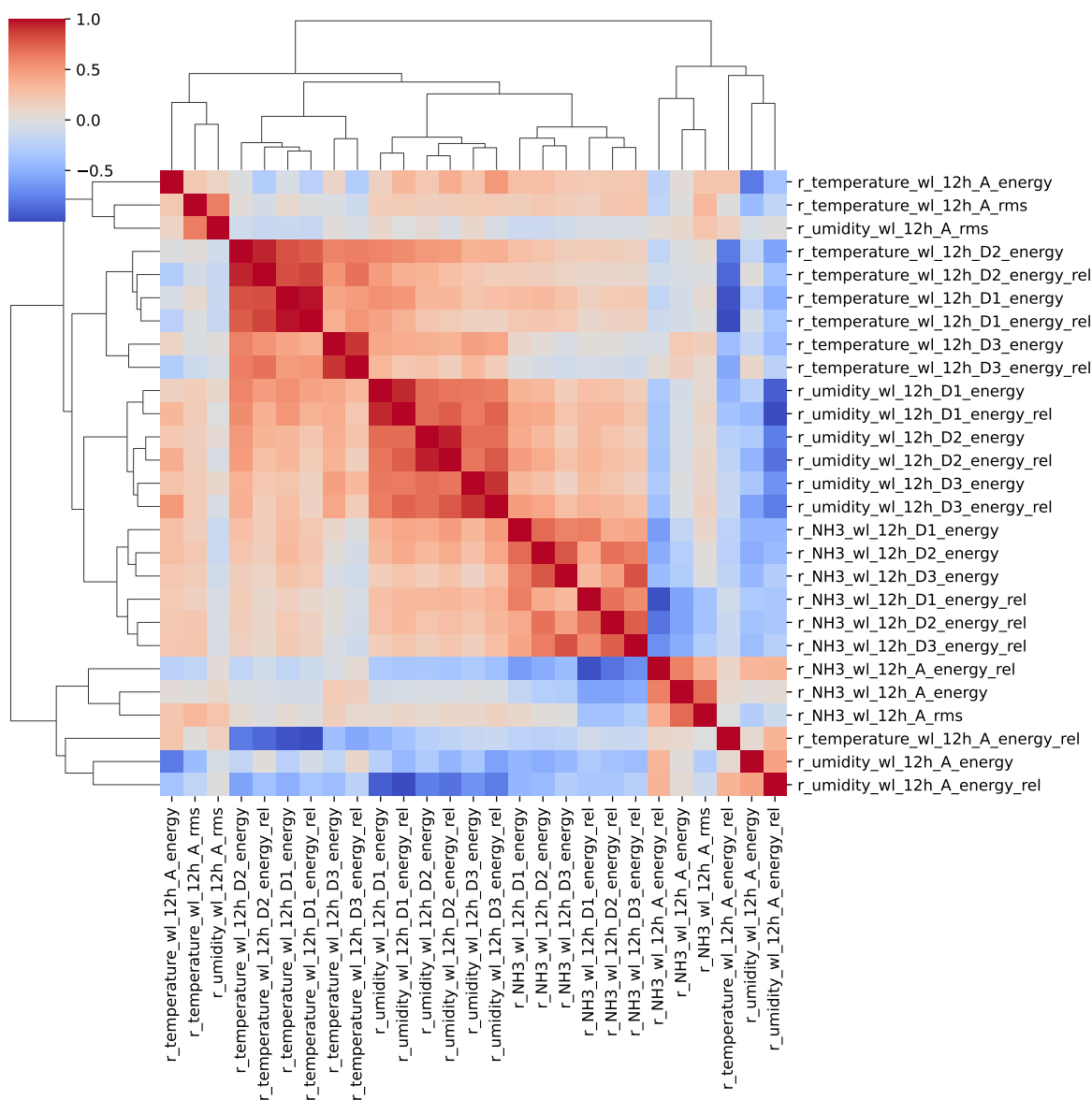
2.2 Features selection

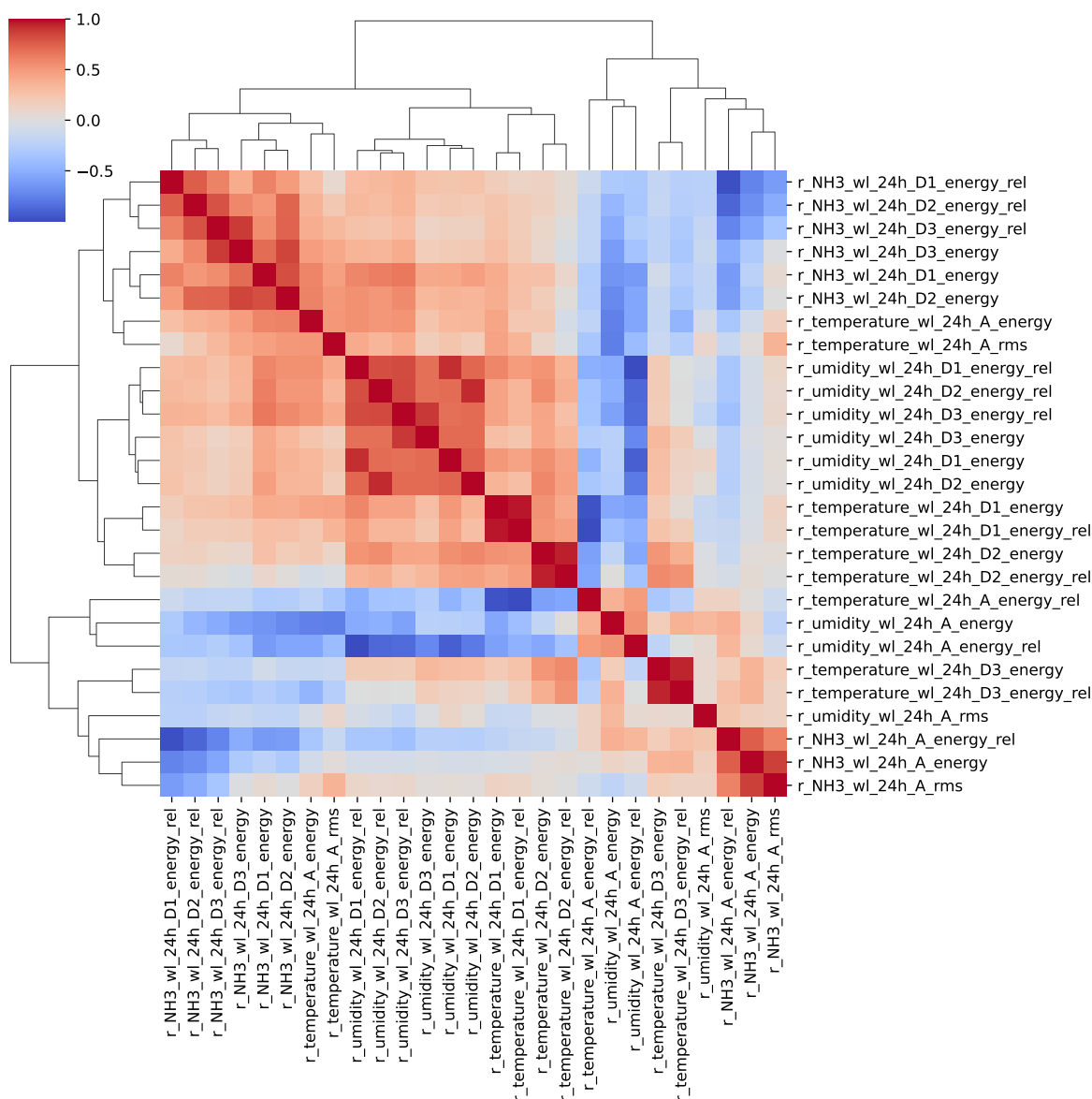
Only the first items of the clusters is choosen for the rest of the study

Chooosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_8h_D1_energy, r_NH3_wl_8h_D2_energy, r_NH3_wl_8h_D3_energy, r_NH3_wl_8h_A_energy,

r_NH3_wl_8h_A_rms, r_NH3_wl_8h_A_energy_rel, r_NH3_wl_8h_D3_en-
ergy_rel, r_temperature_wl_8h_D1_energy_rel, r_temperature_wl_8h_D3_en-
ergy_rel, r_temperature_wl_8h_A_energy, r_temperature_wl_8h_A_rms, r_umid-
ity_wl_8h_D1_energy_rel, r_umidity_wl_8h_D2_energy_rel, r_umidity_wl_8h_D3_en-
ergy, r_umidity_wl_8h_A_rms, r_NH3_wl_12h_D1_energy, r_NH3_wl_12h_D2_energy,
r_NH3_wl_12h_D3_energy_rel, r_NH3_wl_12h_A_energy, r_NH3_wl_12h_A_rms,
r_NH3_wl_12h_D2_energy_rel, r_temperature_wl_12h_A_energy_rel, r_tem-
perature_wl_12h_D3_energy_rel, r_temperature_wl_12h_A_energy, r_tempera-
ture_wl_12h_A_rms, r_umidity_wl_12h_A_energy_rel, r_umidity_wl_12h_D3_energy,
r_umidity_wl_12h_A_energy, r_umidity_wl_12h_A_rms, r_NH3_wl_24h_D2_en-
ergy, r_NH3_wl_24h_A_rms, r_temperature_wl_24h_D1_energy, r_tempera-
ture_wl_24h_D2_energy, r_temperature_wl_24h_D3_energy, r_temperature_wl_24h_A_en-
ergy, r_temperature_wl_24h_A_rms, r_umidity_wl_24h_D3_energy_rel, r_umid-
ity_wl_24h_A_energy, r_umidity_wl_24h_A_rms







3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_8h_D1_energy, r_NH3_wl_8h_D2_energy, r_NH3_wl_8h_D3_energy, r_NH3_wl_8h_A_energy, r_NH3_wl_8h_A_rms, r_NH3_wl_8h_A_energy_rel, r_NH3_wl_8h_D3_en-

ergy_rel, r_temperature_wl_8h_D1_energy_rel, r_temperature_wl_8h_D3_energy_rel, r_temperature_wl_8h_A_energy, r_temperature_wl_8h_A_rms, r_umidity_wl_8h_D1_energy_rel, r_umidity_wl_8h_D2_energy_rel, r_umidity_wl_8h_D3_energy, r_umidity_wl_8h_A_rms, r_NH3_wl_12h_D1_energy, r_NH3_wl_12h_D2_energy, r_NH3_wl_12h_D3_energy_rel, r_NH3_wl_12h_A_energy, r_NH3_wl_12h_A_rms, r_NH3_wl_12h_D2_energy_rel, r_temperature_wl_12h_A_energy_rel, r_temperature_wl_12h_D3_energy_rel, r_temperature_wl_12h_A_energy, r_temperature_wl_12h_A_rms, r_umidity_wl_12h_A_energy_rel, r_umidity_wl_12h_D3_energy, r_umidity_wl_12h_A_energy, r_umidity_wl_12h_A_rms, r_NH3_wl_24h_D2_energy, r_NH3_wl_24h_A_rms, r_temperature_wl_24h_D1_energy, r_temperature_wl_24h_D2_energy, r_temperature_wl_24h_D3_energy, r_temperature_wl_24h_A_energy, r_temperature_wl_24h_A_rms, r_umidity_wl_24h_D3_energy_rel, r_umidity_wl_24h_A_energy, r_umidity_wl_24h_A_rms

Len(Features_X): 44

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_test < 0.95$ (under performance of the test)

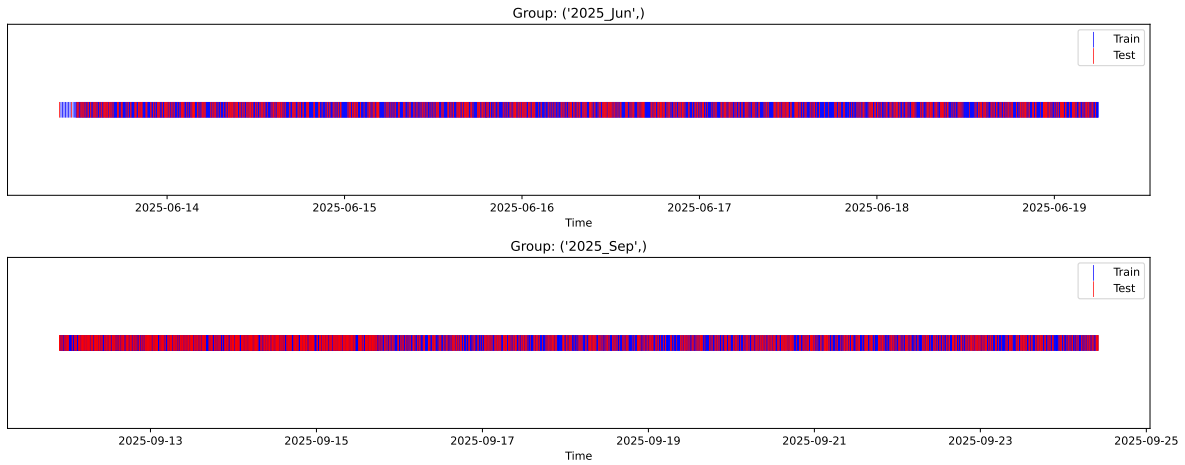
[Green] $r_test > 1.05$ (over performance of the test)

[Red] r_train not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_{train} vs X_{test}

External Split: Train: 5056 samples Test: 2168 samples



3.3.2 Internal CV Folds

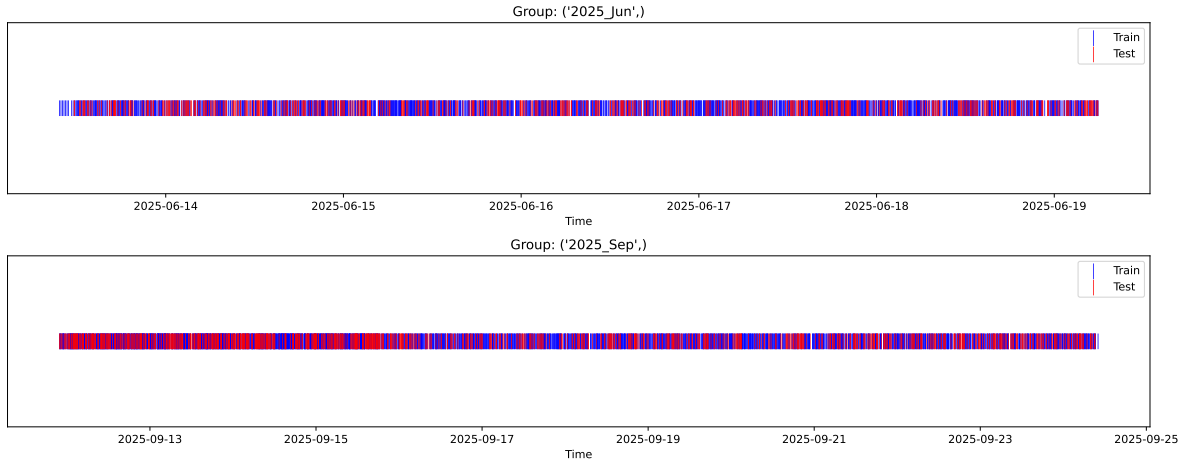
Fold 1/5

Train: 3539 samples Test: 1517 samples Excluded: 0 samples Window exclusion: 0h



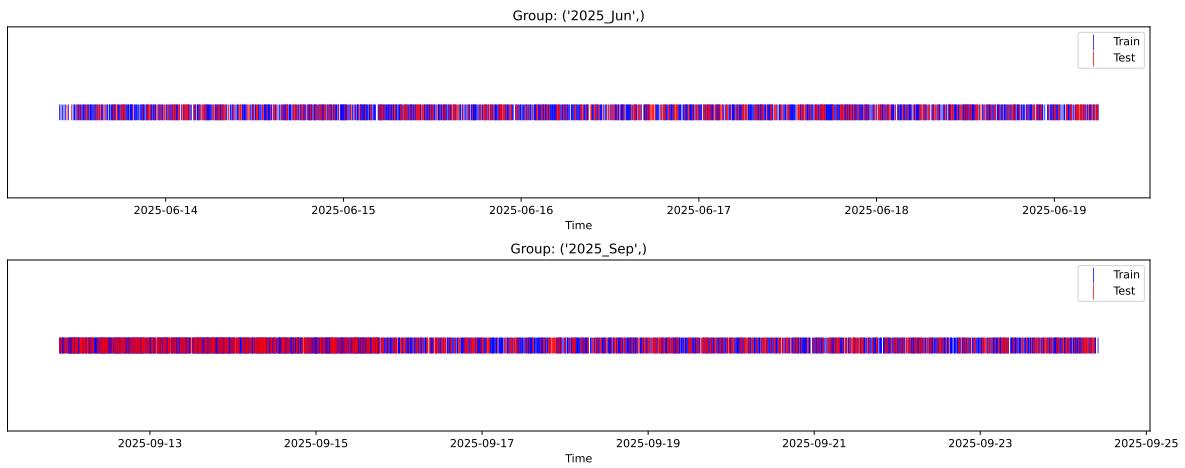
Fold 2/5

Train: 3539 samples Test: 1517 samples Excluded: 0 samples Window exclusion: 0h



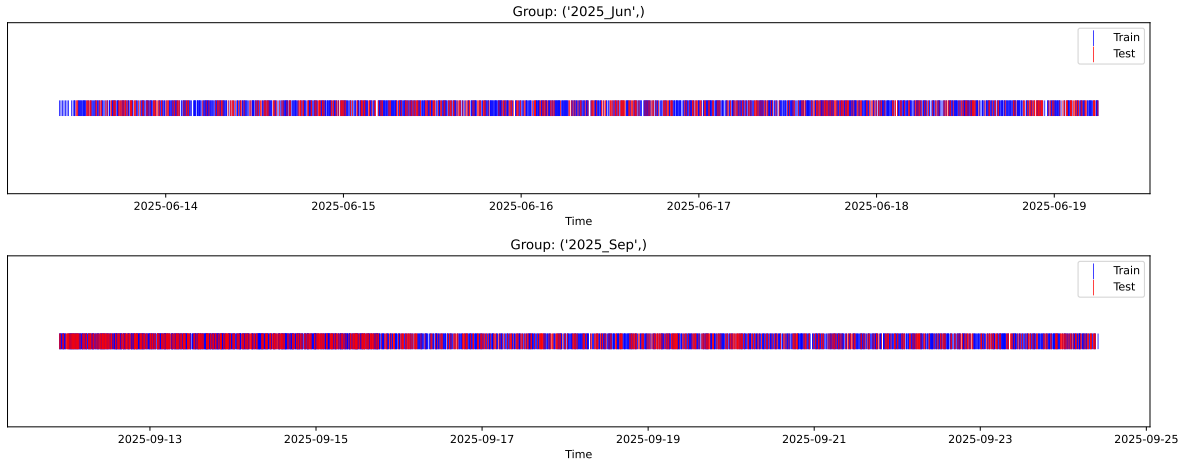
Fold 3/5

Train: 3539 samples Test: 1517 samples Excluded: 0 samples Window exclusion: 0h



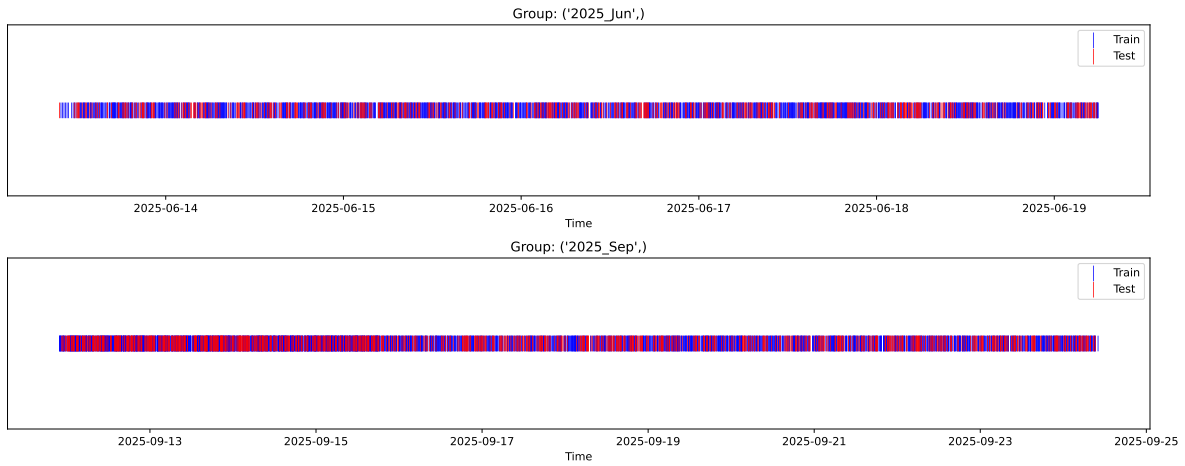
Fold 4/5

Train: 3539 samples Test: 1517 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3539 samples Test: 1517 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.346, std=0.787, min=0.959, max=8.190
 y_{test} : mean=2.382, std=0.828, min=1.071, max=13.845

Fold 0 — y_{test} : mean=2.35, std=0.80, y_{train} : mean=2.34, std=0.78

Fold 1 — y_{test} : mean=2.33, std=0.75, y_{train} : mean=2.35, std=0.80

Fold 2 — y_{test} : mean=2.37, std=0.83, y_{train} : mean=2.34, std=0.77

Fold 3 — y_test: mean=2.34, std=0.78, y_train: mean=2.35, std=0.79

Fold 4 — y_test: mean=2.32, std=0.80, y_train: mean=2.36, std=0.78

train scores CV: [0.62985114 0.62092374 0.62125391 0.62609195 0.62679875]

test scores CV: [0.60221156 0.62361512 0.62082225 0.61087697 0.61009621]

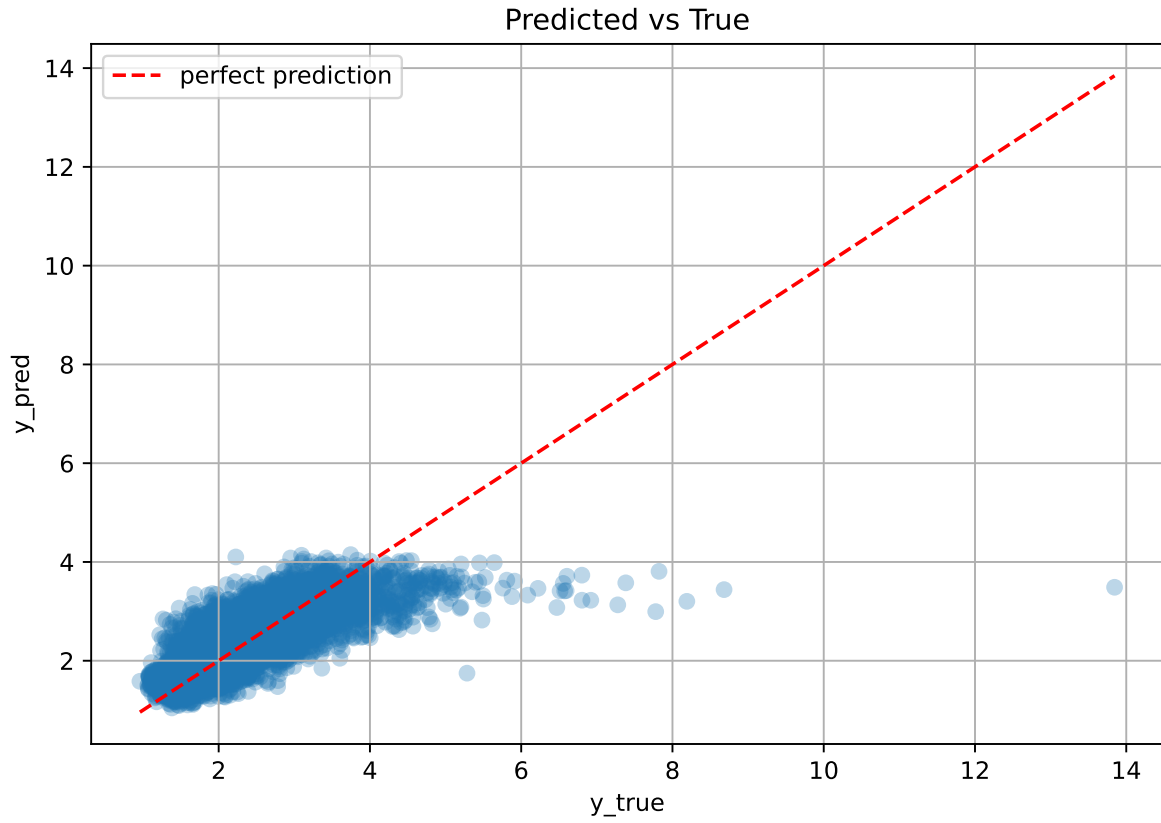
rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.625	0.6135	0.0078	0.5658	0.5679	-0.0456	0.9256	1.0187

Coefficients:

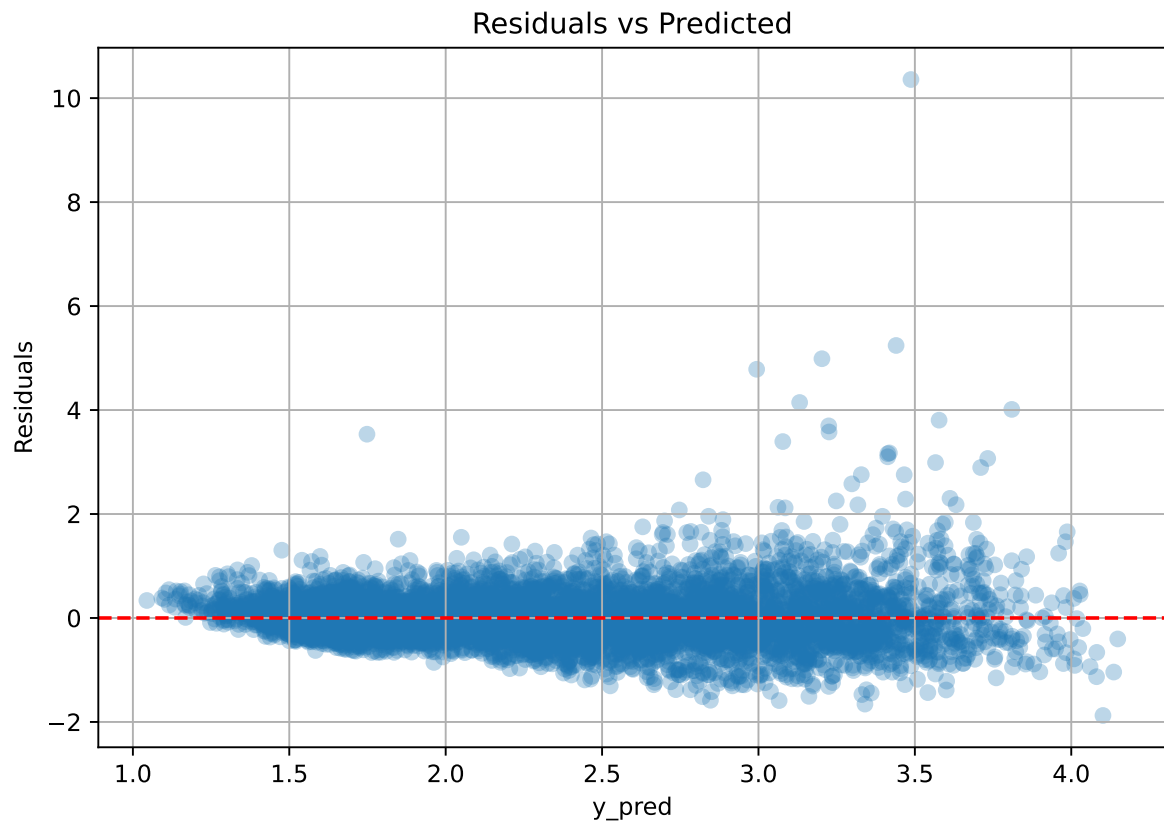
coef_r_CO2 = -0.0320, coef_r_NH3 = 0.0826, coef_r_NH3_wl_12h_A_energy = -0.0661, coef_r_NH3_wl_12h_A_rms = -0.0234, coef_r_NH3_wl_12h_D1_energy = -0.0247, coef_r_NH3_wl_12h_D2_energy = -0.0843, coef_r_NH3_wl_12h_D2_energy_rel = 0.1005, coef_r_NH3_wl_12h_D3_energy_rel = -0.0770, coef_r_NH3_wl_24h_A_rms = 0.0752, coef_r_NH3_wl_24h_D2_energy = 0.1606, coef_r_NH3_wl_8h_A_energy = -0.1148, coef_r_NH3_wl_8h_A_energy_rel = 0.0510, coef_r_NH3_wl_8h_A_rms = -0.0139, coef_r_NH3_wl_8h_D1_energy = 0.0197, coef_r_NH3_wl_8h_D2_energy = 0.0669, coef_r_NH3_wl_8h_D3_energy = -0.0065, coef_r_NH3_wl_8h_D3_energy_rel = -0.0538, coef_r_THI = 0.7109, coef_r_temperature = -0.1090, coef_r_temperature_wl_12h_A_energy = 0.5806, coef_r_temperature_wl_12h_A_energy_rel = 0.0068, coef_r_temperature_wl_12h_A_rms = -0.0522, coef_r_temperature_wl_12h_D3_energy_rel = -0.0260, coef_r_temperature_wl_24h_A_energy = -0.1975, coef_r_temperature_wl_24h_A_rms = 0.1785, coef_r_temperature_wl_24h_D1_energy = -0.0098, coef_r_temperature_wl_24h_D2_energy = -0.0041, coef_r_temperature_wl_24h_D3_energy = 0.0140, coef_r_temperature_wl_8h_A_energy = -0.7574, coef_r_temperature_wl_8h_A_rms = -0.2272, coef_r_temperature_wl_8h_D1_energy_rel = 0.1016, coef_r_temperature_wl_8h_D3_energy_rel = -0.0249, coef_r_umidity = -0.5493, coef_r_umidity_wl_12h_A_energy = -0.0773, coef_r_umidity_wl_12h_A_energy_rel = 0.0340, coef_r_umidity_wl_12h_A_rms = -0.1365, coef_r_umidity_wl_12h_D3_energy = 0.0332, coef_r_umidity_wl_24h_A_energy = 0.1909, coef_r_umidity_wl_24h_A_rms = 0.0427,

coef_r_umidity_wl_24h_D3_energy_rel = 0.0791, coef_r_umidity_wl_8h_A_rms = -0.0472, coef_r_umidity_wl_8h_D1_energy_rel = -0.0671, coef_r_umidity_wl_8h_D2_energy_rel = -0.0257, coef_r_umidity_wl_8h_D3_energy = 0.0156, intercept = 2.3462,

3.4.2 Scatter



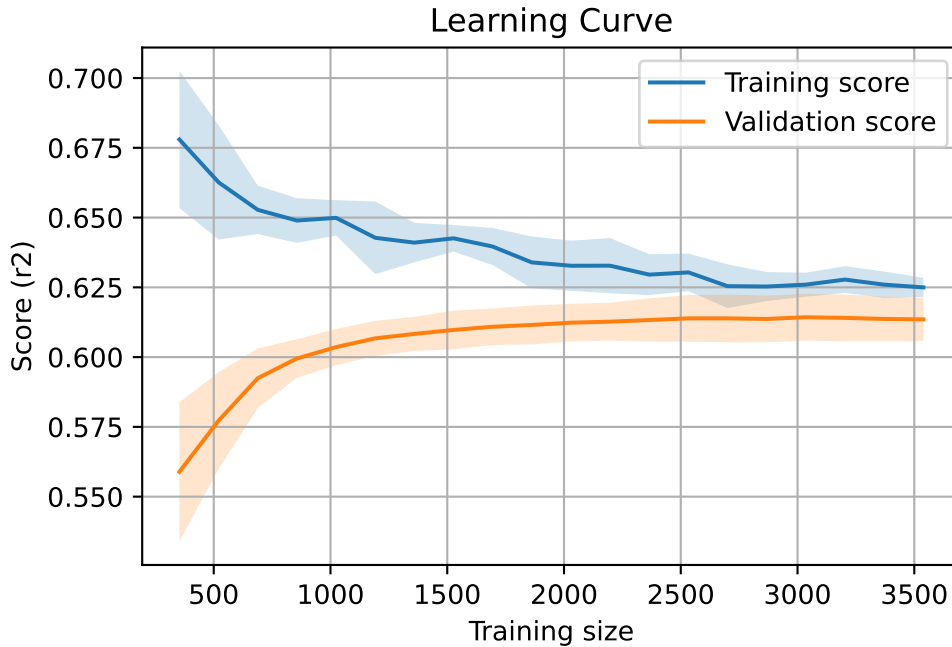
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 7224, `Len(training)` : 5057,

`Len(training_real)` : 3540, `Len(test_of_training)` : 1517, `Len(test)` : 2167



3.5 Model : LGBM

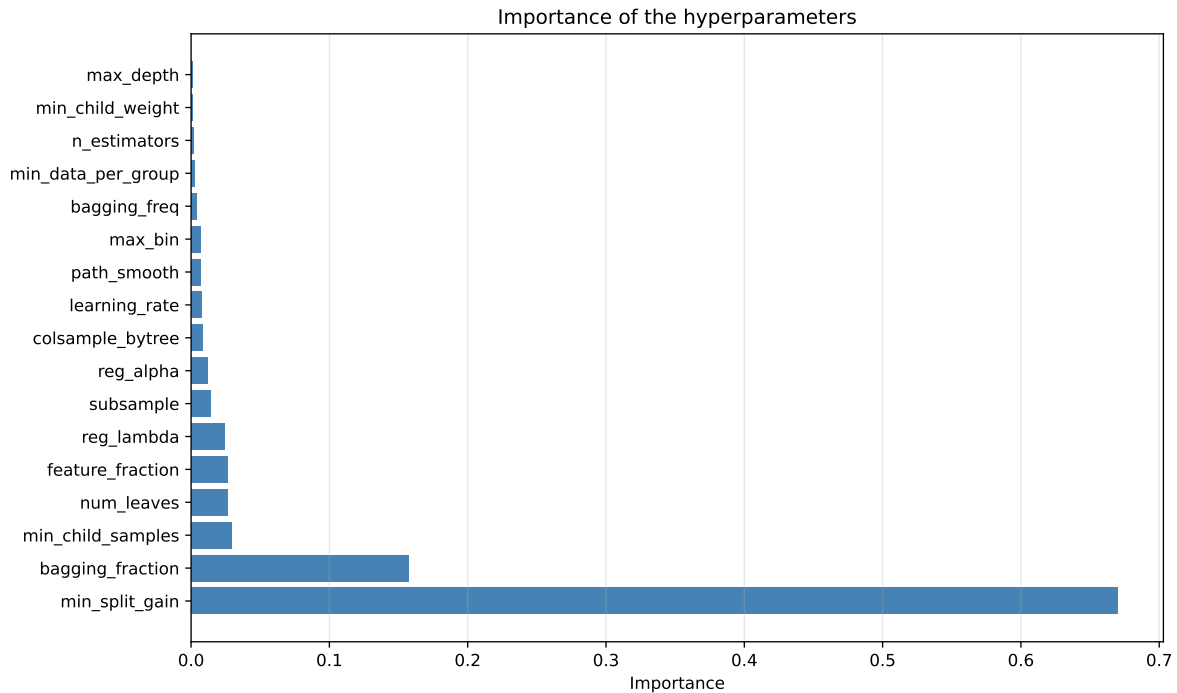
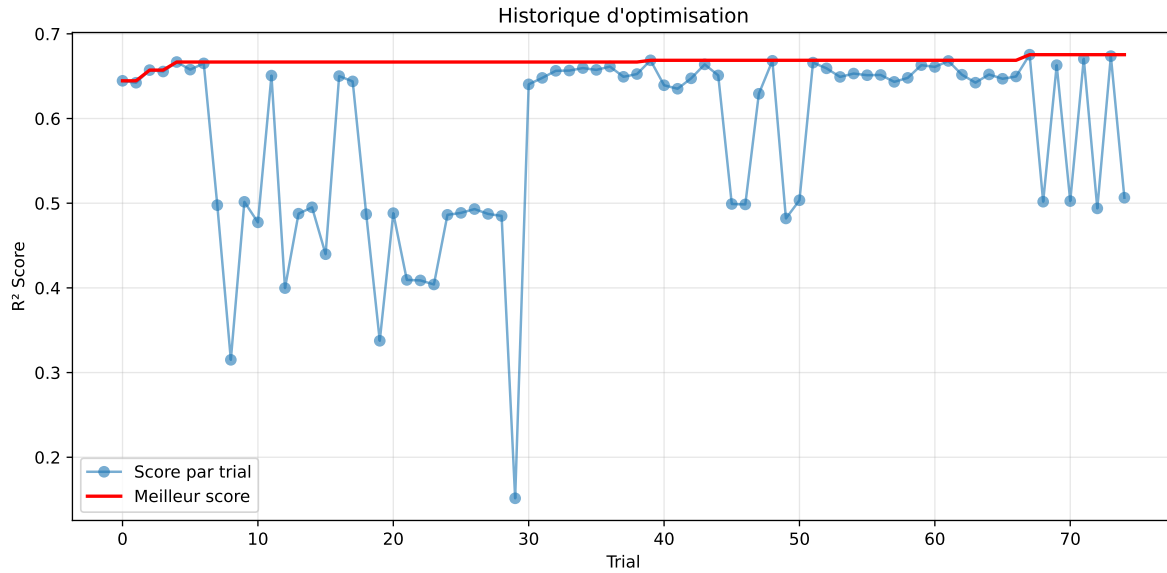
3.5.1 Gridsearch

Distribution y_train vs y_test: y_train: mean=2.346, std=0.787, min=0.959, max=8.190
y_test: mean=2.382, std=0.828, min=1.071, max=13.845

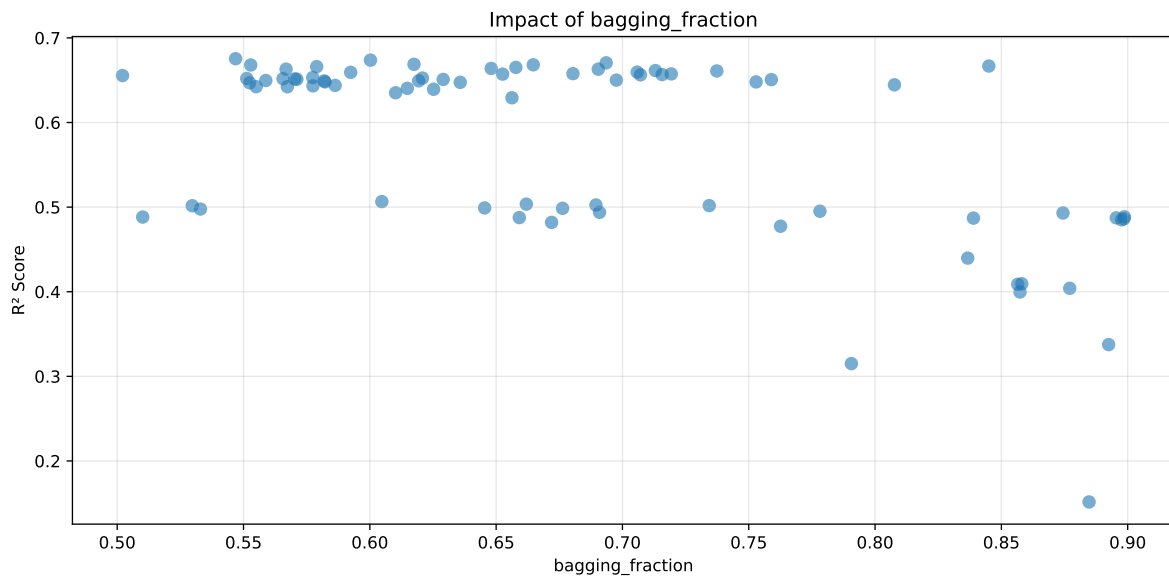
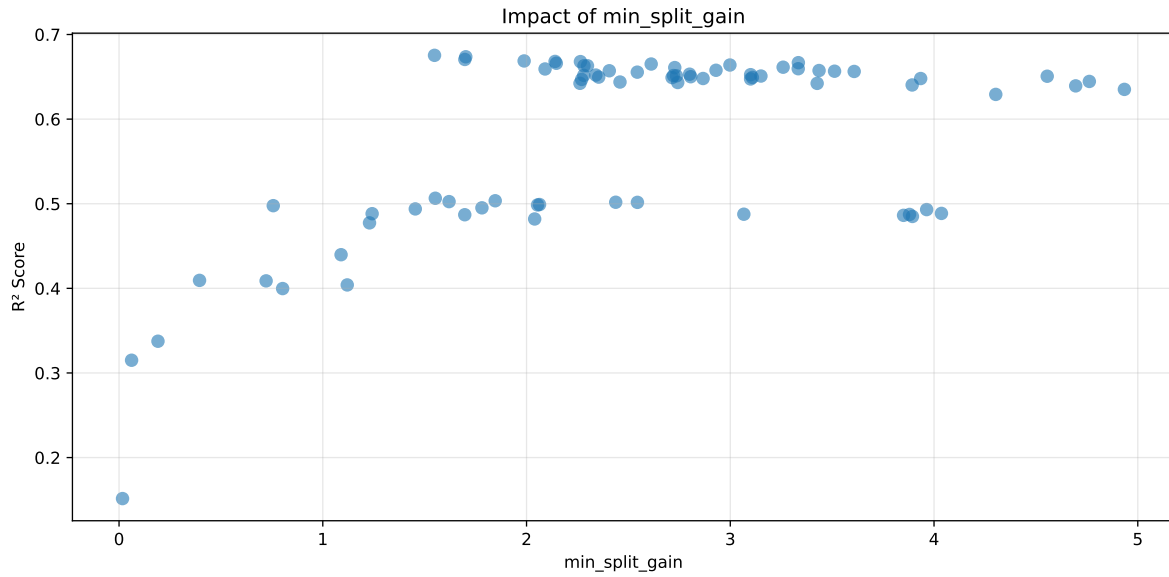
===== TOP Importance FEATURES =====

feature	importance
r_umidity	194
r_CO2	192
r_temperature_wl_12h_A_rms	119
r_umidity_wl_24h_A_energy	108
r_temperature_wl_12h_A_energy	105
r_umidity_wl_12h_A_rms	84
r_temperature_wl_8h_D1_energy_rel	81
r_temperature	81
r_umidity_wl_24h_D3_energy_rel	80
r_temperature_wl_8h_A_energy	78
r_NH3	66
r_NH3_wl_8h_D3_energy_rel	59
r_NH3_wl_8h_A_energy	54
r_NH3_wl_8h_D1_energy	53
r_THI	50
r_NH3_wl_12h_D1_energy	45
r_NH3_wl_12h_A_energy	42
r_temperature_wl_12h_A_energy_rel	39
r_NH3_wl_24h_D2_energy	39
r_umidity_wl_12h_A_energy_rel	33
r_temperature_wl_24h_A_rms	32
r_NH3_wl_24h_A_rms	26
r_NH3_wl_8h_D2_energy	26
r_temperature_wl_24h_A_energy	24
r_umidity_wl_24h_A_rms	24
r_NH3_wl_8h_D3_energy	24
r_NH3_wl_12h_D2_energy	22
r_umidity_wl_8h_D1_energy_rel	19
r_NH3_wl_12h_D3_energy_rel	19
r_NH3_wl_12h_A_rms	18
r_temperature_wl_8h_A_rms	17
r_NH3_wl_8h_A_energy_rel	17
r_umidity_wl_8h_D2_energy_rel	16
r_umidity_wl_12h_A_energy	13
r_NH3_wl_12h_D2_energy_rel	13
r_umidity_wl_8h_A_rms	11
r_NH3_wl_8h_A_rms	11

9 r_umidity_wl_8h_D3_energy 7 r_temperature_wl_24h_D2_energy 5 r_temperature_wl_8h_D3_energy_rel 4 r_umidity_wl_12h_D3_energy 1 r_temperature_wl_24h_D1_energy 1 r_temperature_wl_24h_D3_energy 0 r_temperature_wl_12h_D3_energy_rel 0

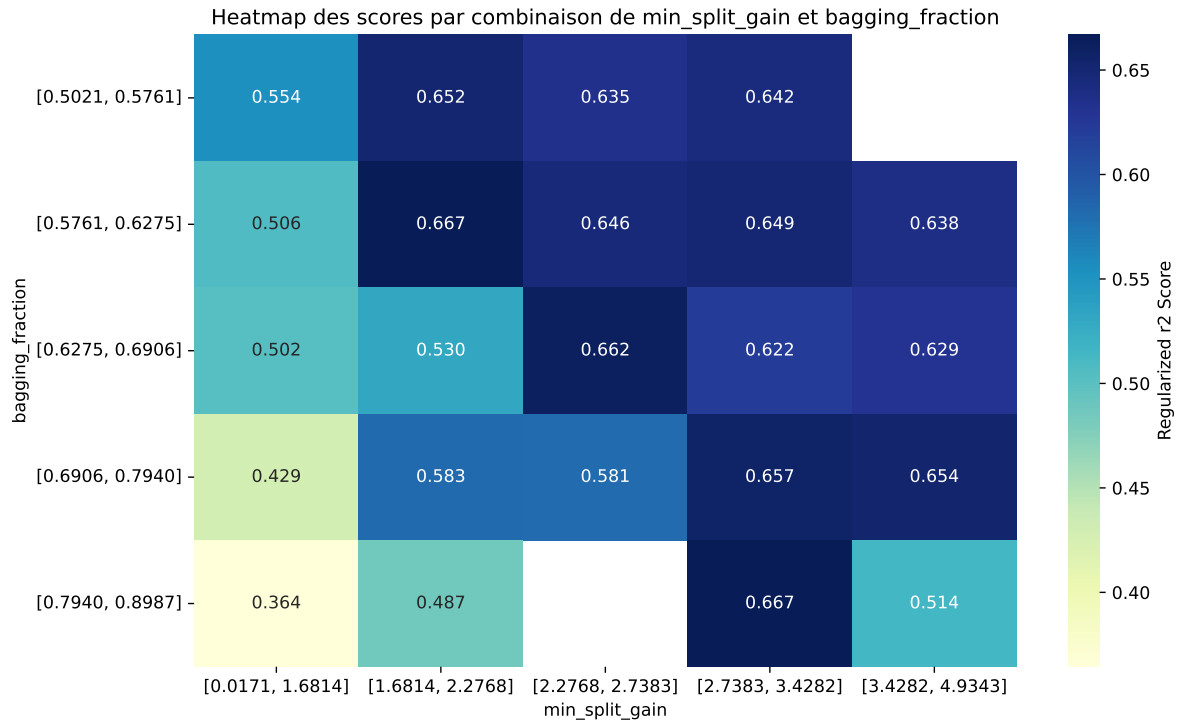


Paramètres avec >5% d'importance : ['min_split_gain', 'bagging_fraction']



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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7085	0.6754	0.618	0.0108	0.6754	0.6429	-0.0325	0.9519	1.0491
2	0.7052	0.6737	0.6154	0.0103	0.6737	0.6391	-0.0346	0.9487	1.0468
3	0.7029	0.6705	0.6126	0.0087	0.6705	0.6306	-0.0399	0.9404	1.0483
4	0.697	0.6687	0.6111	0.0113	0.6687	0.6298	-0.039	0.9417	1.0423
5	0.698	0.6682	0.6103	0.0116	0.6682	0.6339	-0.0343	0.9486	1.0446
6	0.6983	0.668	0.6098	0.0119	0.668	0.6291	-0.0389	0.9417	1.0454
7	0.6994	0.6667	0.6085	0.0134	0.6667	0.6305	-0.0362	0.9456	1.049
8	0.695	0.666	0.6079	0.0113	0.666	0.6267	-0.0393	0.941	1.0435
9	0.6958	0.6651	0.6079	0.0127	0.6651	0.6199	-0.0452	0.9321	1.0462
10	0.6927	0.6639	0.608	0.0086	0.6639	0.6246	-0.0393	0.9407	1.0434
---	---	---	---	---	---	---	---	---	---
75	0.9206	0.7925	0.7401	0.0128	0.1515	0.7696	-0.0229	0.9711	1.1618

Hyperparameters:

Rank 1: {'reg__n_estimators': 400, 'reg__max_depth': 4, 'reg__learning_rate': 0.01955755065090801, 'reg__num_leaves': 57, 'reg__subsample': 0.8286541292901002, 'reg__colsample_bytree': 0.5689164746334457, 'reg__min_child_samples': 48, 'reg__reg_alpha': 6.787850318206774, 'reg__reg_lambda': 8.77409502216203, 'reg__min_split_gain': 1.5483837216068346, 'reg__path_smooth': 0.714632860827688, 'reg__min_child_weight':

0.006989907553349662, 'reg__feature_fraction': 0.6268028318734362, 'reg__bagging_fraction': 0.5468629983002502, 'reg__bagging_freq': 6, 'reg__max_bin': 129, 'reg__min_data_per_group': 130}

Rank 2: {'reg__n_estimators': 500, 'reg__max_depth': 6, 'reg__learning_rate': 0.02921058320542682, 'reg__num_leaves': 30, 'reg__subsample': 0.7006136626571554, 'reg__colsample_bytree': 0.570103546428222, 'reg__min_child_samples': 48, 'reg__reg_alpha': 6.34950091828191, 'reg__reg_lambda': 27.310143757645967, 'reg__min_split_gain': 1.70179545308083, 'reg__path_smooth': 2.468574110263346, 'reg__min_child_weight': 0.004530219823814064, 'reg__feature_fraction': 0.686452097975975, 'reg__bagging_fraction': 0.6002521900246548, 'reg__bagging_freq': 6, 'reg__max_bin': 131, 'reg__min_data_per_group': 58}

Rank 3: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.02877202104895653, 'reg__num_leaves': 51, 'reg__subsample': 0.7028042273966336, 'reg__colsample_bytree': 0.5732954096762564, 'reg__min_child_samples': 42, 'reg__reg_alpha': 6.74200480951502, 'reg__reg_lambda': 22.877654065556378, 'reg__min_split_gain': 1.6972027734009503, 'reg__path_smooth': 2.3253151397096463, 'reg__min_child_weight': 0.03724686464130108, 'reg__feature_fraction': 0.5775151935591519, 'reg__bagging_fraction': 0.6935654361254774, 'reg__bagging_freq': 6, 'reg__max_bin': 238, 'reg__min_data_per_group': 113}

Rank 4: {'reg__n_estimators': 300, 'reg__max_depth': 5, 'reg__learning_rate': 0.04390916865384773, 'reg__num_leaves': 54, 'reg__subsample': 0.7488650183555967, 'reg__colsample_bytree': 0.6224601941813068, 'reg__min_child_samples': 36, 'reg__reg_alpha': 5.641065219718193, 'reg__reg_lambda': 49.93298807048295, 'reg__min_split_gain': 1.9882224815599359, 'reg__path_smooth': 3.488255032451333, 'reg__min_child_weight': 8.056018524034268, 'reg__feature_fraction': 0.6103997774047966, 'reg__bagging_fraction': 0.617488343170582, 'reg__bagging_freq': 3, 'reg__max_bin': 64, 'reg__min_data_per_group': 75}

Rank 5: {'reg__n_estimators': 200, 'reg__max_depth': 3, 'reg__learning_rate': 0.0384619998869386, 'reg__num_leaves': 48, 'reg__subsample': 0.7177496084864438, 'reg__colsample_bytree': 0.5292747444348976, 'reg__min_child_samples': 47, 'reg__reg_alpha': 3.940530892996456, 'reg__reg_lambda': 38.66120692495577, 'reg__min_split_gain': 2.1400476274911613, 'reg__path_smooth': 4.677084771472463, 'reg__min_child_weight': 0.025300270145056095, 'reg__feature_fraction': 0.6340664169013972, 'reg__bagging_fraction': 0.6646903872133613, 'reg__bagging_freq': 5, 'reg__max_bin': 117, 'reg__min_data_per_group': 101}

Rank 6: {'reg__n_estimators': 400, 'reg__max_depth': 7, 'reg__learning_rate': 0.022079039404021, 'reg__num_leaves': 39, 'reg__subsample': 0.842326078785911, 'reg__colsample_bytree': 0.5144505649593003, 'reg__min_child_samples': 33, 'reg__reg_alpha': 6.988091260196299, 'reg__reg_lambda': 7.447434955305113, 'reg__min_split_gain': 2.2645102040076734, 'reg__path_smooth': 2.888476476014901, 'reg__min_child_weight':

0.009678909251733325, 'reg__feature_fraction': 0.6403976556956051, 'reg__bagging_fraction': 0.5528610360380651, 'reg__bagging_freq': 5, 'reg__max_bin': 131, 'reg__min_data_per_group': 53}

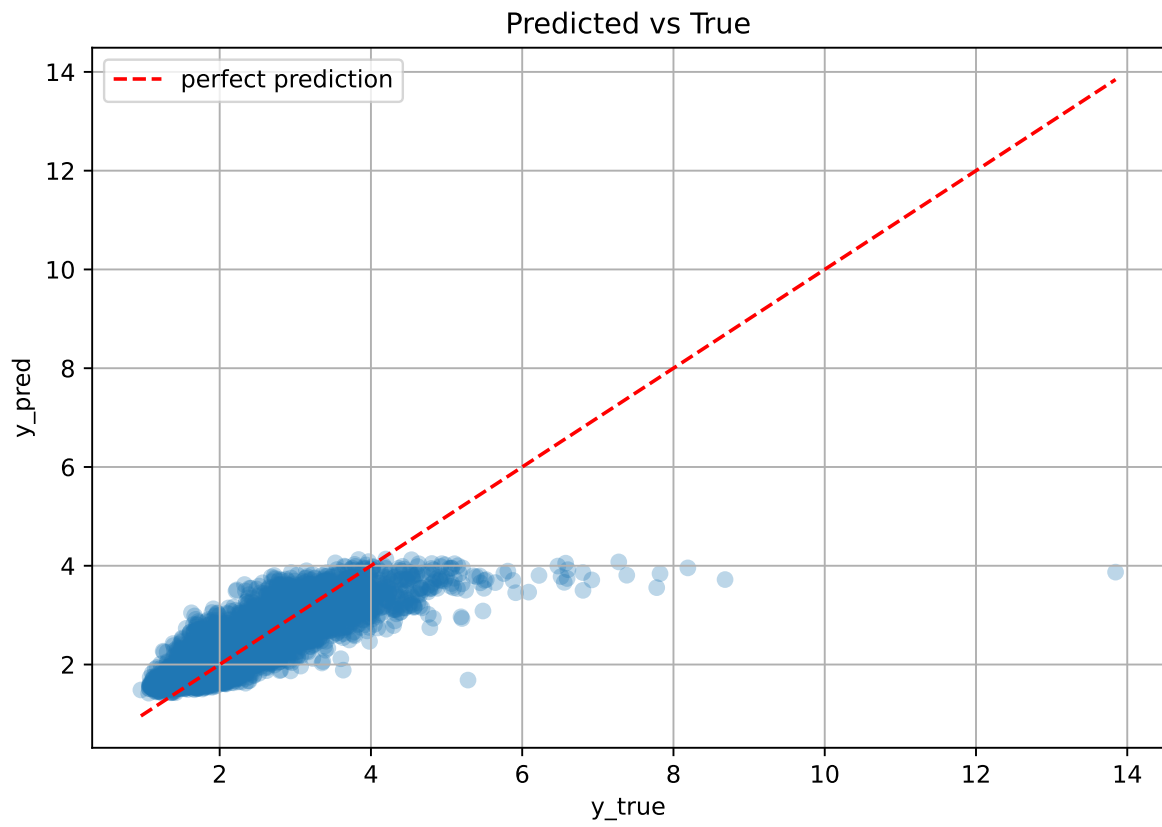
Rank 7: {'reg__n_estimators': 800, 'reg__max_depth': 8, 'reg__learning_rate': 0.022131307661033193, 'reg__num_leaves': 45, 'reg__subsample': 0.8916232920683076, 'reg__colsample_bytree': 0.5568420357795278, 'reg__min_child_samples': 50, 'reg__reg_alpha': 4.128536888212107, 'reg__reg_lambda': 34.61864890093662, 'reg__min_split_gain': 3.3341453657970535, 'reg__path_smooth': 2.2770294259067825, 'reg__min_child_weight': 0.43854851078304663, 'reg__feature_fraction': 0.6406122752172648, 'reg__bagging_fraction': 0.8450262745209445, 'reg__bagging_freq': 6, 'reg__max_bin': 249, 'reg__min_data_per_group': 113}

Rank 8: {'reg__n_estimators': 400, 'reg__max_depth': 3, 'reg__learning_rate': 0.02338534150677764, 'reg__num_leaves': 48, 'reg__subsample': 0.7693030927512553, 'reg__colsample_bytree': 0.5295030378731166, 'reg__min_child_samples': 45, 'reg__reg_alpha': 5.719497773865469, 'reg__reg_lambda': 1.3054264755047011, 'reg__min_split_gain': 2.146265802014016, 'reg__path_smooth': 0.5789856321741418, 'reg__min_child_weight': 0.016419581237349146, 'reg__feature_fraction': 0.639250504700366, 'reg__bagging_fraction': 0.5789679596851188, 'reg__bagging_freq': 5, 'reg__max_bin': 118, 'reg__min_data_per_group': 68}

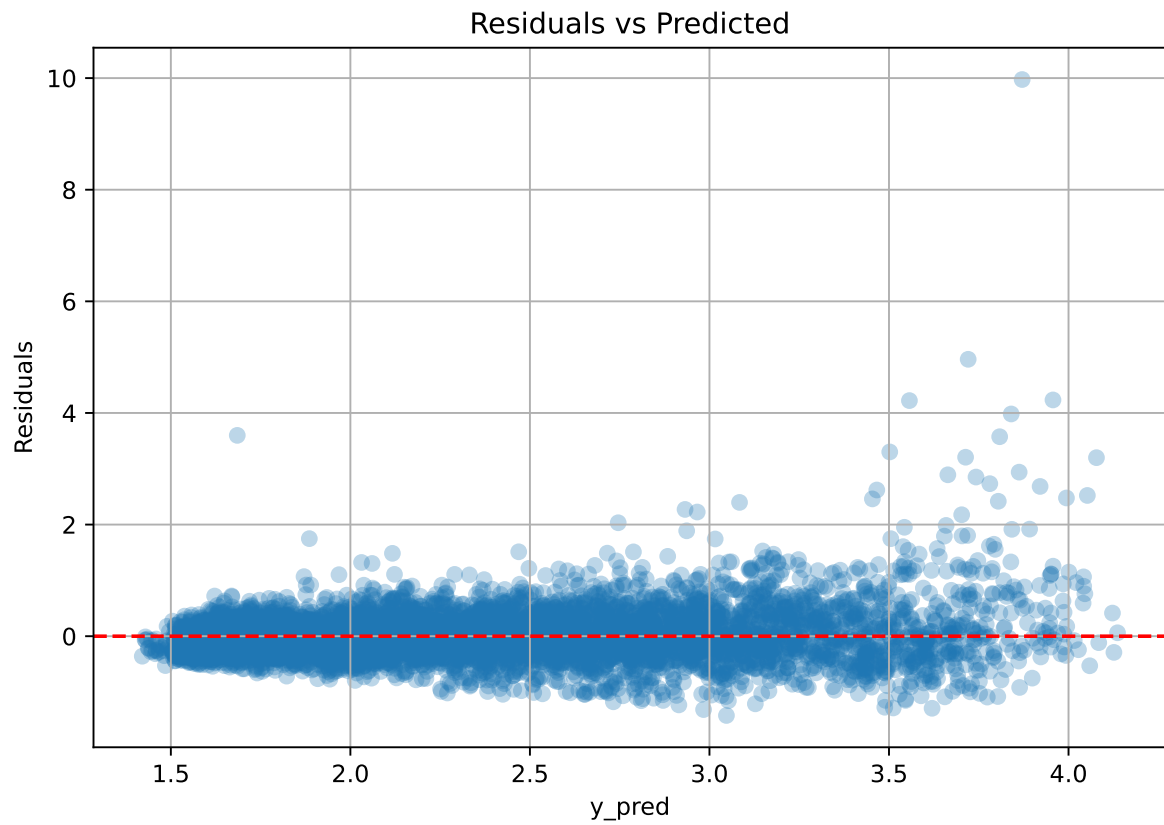
Rank 9: {'reg__n_estimators': 300, 'reg__max_depth': 3, 'reg__learning_rate': 0.07156115866187829, 'reg__num_leaves': 53, 'reg__subsample': 0.7115166437826026, 'reg__colsample_bytree': 0.8857992318939623, 'reg__min_child_samples': 28, 'reg__reg_alpha': 3.488628272911607, 'reg__reg_lambda': 13.184500962783963, 'reg__min_split_gain': 2.611795222425658, 'reg__path_smooth': 0.1798607547482478, 'reg__min_child_weight': 0.011550444593548372, 'reg__feature_fraction': 0.8916992067294061, 'reg__bagging_fraction': 0.6578092390583112, 'reg__bagging_freq': 6, 'reg__max_bin': 117, 'reg__min_data_per_group': 128}

Rank 10: {'reg__n_estimators': 300, 'reg__max_depth': 3, 'reg__learning_rate': 0.03839487835361631, 'reg__num_leaves': 54, 'reg__subsample': 0.7399115115039465, 'reg__colsample_bytree': 0.7282854811857375, 'reg__min_child_samples': 46, 'reg__reg_alpha': 0.0071231598264622775, 'reg__reg_lambda': 36.91946098257768, 'reg__min_split_gain': 2.9982407971001255, 'reg__path_smooth': 1.6438966037161225, 'reg__min_child_weight': 0.09181459034450044, 'reg__feature_fraction': 0.6148969581201819, 'reg__bagging_fraction': 0.6480422079230277, 'reg__bagging_freq': 6, 'reg__max_bin': 83, 'reg__min_data_per_group': 77}

3.5.2 Scatter



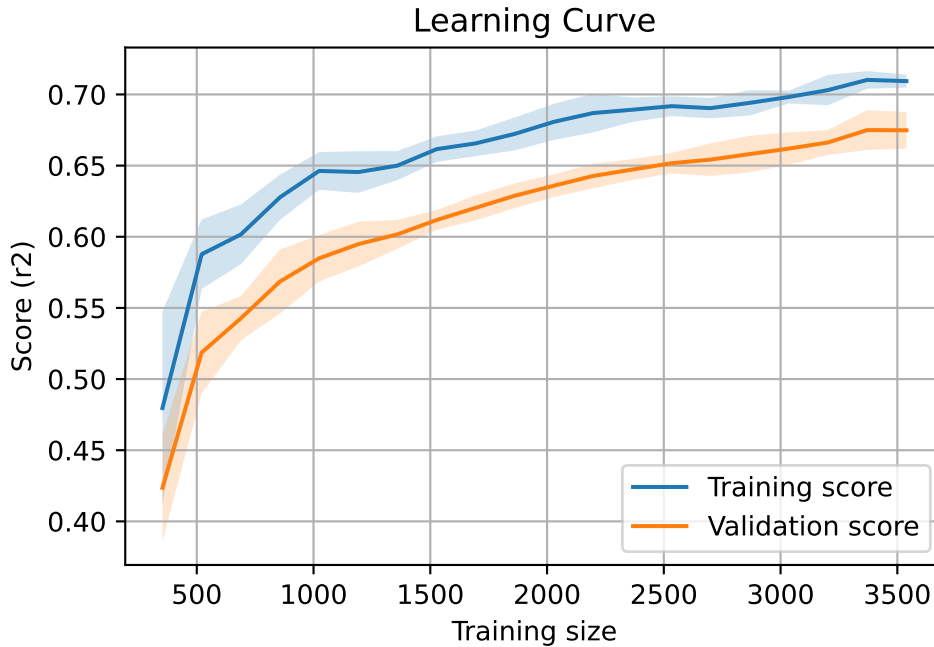
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 7224, `Len(training)` : 5057,

`Len(training_real)` : 3540, `Len(test_of_training)` : 1517, `Len(test)` : 2167



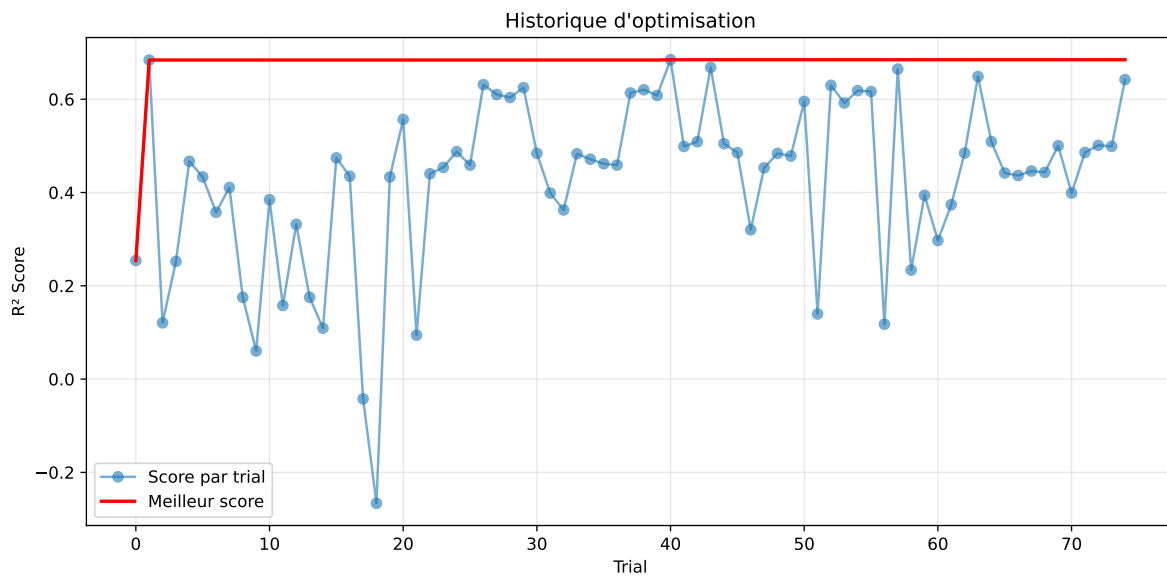
3.6 Model : CatBoost

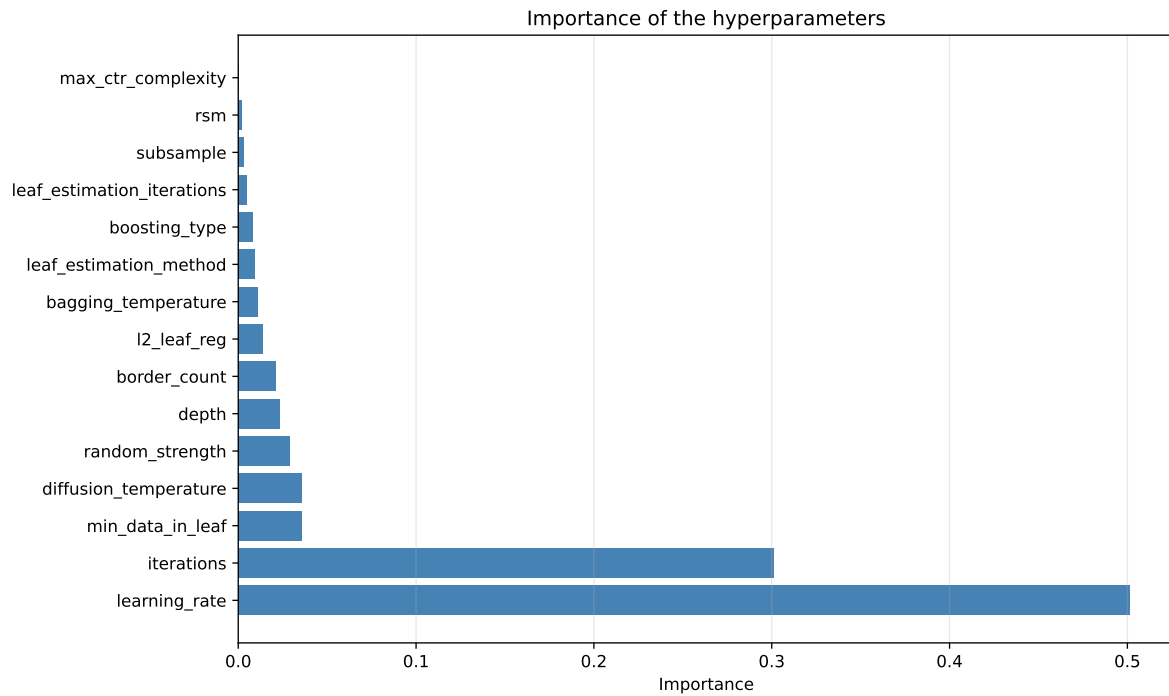
3.6.1 Gridsearch

Distribution y_train vs y_test: y_train: mean=2.346, std=0.787, min=0.959, max=8.190
y_test: mean=2.382, std=0.828, min=1.071, max=13.845

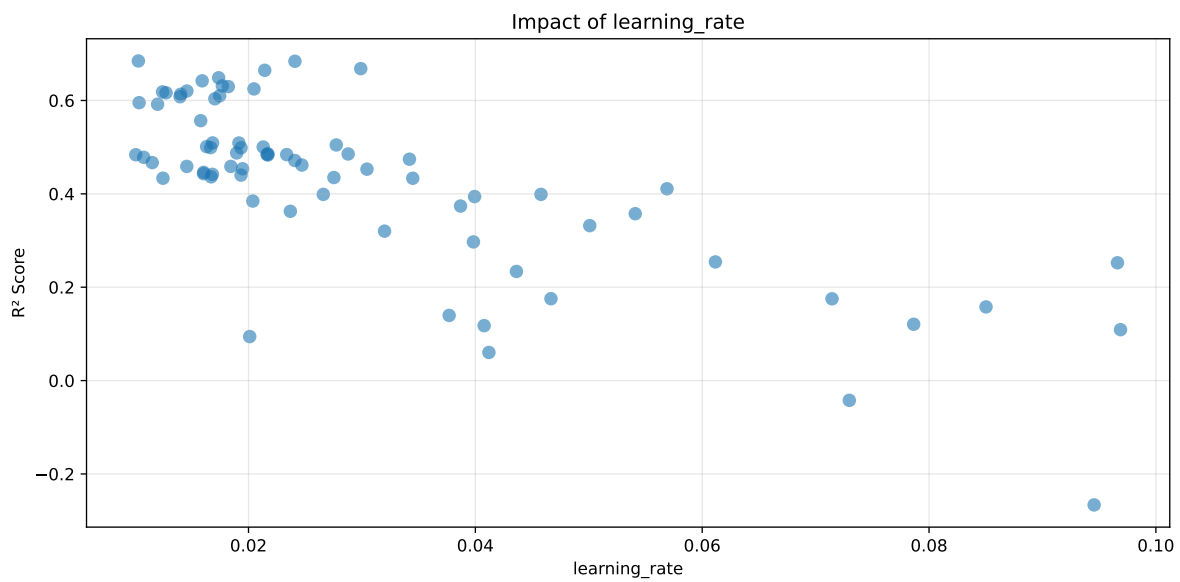
```
===== TOP Importance FEATURES =====
feature importance r_umidity 24.417153 r_temperature_wl_8h_A_energy 9.709364
r_temperature 9.209891 r_temperature_wl_12h_A_energy 7.290870 r_tempera-
ture_wl_12h_A_rms 6.102185 r_temperature_wl_8h_D1_energy_rel 5.291382
r_CO2 4.185290 r_umidity_wl_12h_A_rms 4.118405 r_temperature_wl_12h_A_en-
ergy_rel 2.876244 r_NH3 2.442130 r_THI 2.346973 r_umidity_wl_24h_D3_en-
ergy_rel 2.005105 r_NH3_wl_8h_D1_energy 1.940407 r_umidity_wl_24h_A_en-
ergy 1.789880 r_NH3_wl_24h_A_rms 1.746129 r_NH3_wl_12h_D1_energy 1.522668
r_NH3_wl_8h_A_energy 1.415413 r_NH3_wl_24h_D2_energy 1.354400 r_NH3_wl_8h_D2_en-
ergy 1.292755 r_NH3_wl_8h_D3_energy 0.919176 r_NH3_wl_12h_D2_energy
0.848292 r_NH3_wl_12h_A_energy 0.775847 r_NH3_wl_8h_D3_energy_rel 0.721256
r_NH3_wl_12h_D3_energy_rel 0.650641 r_umidity_wl_8h_D1_energy_rel 0.612460
r_NH3_wl_12h_D2_energy_rel 0.575163 r_umidity_wl_12h_A_energy_rel 0.523648
r_temperature_wl_24h_A_energy 0.515225 r_umidity_wl_12h_A_energy 0.426045
r_umidity_wl_8h_D2_energy_rel 0.358427 r_umidity_wl_8h_A_rms 0.331907
```

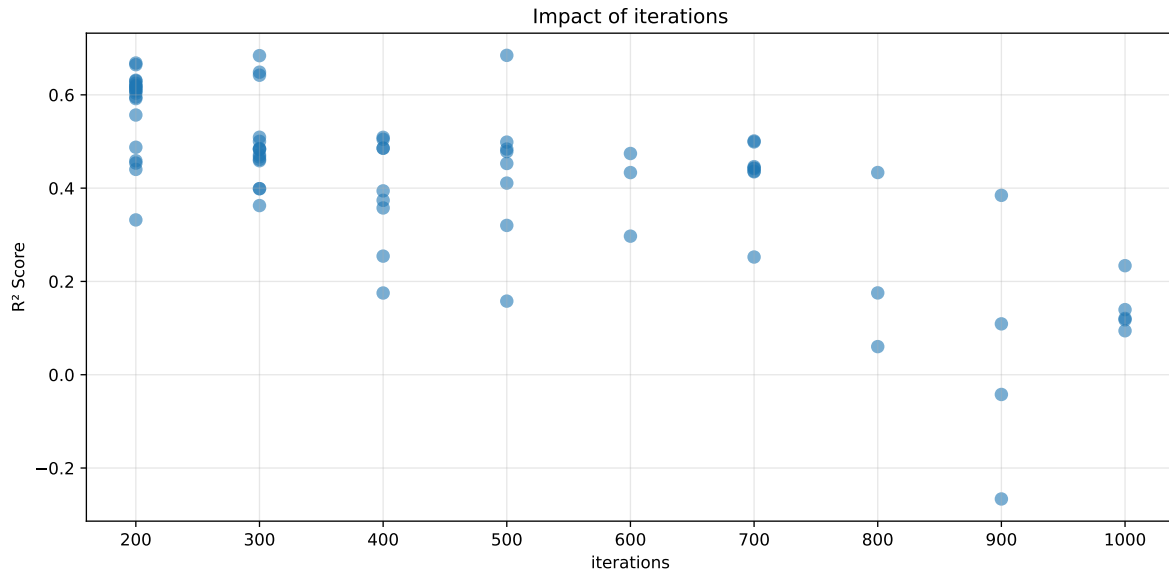
r_NH3_wl_8h_A_energy_rel 0.326073 r_temperature_wl_24h_A_rms 0.322386
 r_NH3_wl_12h_A_rms 0.223476 r_temperature_wl_8h_A_rms 0.211512 r_NH3_wl_8h_A_rms
 0.181195 r_umidity_wl_24h_A_rms 0.101139 r_temperature_wl_24h_D1_energy 0.078594
 r_umidity_wl_12h_D3_energy 0.077770 r_temperature_wl_24h_D3_energy 0.060829
 r_temperature_wl_24h_D2_energy 0.040208 r_temperature_wl_8h_D3_energy_rel
 0.032776 r_umidity_wl_8h_D3_energy 0.029312 r_temperature_wl_12h_D3_energy_rel
 0.000000





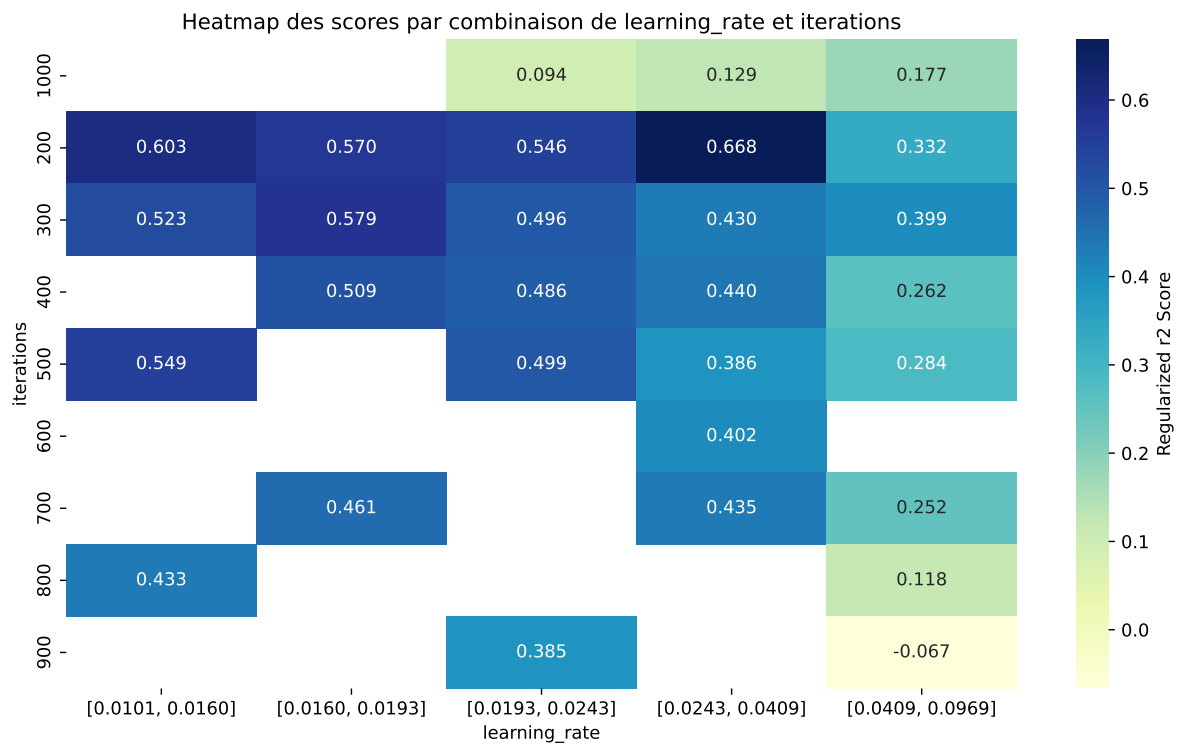
Paramètres avec >5% d'importance : ['learning_rate', 'iterations']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7164	0.6847	0.6281	0.0077	0.6847	0.6359	-0.0487	0.9288	1.0464
2	0.7178	0.684	0.6266	0.0081	0.684	0.6323	-0.0517	0.9244	1.0494
3	0.6976	0.6682	0.6083	0.0096	0.6682	0.6171	-0.0511	0.9235	1.044
4	0.6965	0.6646	0.6068	0.0089	0.6646	0.6121	-0.0526	0.9209	1.0479
5	0.671	0.6486	0.5911	0.01	0.6486	0.5975	-0.0511	0.9212	1.0345
6	0.6639	0.6421	0.5853	0.011	0.6421	0.5962	-0.0458	0.9286	1.0341
7	0.6596	0.6313	0.5803	0.0082	0.6313	0.5816	-0.0496	0.9214	1.0448
8	0.6511	0.6297	0.5731	0.0115	0.6297	0.5831	-0.0466	0.926	1.0339
9	0.6435	0.6248	0.5679	0.0108	0.6248	0.5772	-0.0476	0.9239	1.03
10	0.6396	0.6203	0.5638	0.0114	0.6203	0.5732	-0.0471	0.9241	1.0312
---	---	---	---	---	---	---	---	---	---
75	0.9228	0.7246	0.6823	0.0061	-0.2663	0.7191	-0.0055	0.9925	1.2735

Hyperparameters:

Rank 1: {'reg__iterations': 500, 'reg__depth': 4, 'reg__learning_rate': 0.01028977491275845, 'reg__l2_leaf_reg': 27.790150415327073, 'reg__bagging_temperature': 0.1970625340217248, 'reg__random_strength': 0.6320266781771524, 'reg__subsample': 0.7115412478813403, 'reg__min_data_in_leaf': 43, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.7522435405609199, 'reg__border_count': 98, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 5298.7112778043265}

Rank 2: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.02409331049498687, 'reg__l2_leaf_reg': 16.92221318659996, 'reg__bagging_temperature': 0.06707732500130292, 'reg__random_strength': 1.6277174927287148, 'reg__subsample': 0.6839799650438366, 'reg__min_data_in_leaf': 33, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.6366718997658437, 'reg__border_count': 50, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 7574.244819642651}

Rank 3: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.029896566121975422, 'reg__l2_leaf_reg': 27.456147909433017, 'reg__bagging_temperature': 0.037646347609402886, 'reg__random_strength': 2.6784035985788304, 'reg__subsample': 0.6233494723537432, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.7408679664248082, 'reg__border_count': 97, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 7637.856888386788}

Rank 4: {'reg__iterations': 200, 'reg__depth': 5, 'reg__learning_rate': 0.02143197923589406, 'reg__l2_leaf_reg': 32.11705715328538, 'reg__bagging_temperature': 4.436966076218651, 'reg__random_strength': 2.7263237679510777, 'reg__subsample': 0.6471811093841363, 'reg__min_data_in_leaf': 20, 'reg__leaf_estimation_iterations': 8, 'reg__rsm':

0.6463722162864409, 'reg__border_count': 41, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 7573.333903134296}

Rank 5: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.01736805341637696, 'reg__l2_leaf_reg': 36.946795678792824, 'reg__bagging_temperature': 4.50938891206142, 'reg__random_strength': 3.088785837038822, 'reg__subsample': 0.6457390718785889, 'reg__min_data_in_leaf': 20, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.7813398586767111, 'reg__border_count': 36, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 7257.807529712325}

Rank 6: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.01591464983668851, 'reg__l2_leaf_reg': 22.29904500726465, 'reg__bagging_temperature': 4.580421854302346, 'reg__random_strength': 3.2481154100023093, 'reg__subsample': 0.6735579658382307, 'reg__min_data_in_leaf': 14, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.6491498860197665, 'reg__border_count': 48, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 7834.267717538533}

Rank 7: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.017693908766981768, 'reg__l2_leaf_reg': 12.352370821790995, 'reg__bagging_temperature': 3.9699370767211972, 'reg__random_strength': 0.07544736642670591, 'reg__subsample': 0.6071261253999356, 'reg__min_data_in_leaf': 29, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.7450600957458788, 'reg__border_count': 125, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 6942.806697685359}

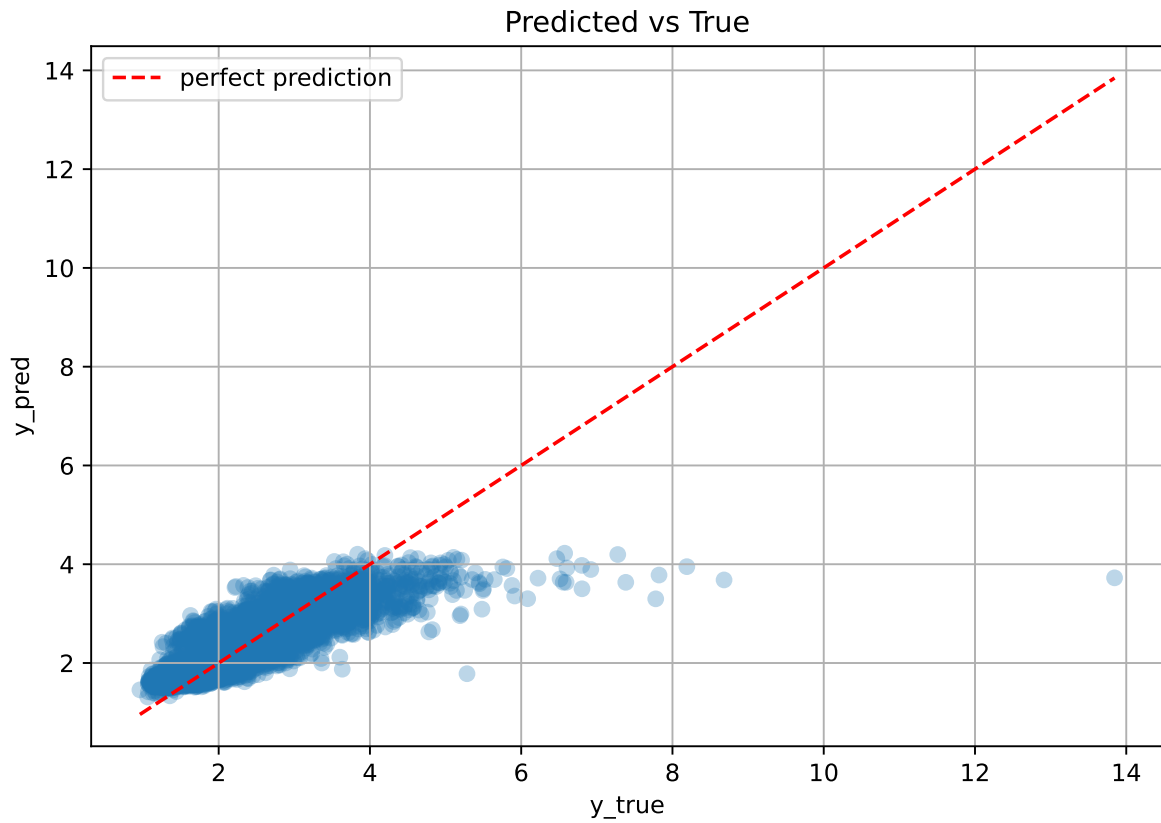
Rank 8: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.018215506724620873, 'reg__l2_leaf_reg': 29.685251537983987, 'reg__bagging_temperature': 3.935416865873433, 'reg__random_strength': 2.7557501478322775, 'reg__subsample': 0.7593188705544174, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.8368839972524387, 'reg__border_count': 40, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 6790.34206026599}

Rank 9: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.020483001657219134, 'reg__l2_leaf_reg': 11.164285669567244, 'reg__bagging_temperature': 4.0394783234550875, 'reg__random_strength': 1.4867920586937737, 'reg__subsample': 0.5827898015294969, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.7061146695883412, 'reg__border_count': 121, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 6165.559488688744}

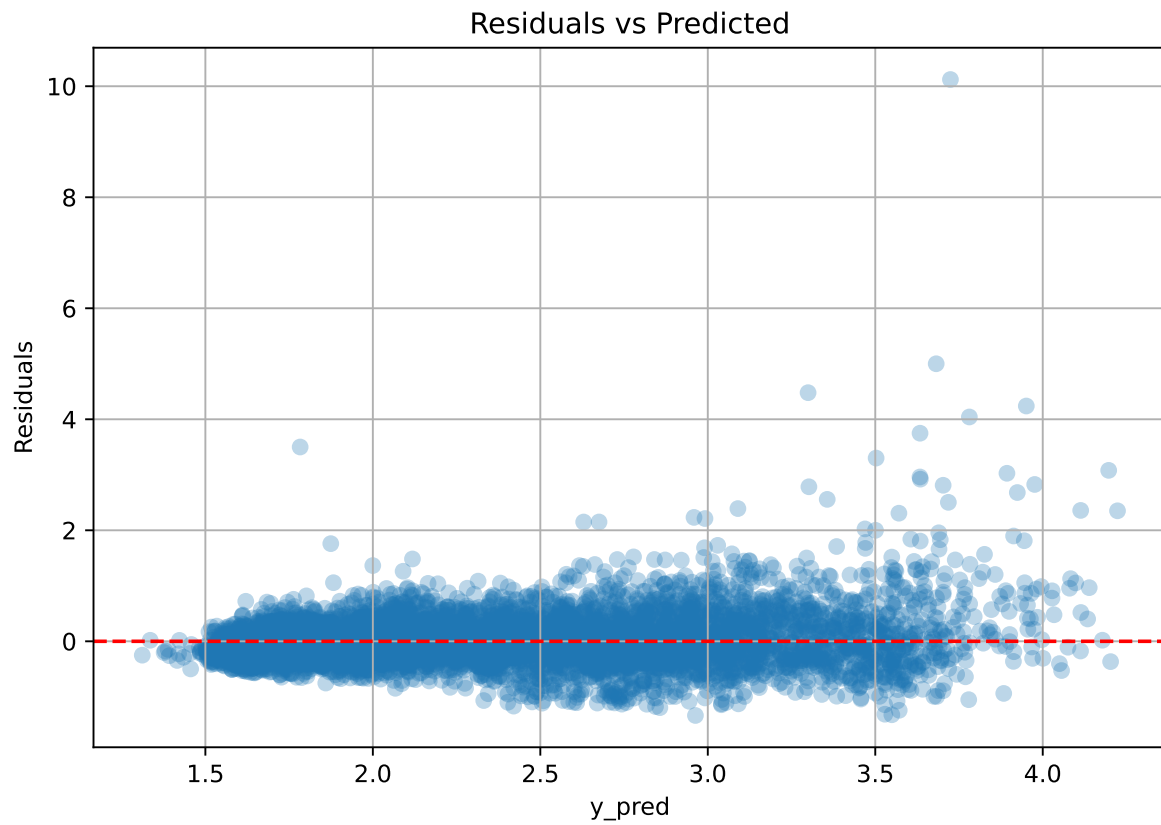
Rank 10: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.014568293442371386, 'reg__l2_leaf_reg': 16.68824638491471, 'reg__bagging_temperature': 3.342297020572072,

'reg__random_strength': 2.287643145481647, 'reg__subsample': 0.7434756177931281,
'reg__min_data_in_leaf': 22, 'reg__leaf_estimation_iterations': 6, 'reg__rsm':
0.8091036651160336, 'reg__border_count': 78, 'reg__leaf_estimation_method': 'Gradient',
'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature':
6675.234544018319}

3.6.2 Scatter



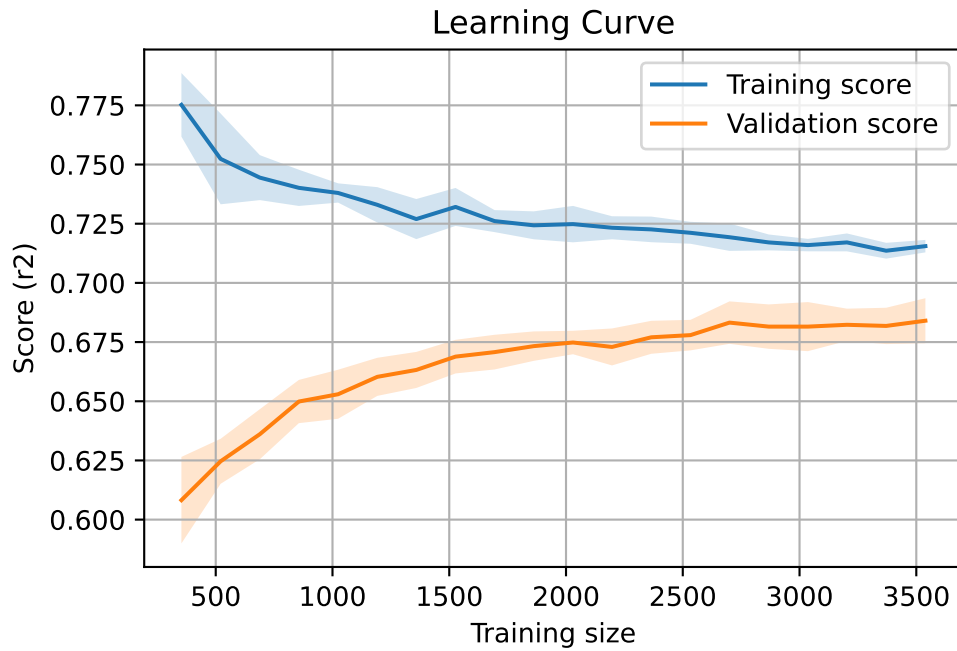
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 7224, `Len(training)` : 5057,

`Len(training_real)` : 3540, `Len(test_of_training)` : 1517, `Len(test)` : 2167



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.625	0.614	0.008	0.566	0.568	-0.046	0.926	1.019	nan
LGBM	0.708	0.675	0.011	0.618	0.643	-0.032	0.952	1.049	0.6754
CatBoost	0.716	0.685	0.008	0.628	0.636	-0.049	0.929	1.046	0.6847

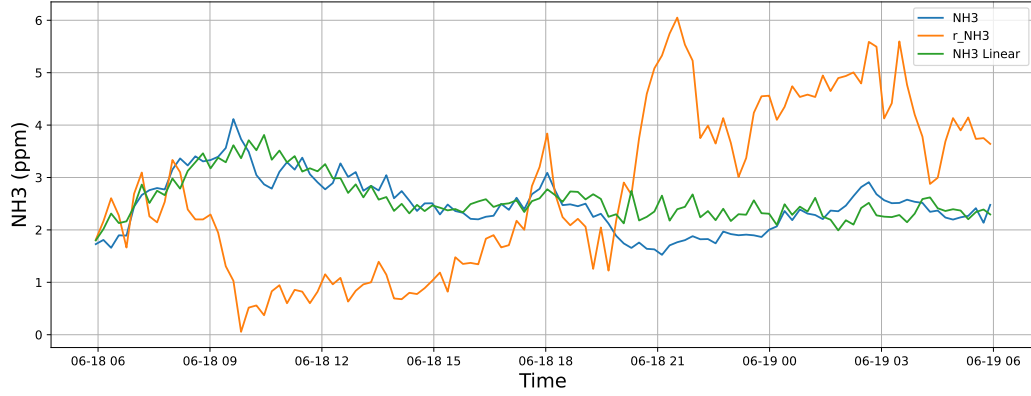
4 Plot

4.1 Standalone plots

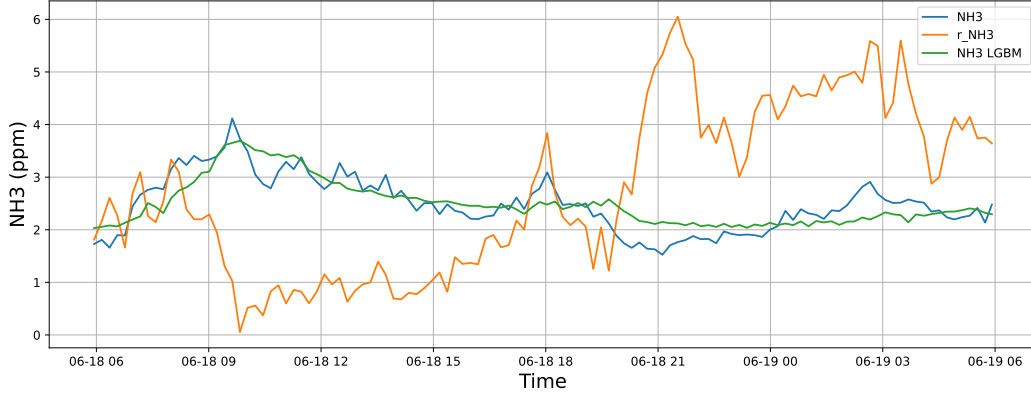
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

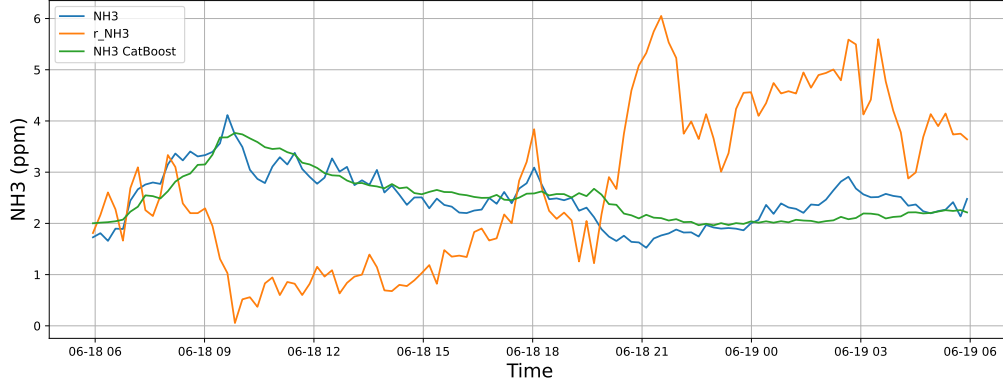
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

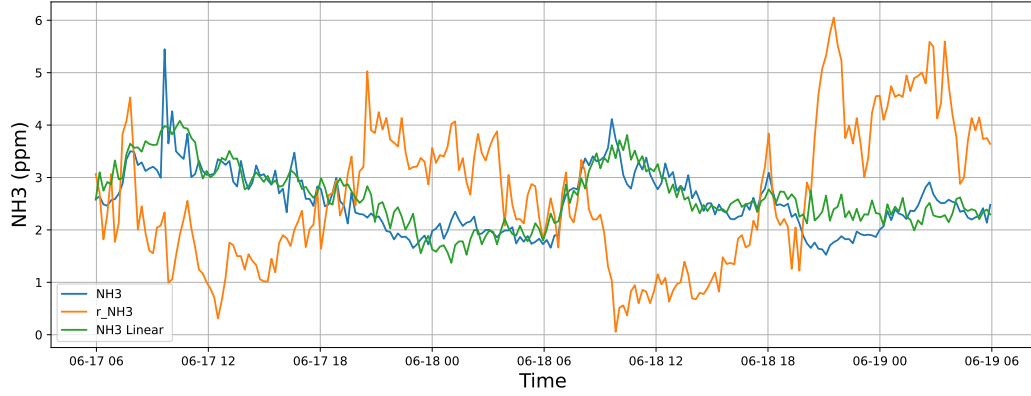


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

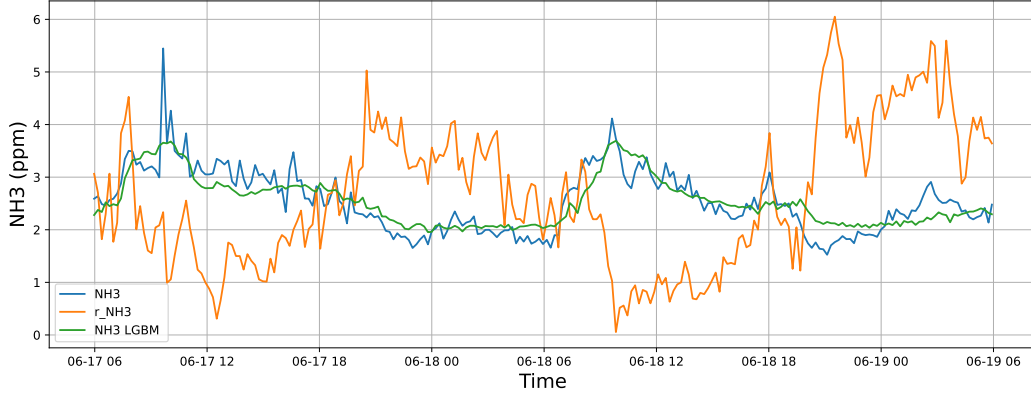


4.1.1.2 Time window : 48

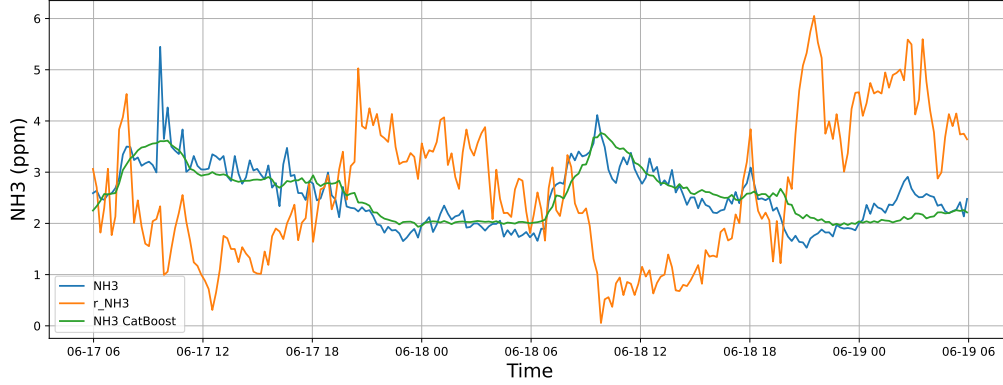
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

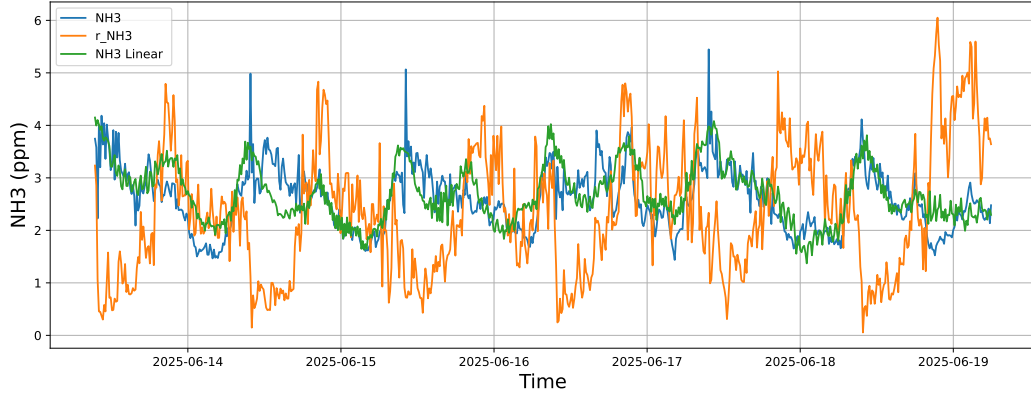


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

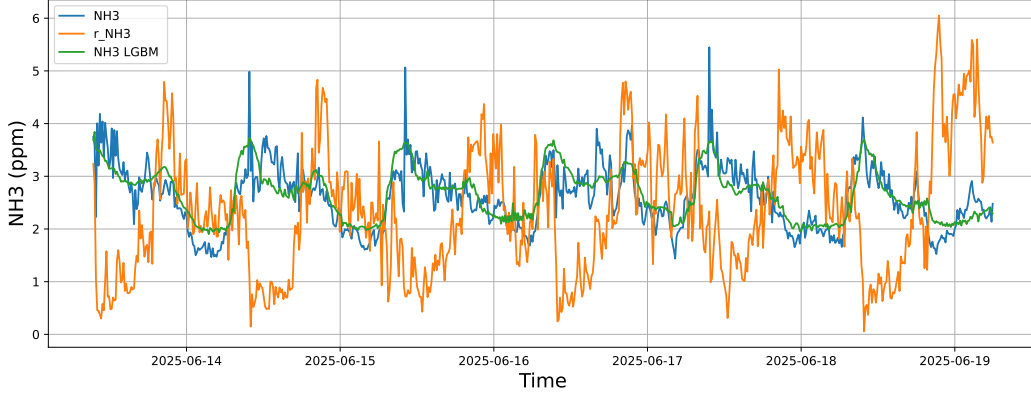


4.1.1.3 Time window : ALL

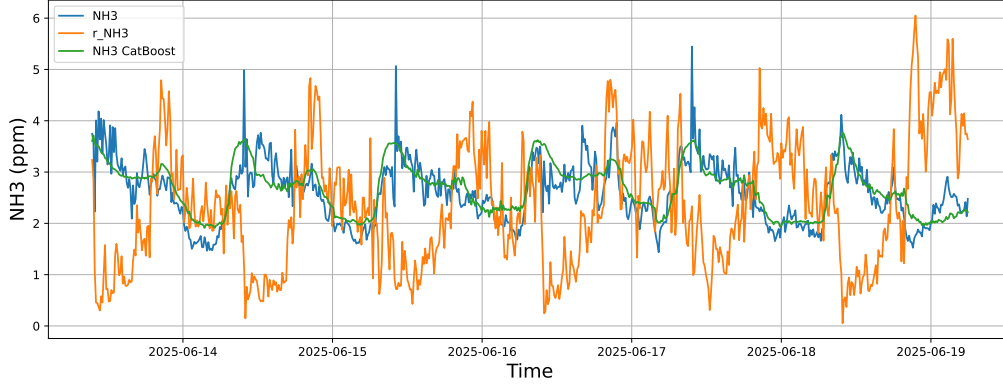
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-13 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-13 → 2025-06-19



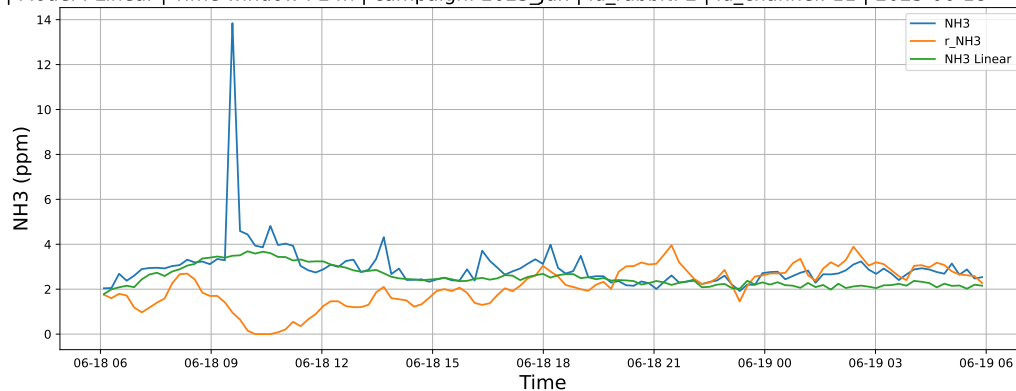
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-13 → 2025-06-19



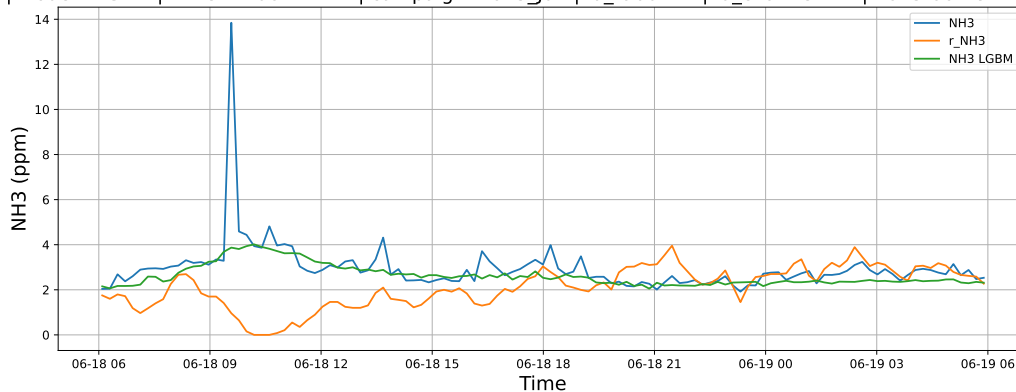
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

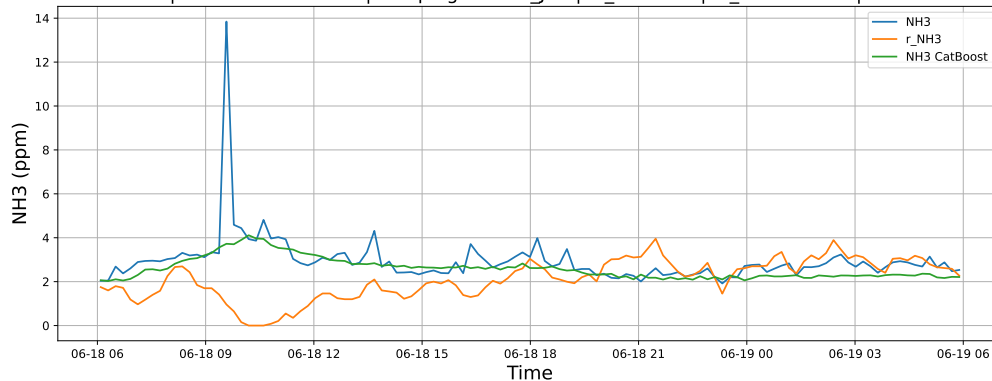
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

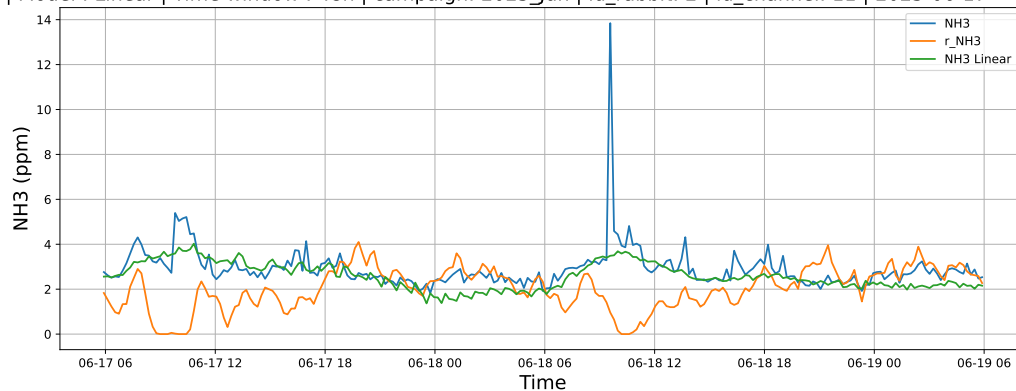


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

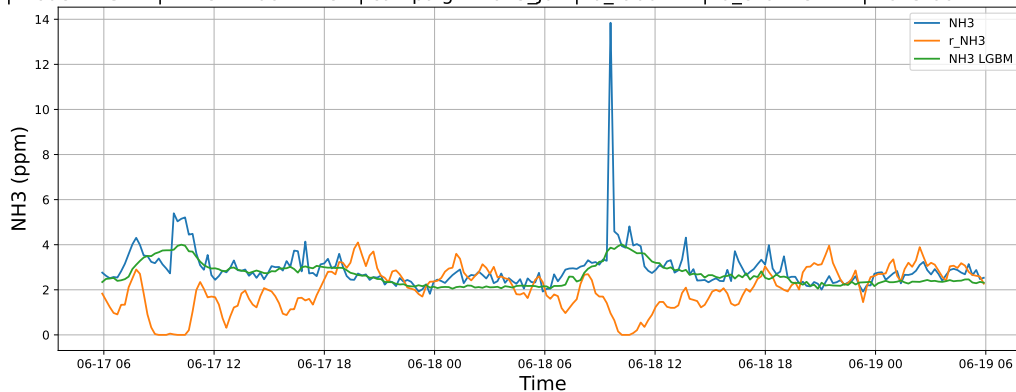


4.1.2.2 Time window : 48

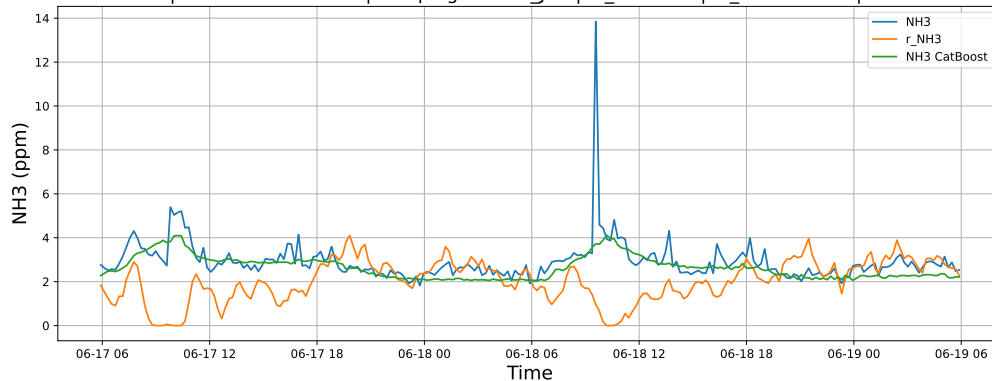
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

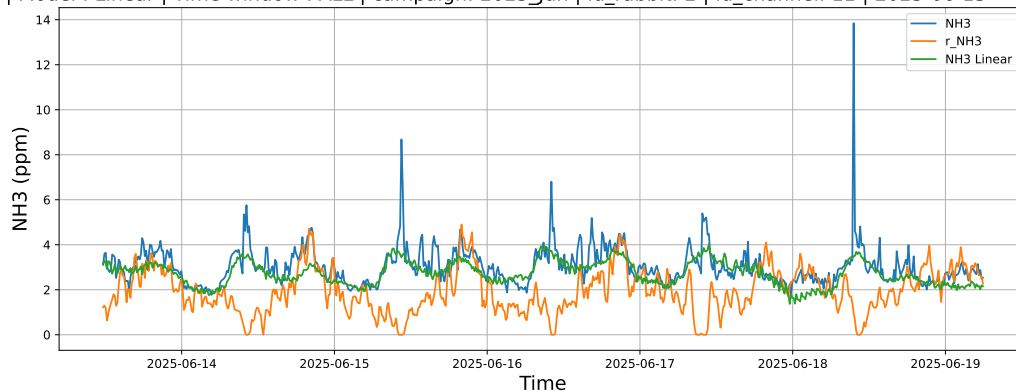


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

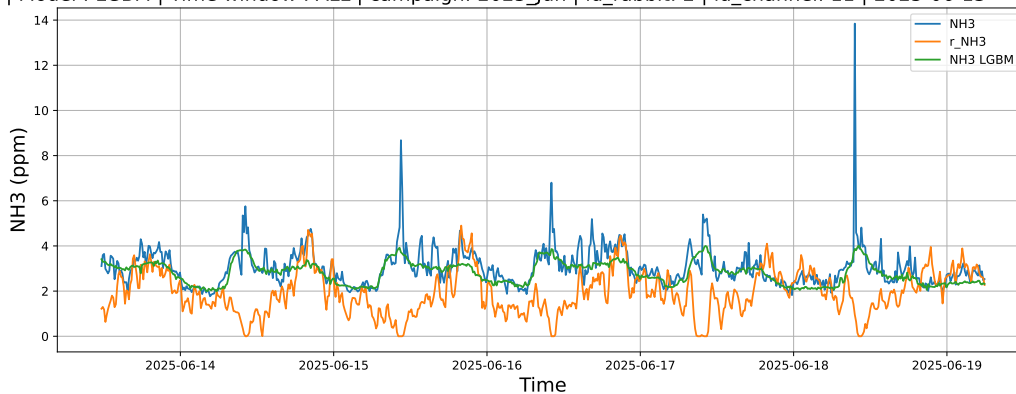


4.1.2.3 Time window : ALL

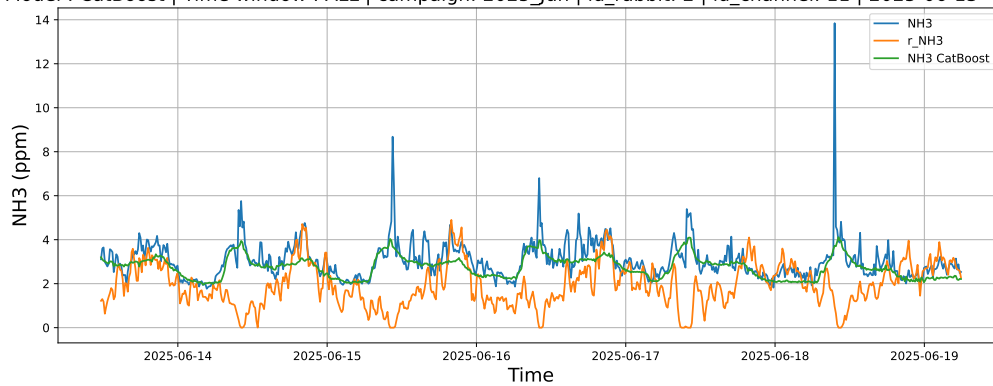
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-13 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-13 → 2025-06-19



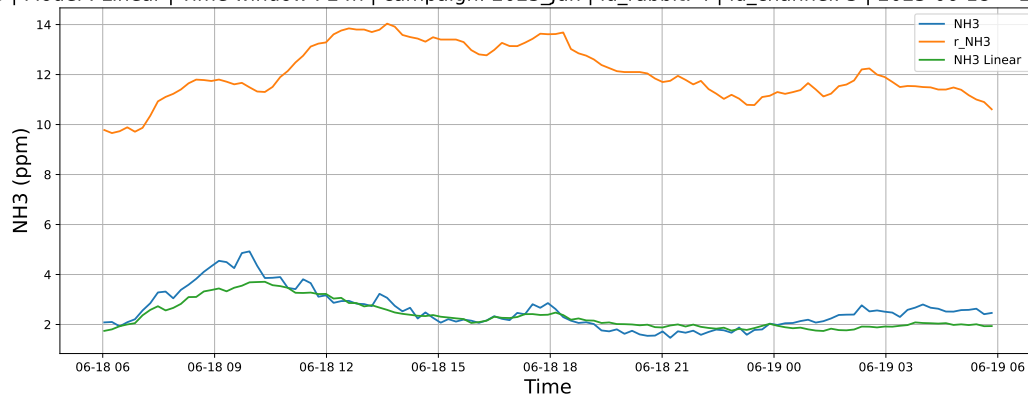
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-13 → 2025-06-19



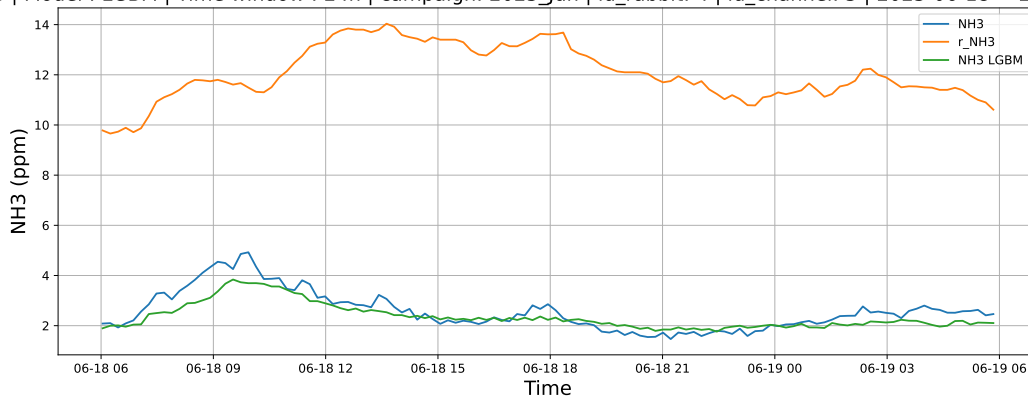
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

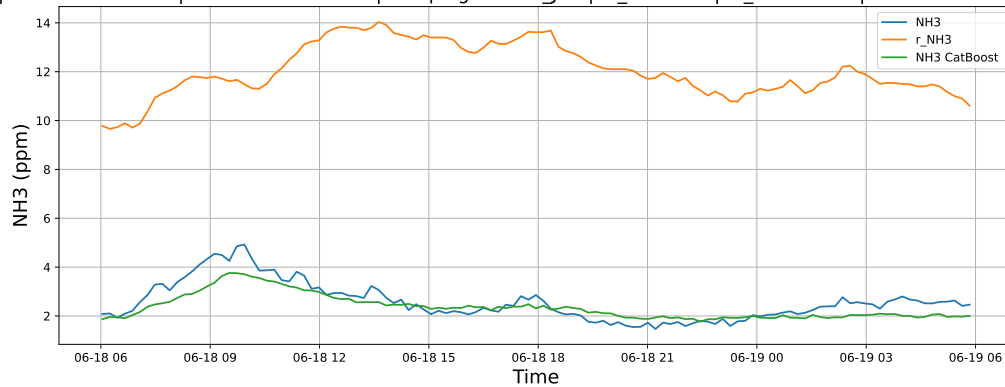
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

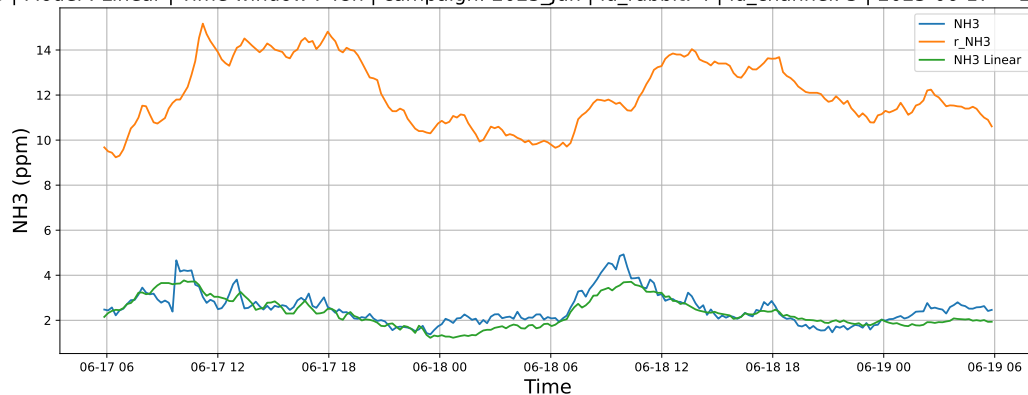


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

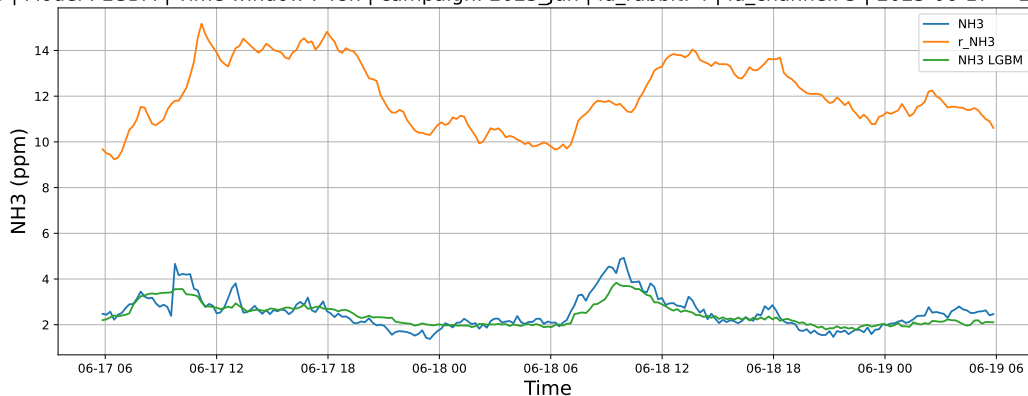


4.1.3.2 Time window : 48

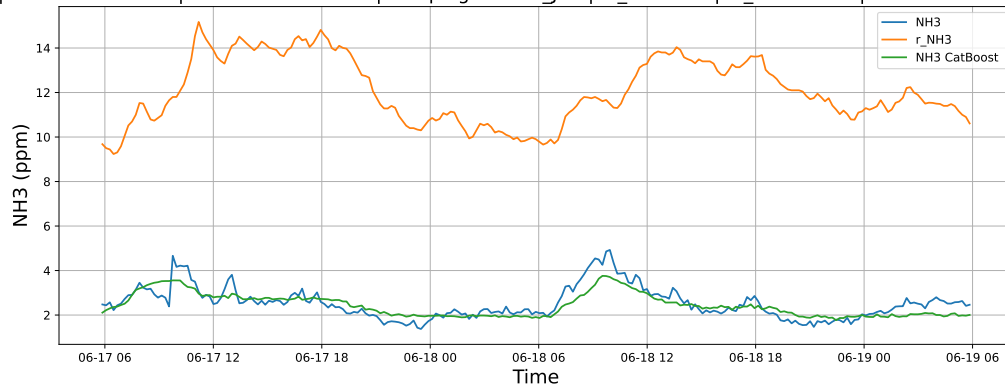
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

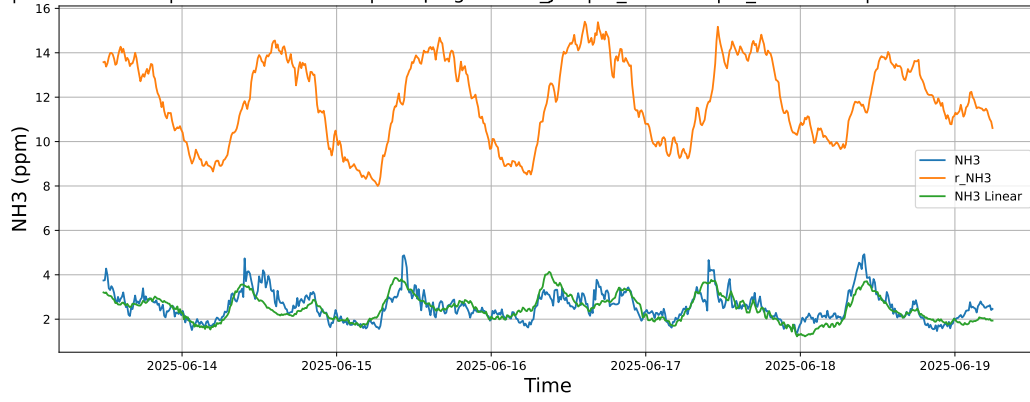


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

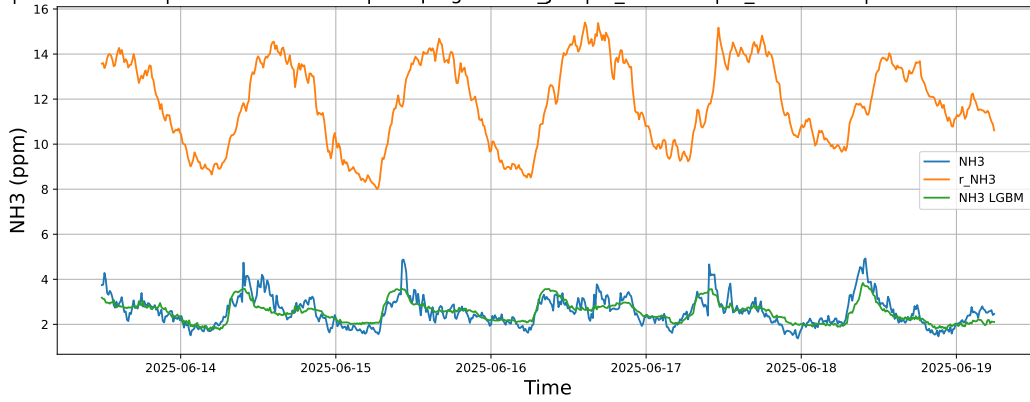


4.1.3.3 Time window : ALL

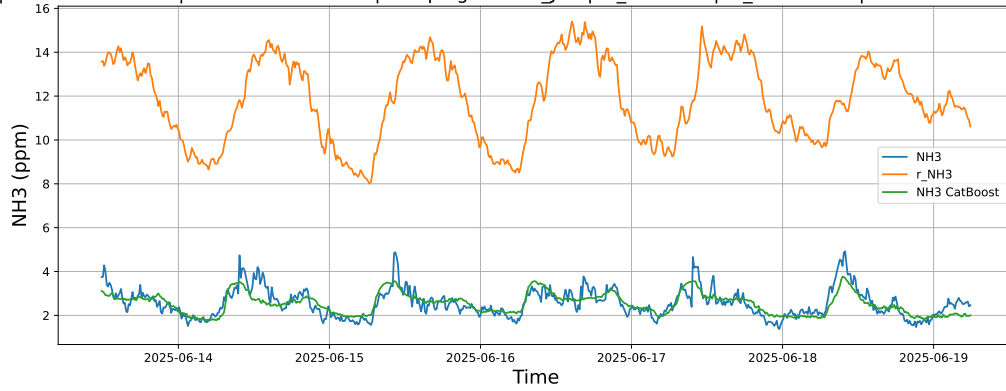
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-13 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-13 → 2025-06-19



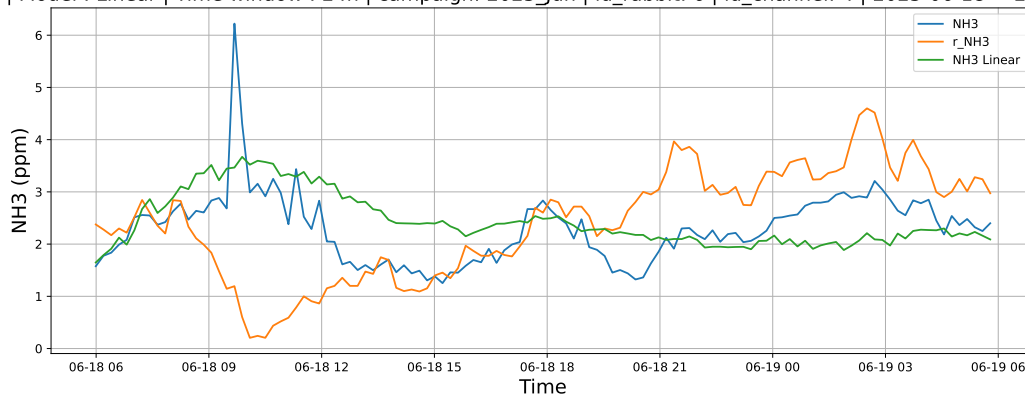
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-13 → 2025-06-19



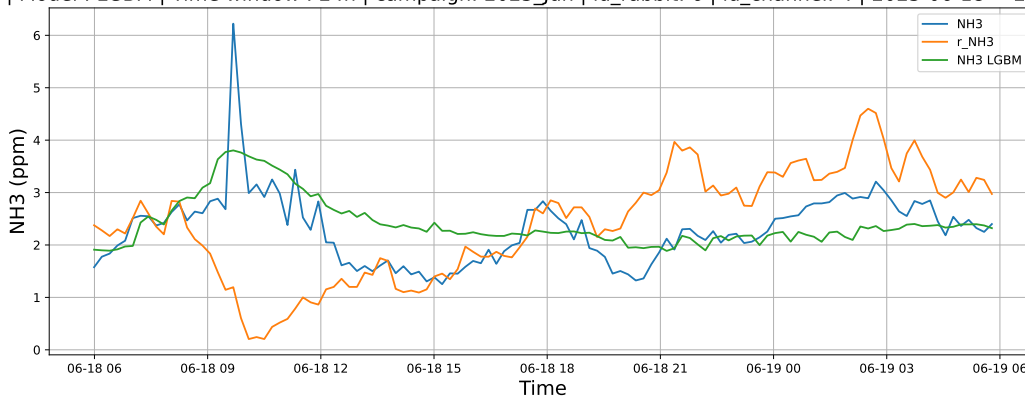
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

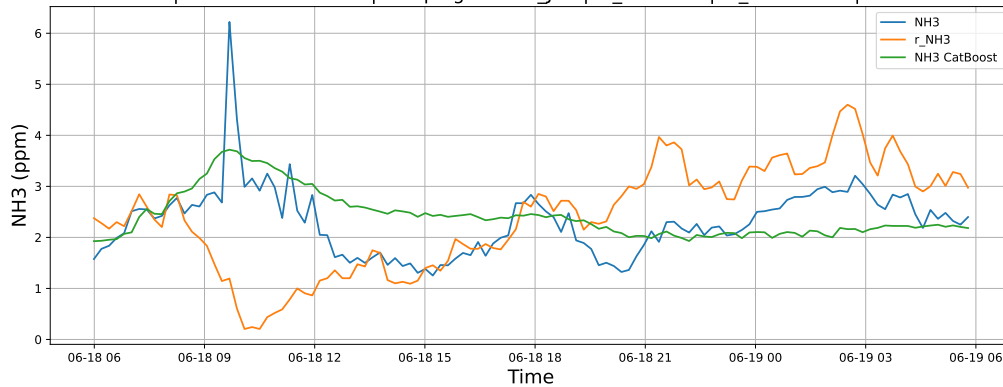
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

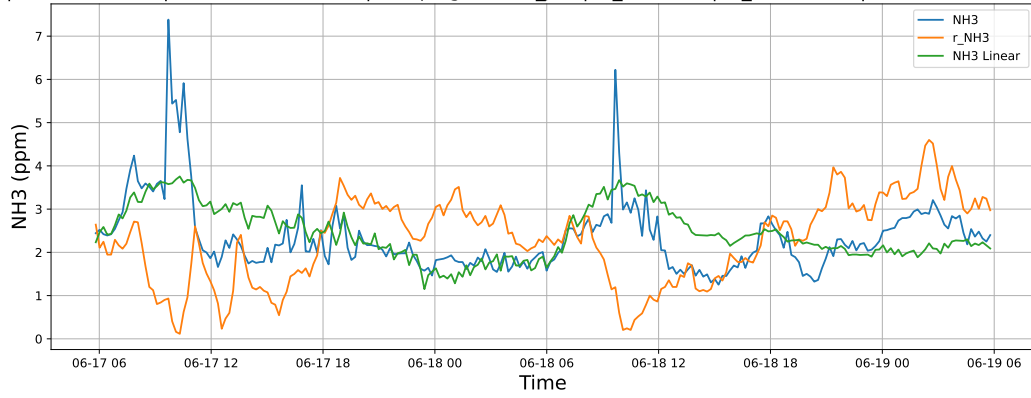


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

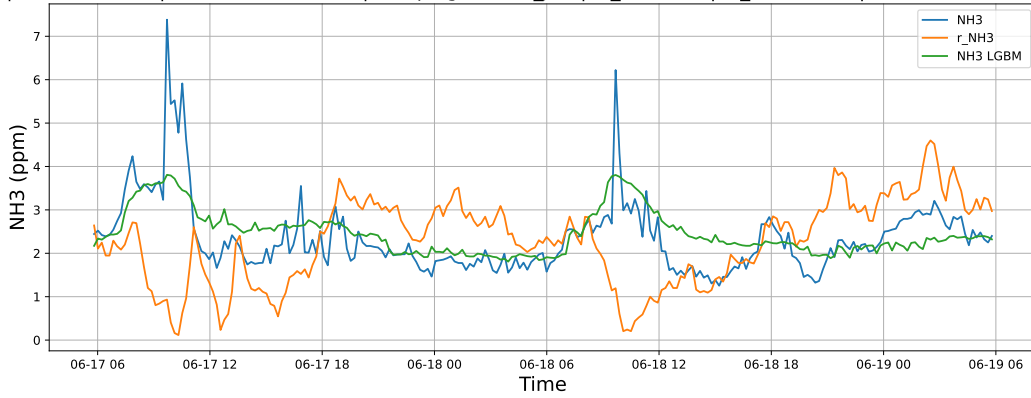


4.1.4.2 Time window : 48

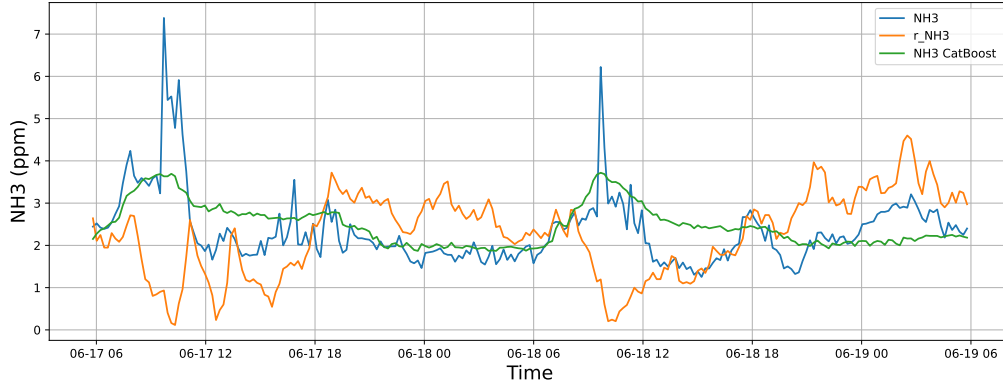
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

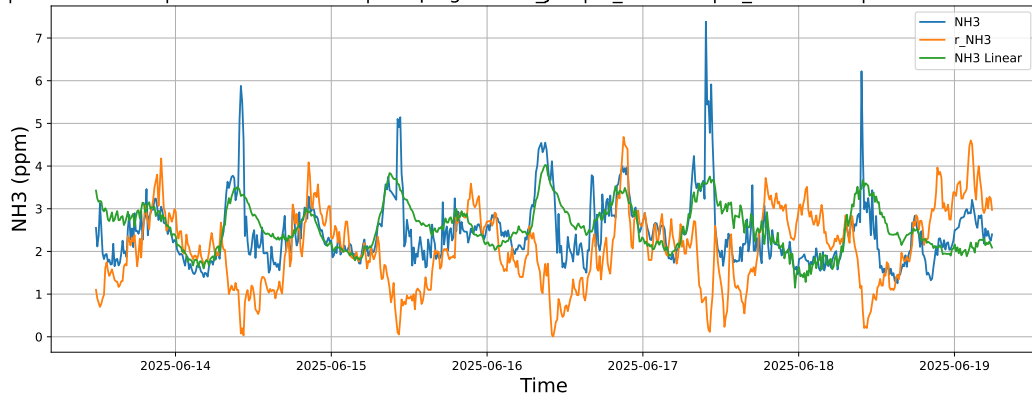


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

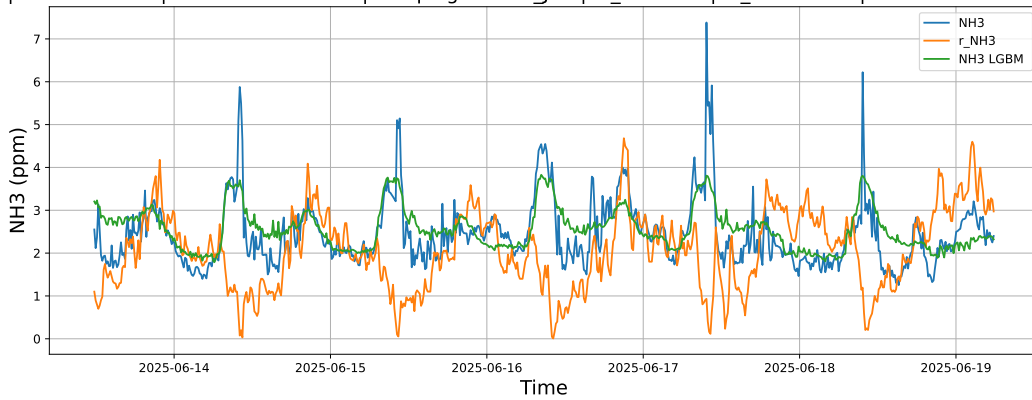


4.1.4.3 Time window : ALL

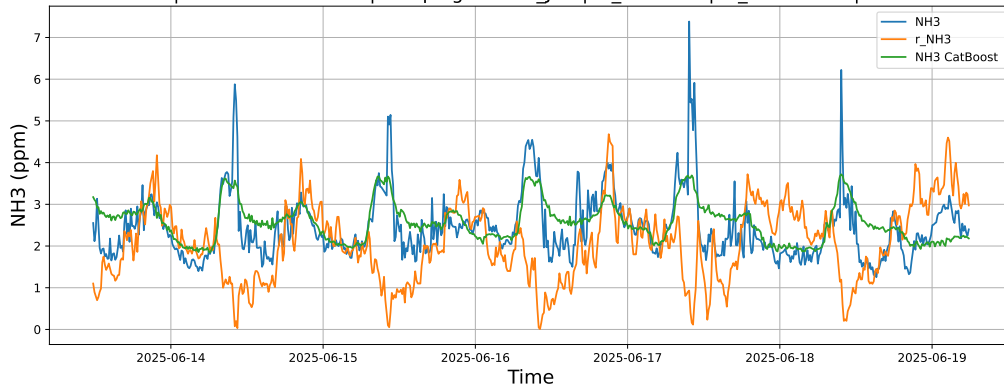
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-13 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-13 → 2025-06-19



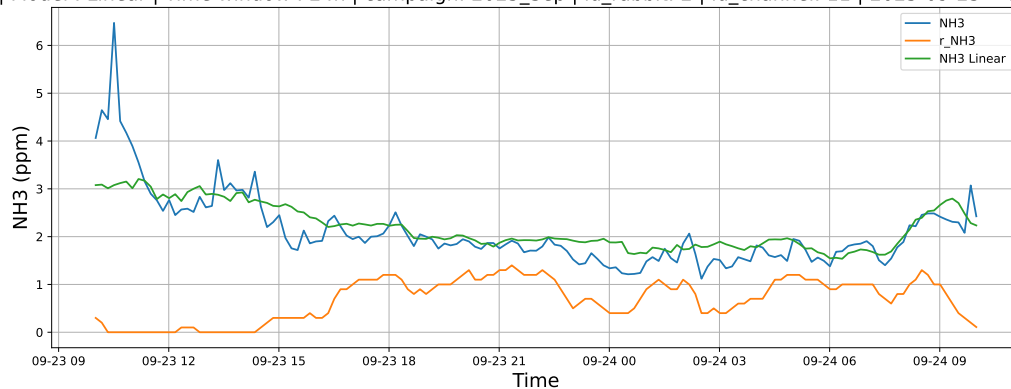
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-13 → 2025-06-19



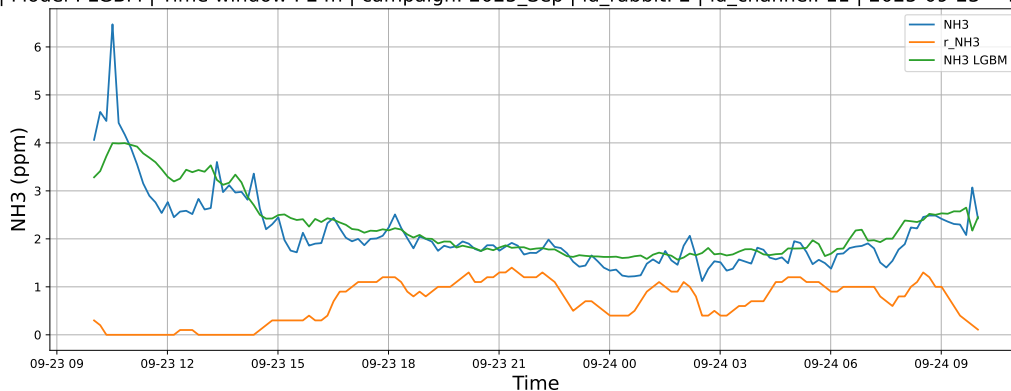
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

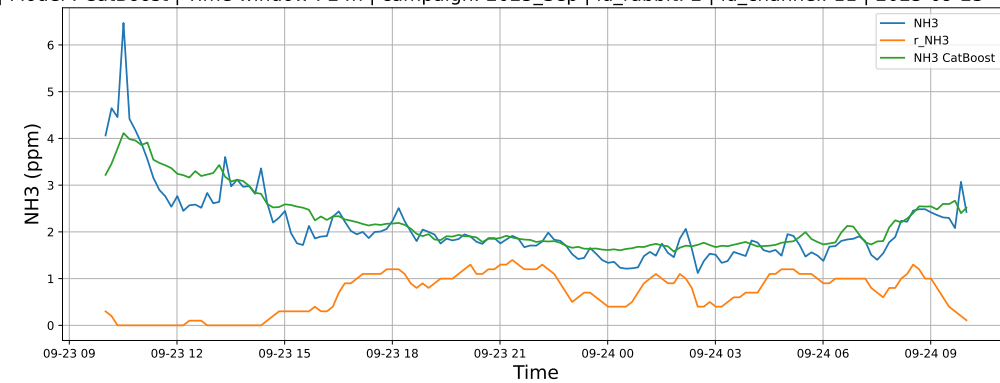
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

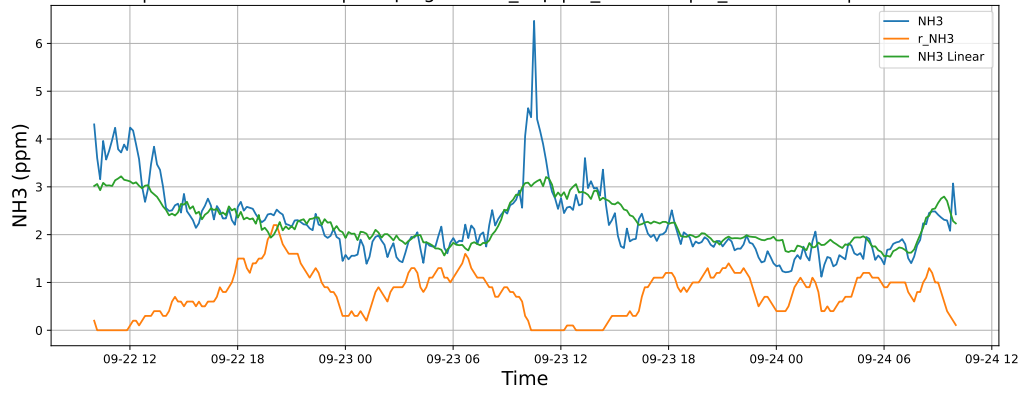


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

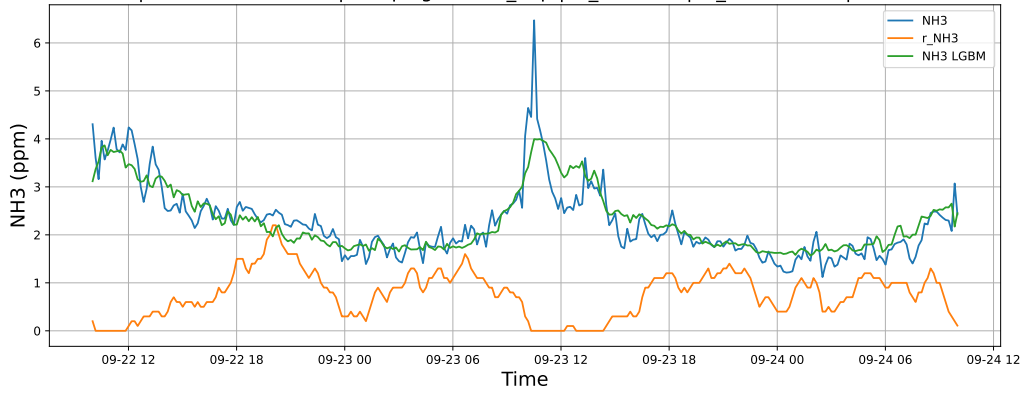


4.1.5.2 Time window : 48

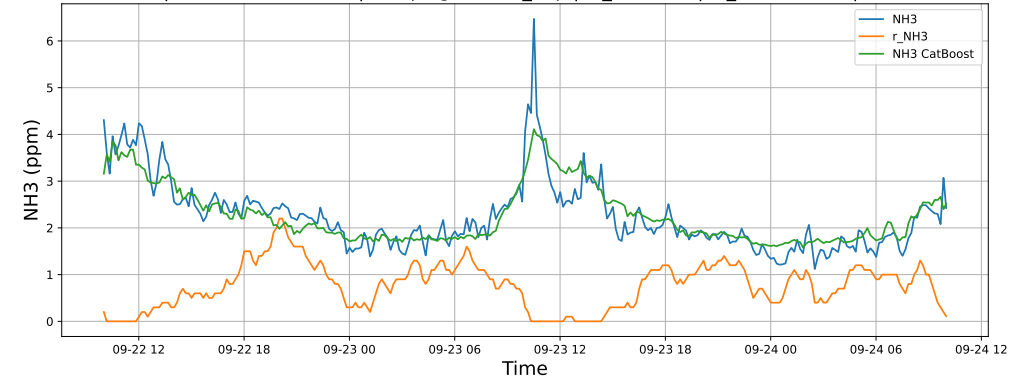
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

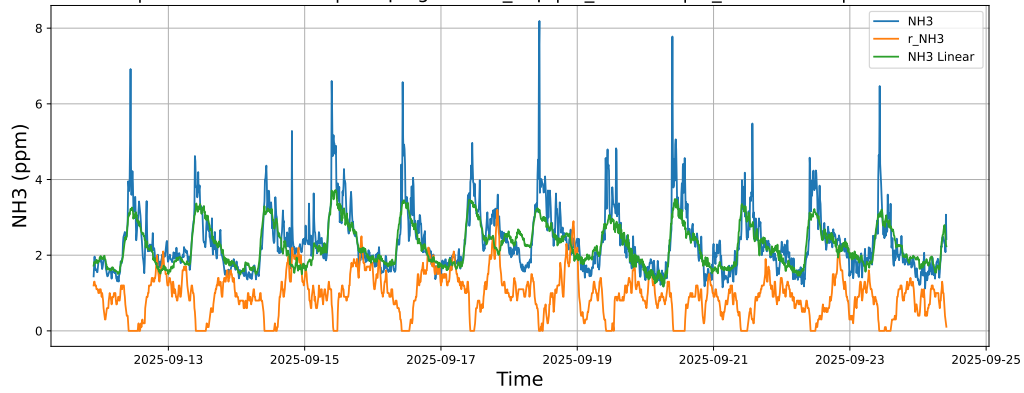


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

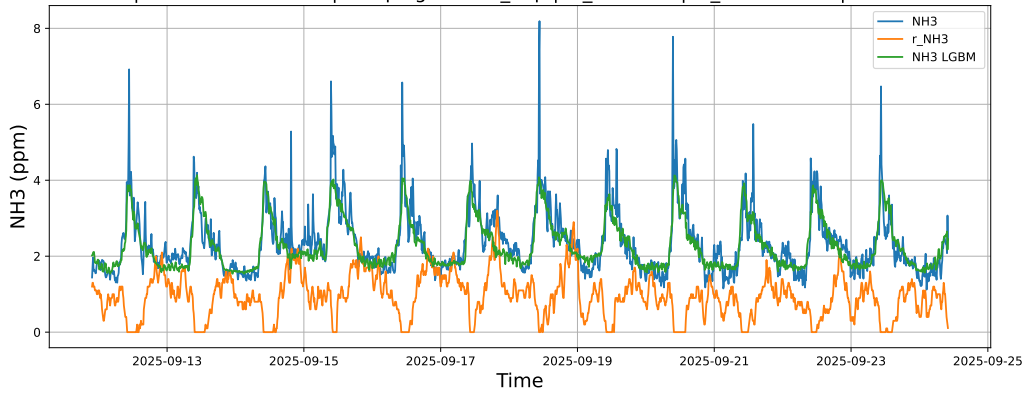


4.1.5.3 Time window : ALL

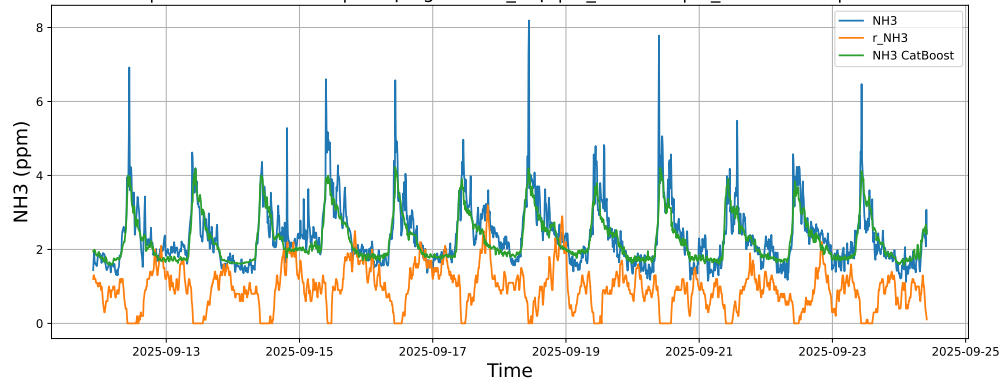
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



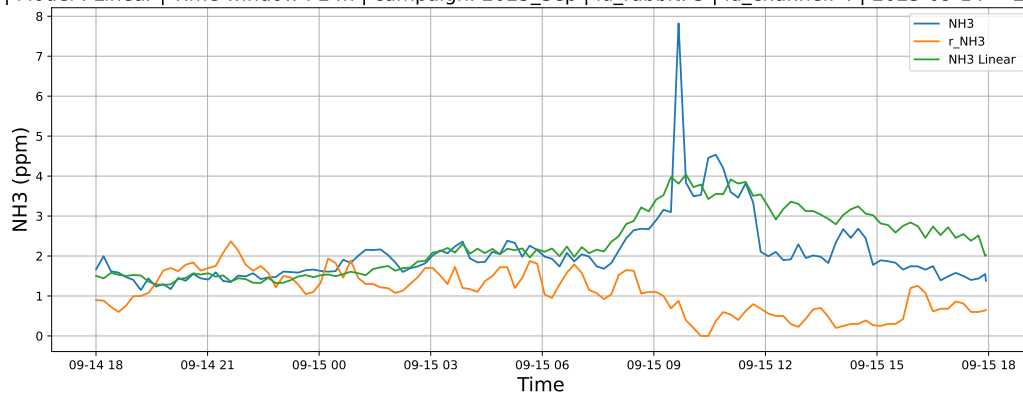
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



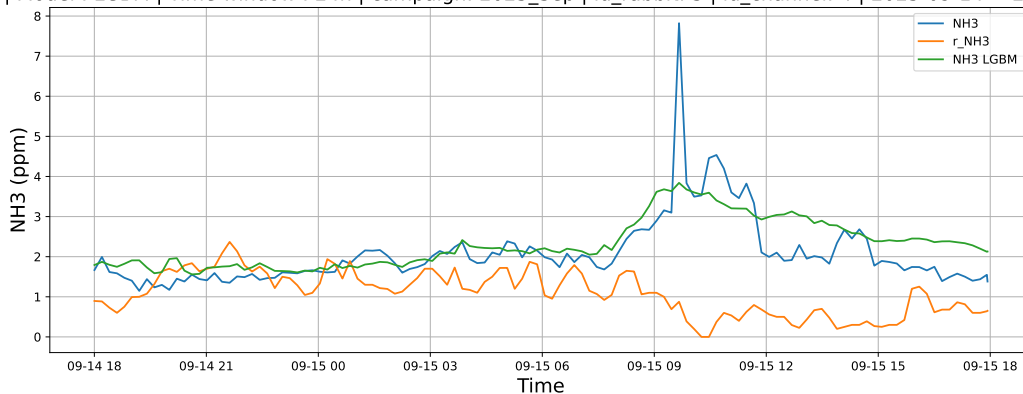
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

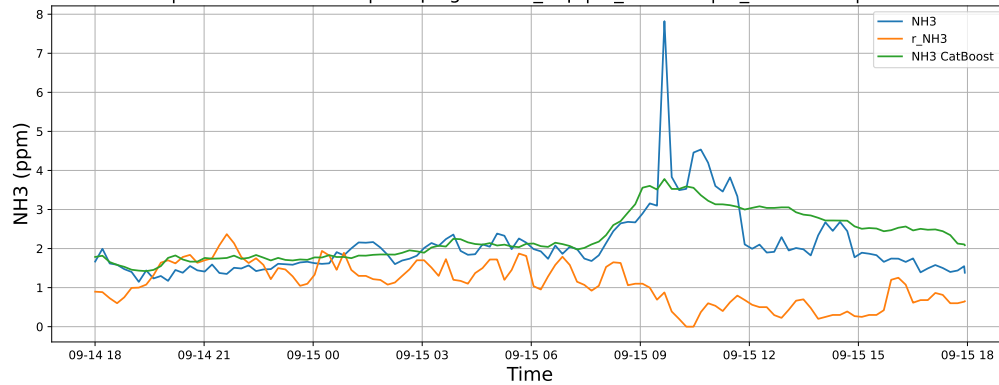
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

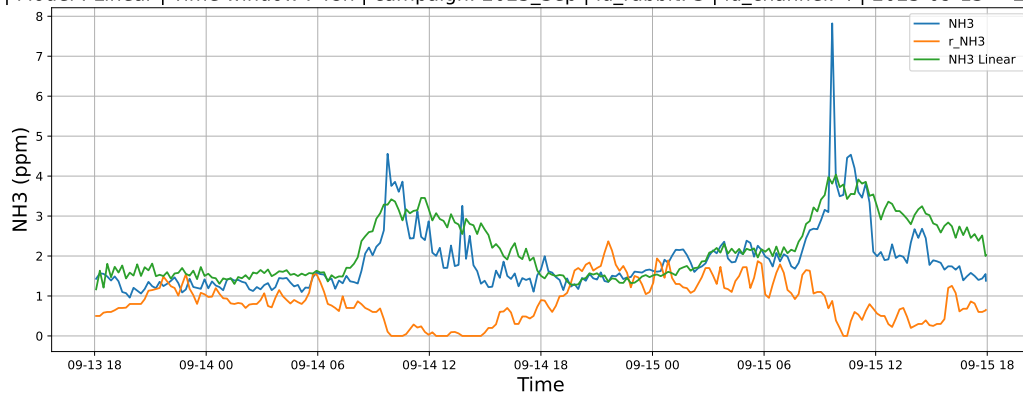


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

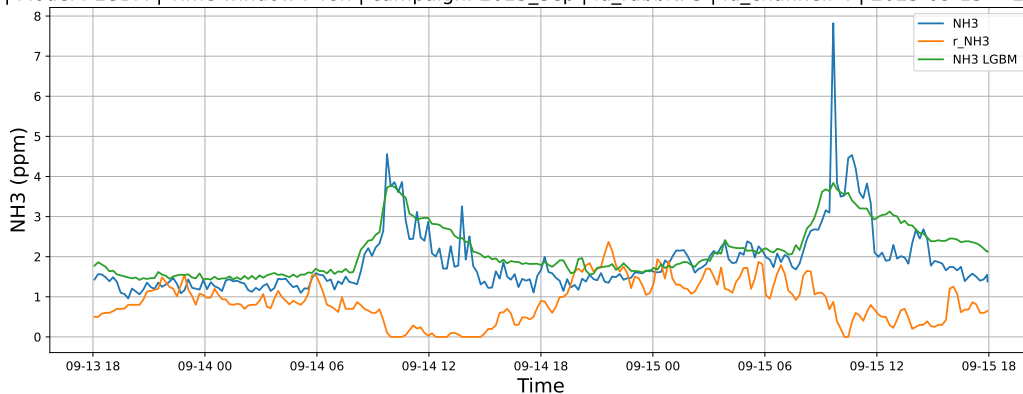


4.1.6.2 Time window : 48

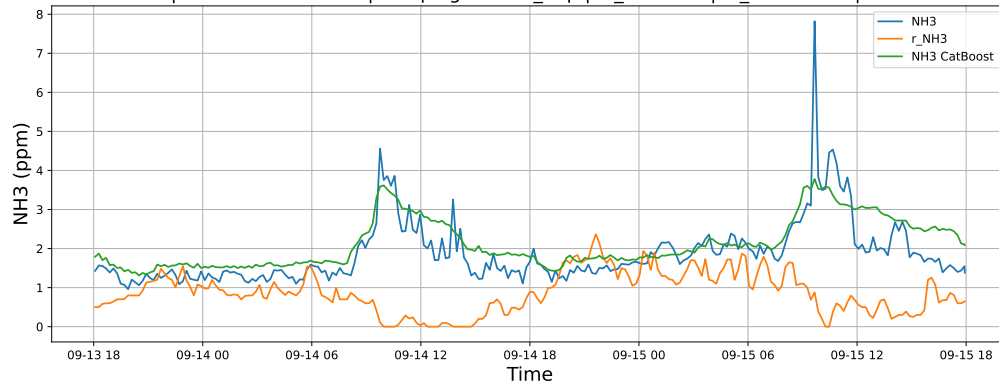
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

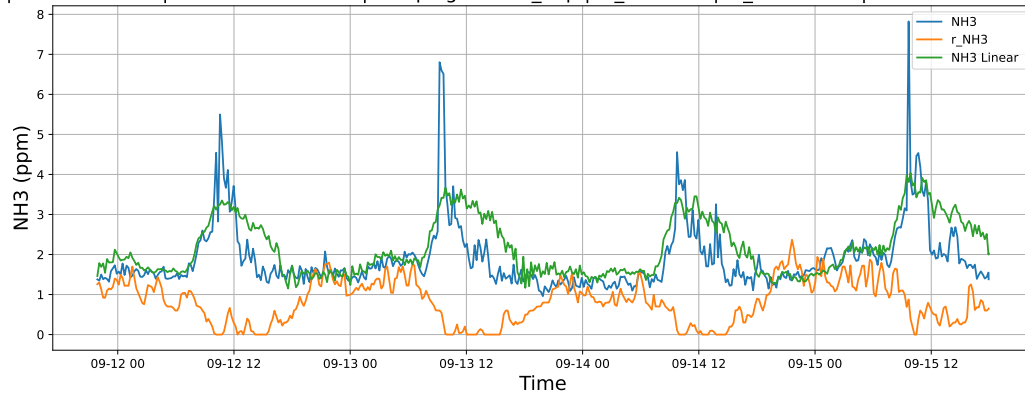


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

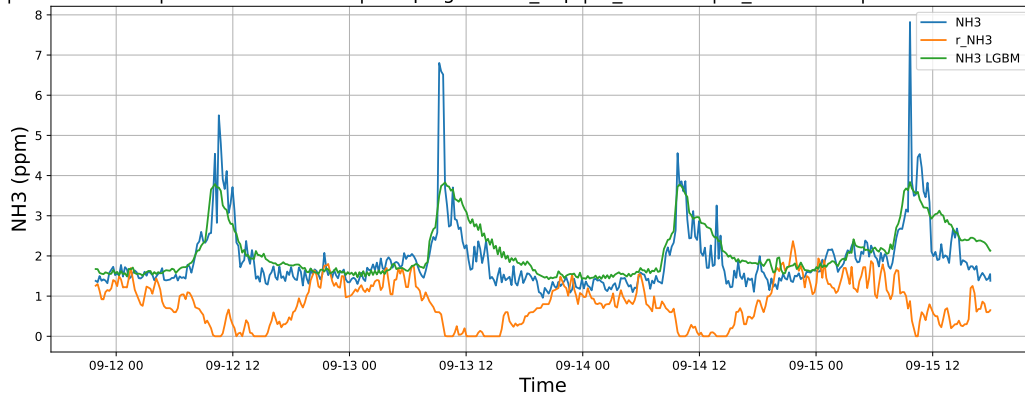


4.1.6.3 Time window : ALL

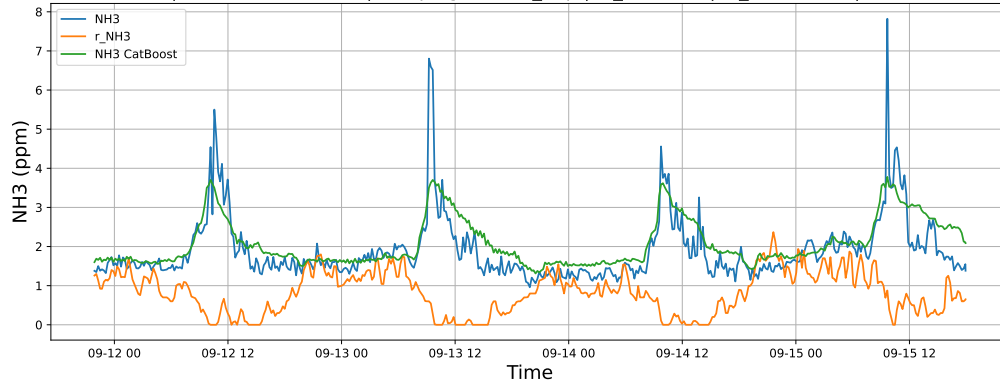
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



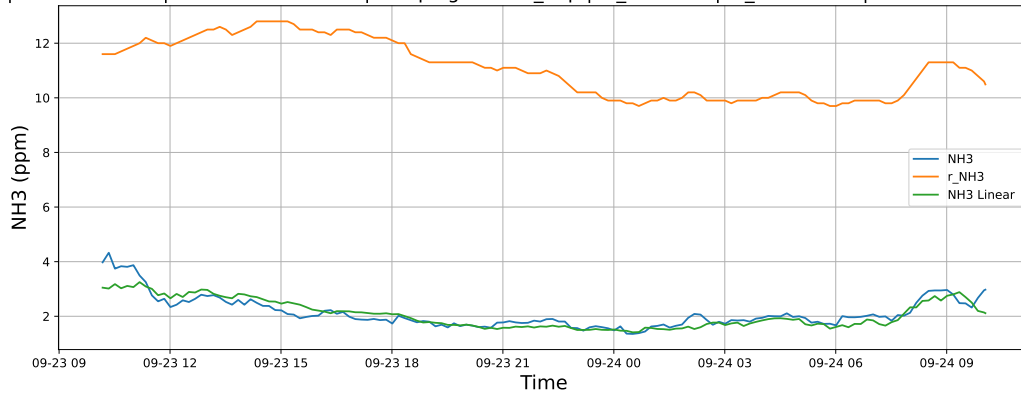
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



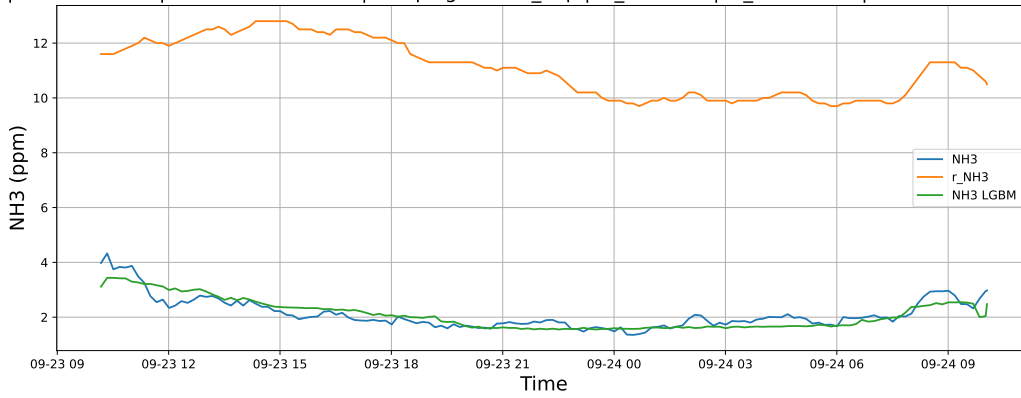
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

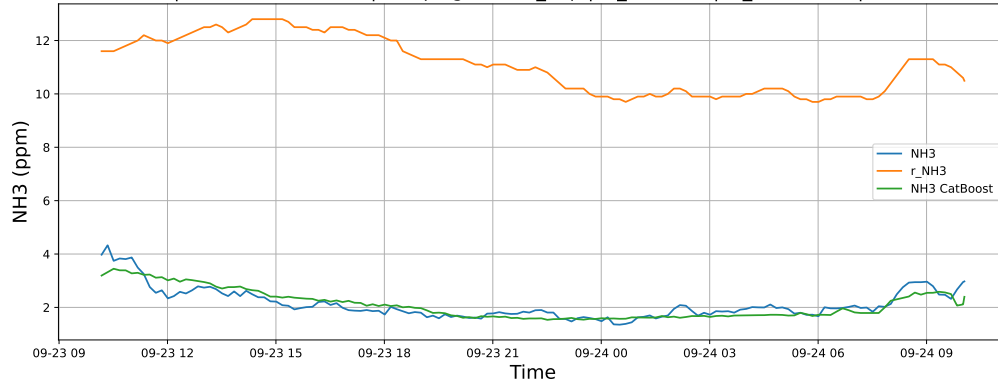
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

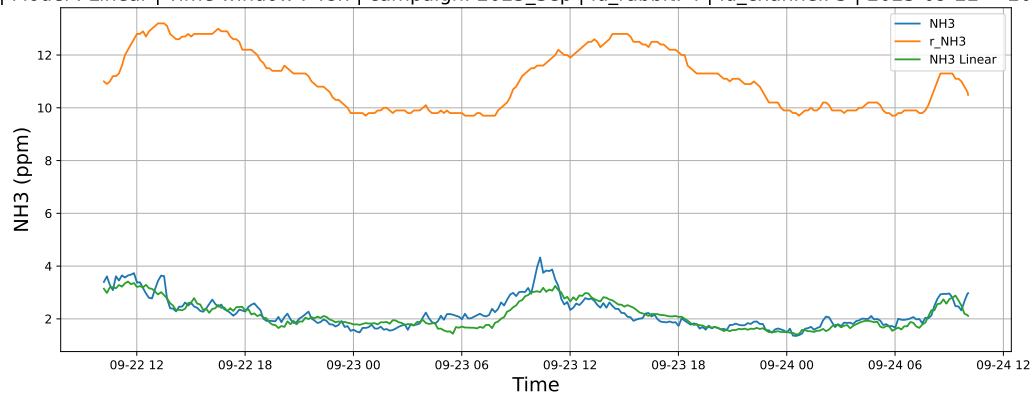


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

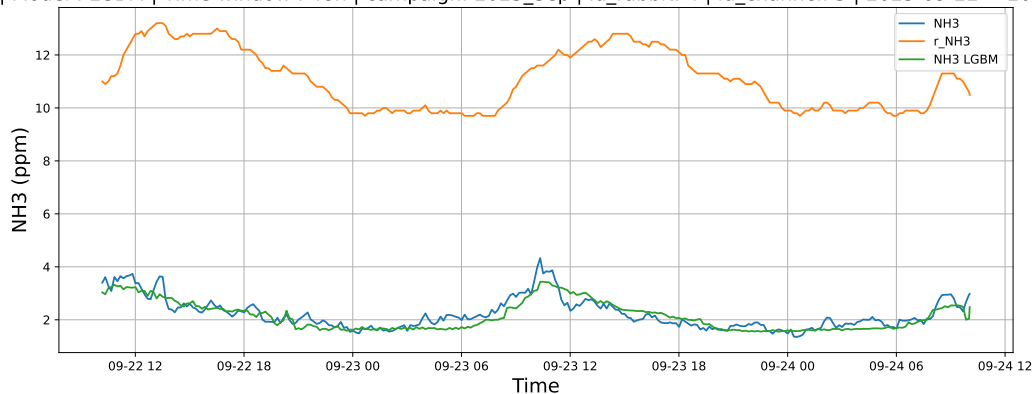


4.1.7.2 Time window : 48

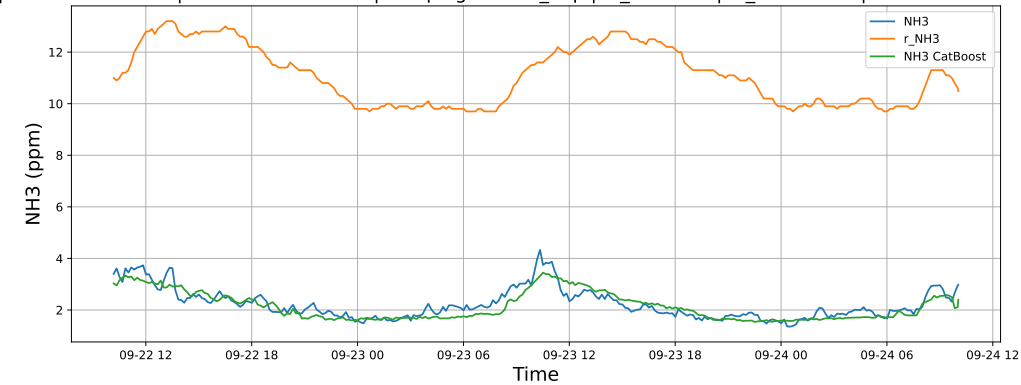
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

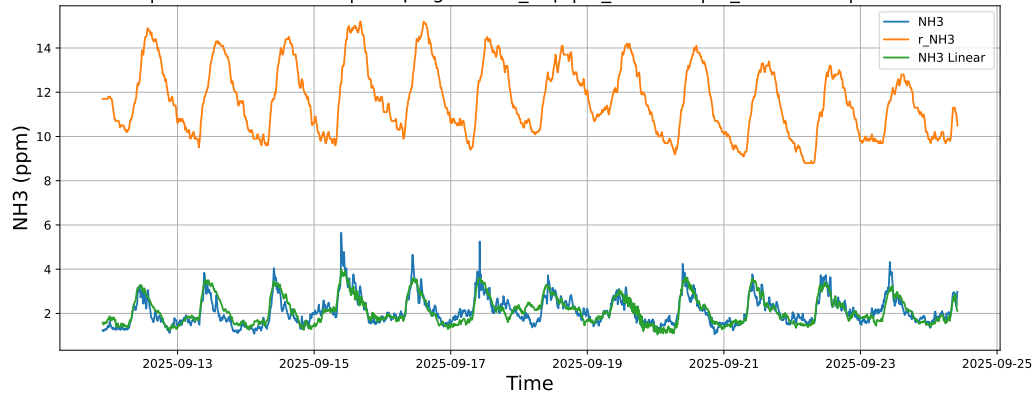


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

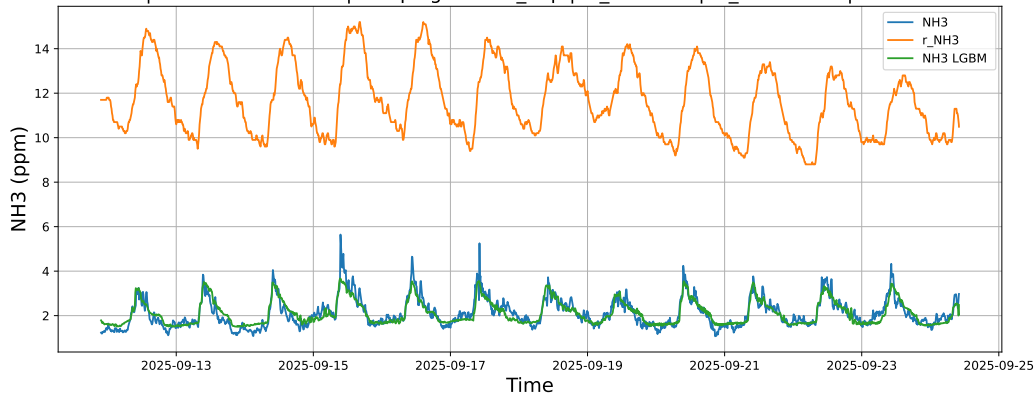


4.1.7.3 Time window : ALL

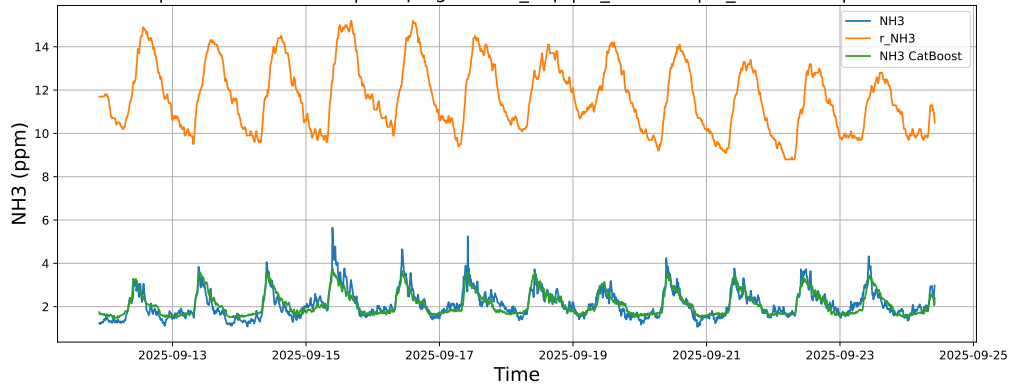
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



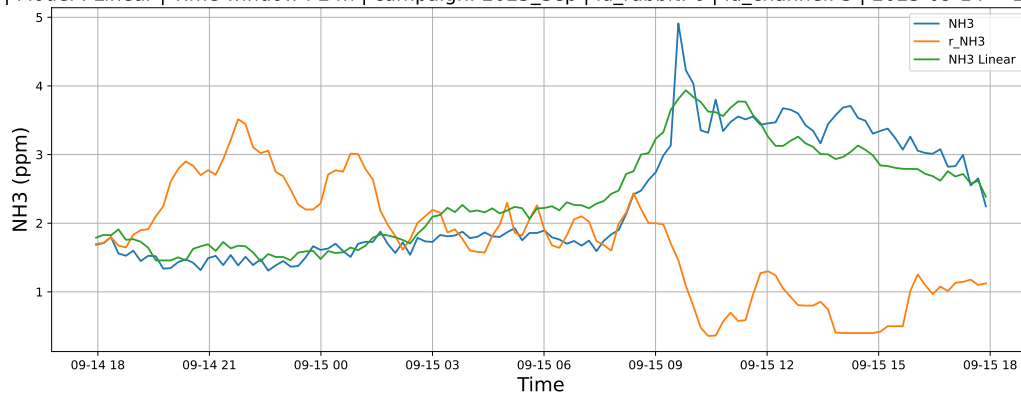
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



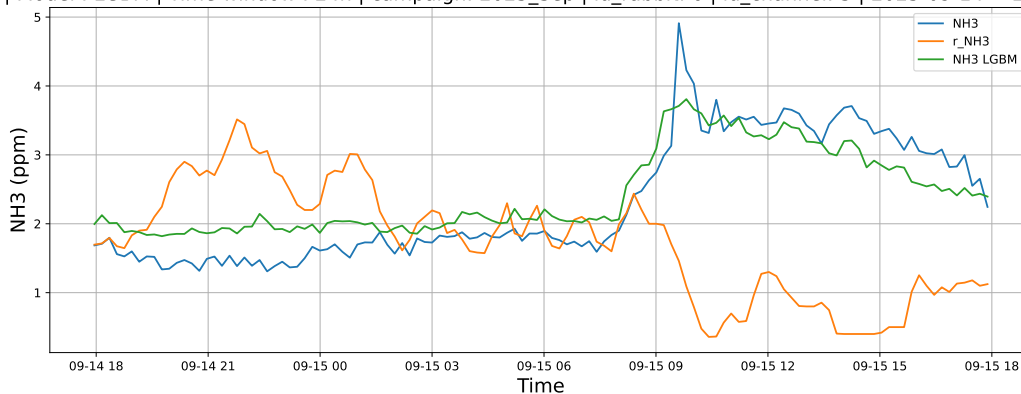
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

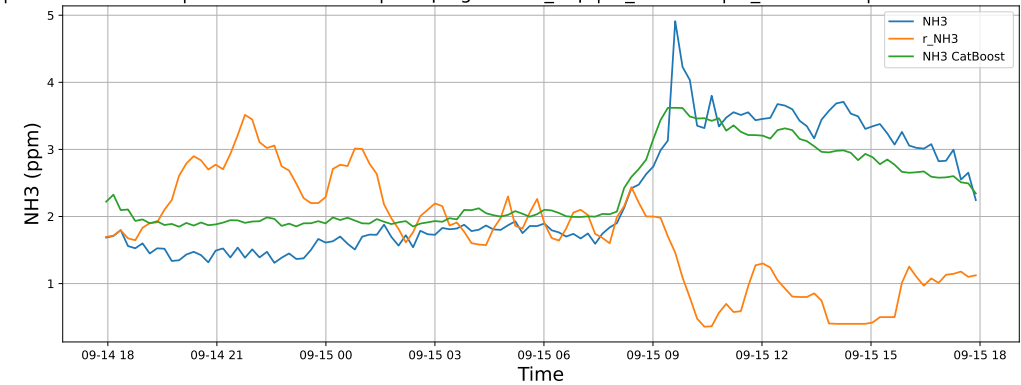
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

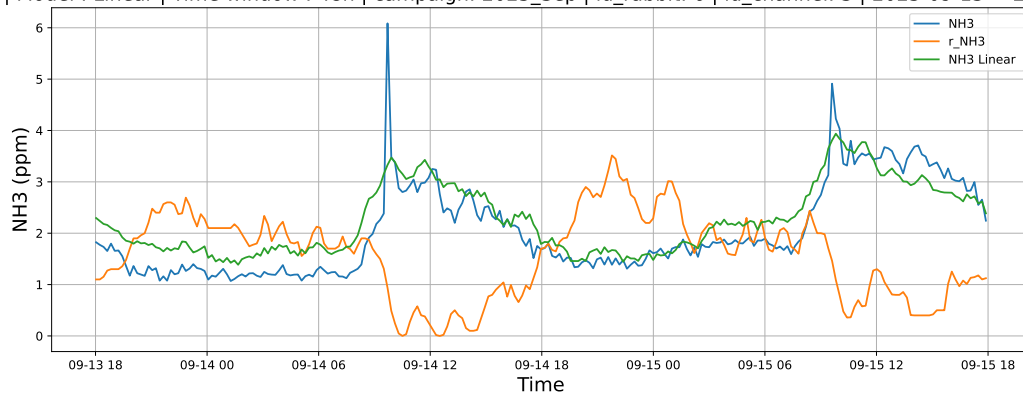


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

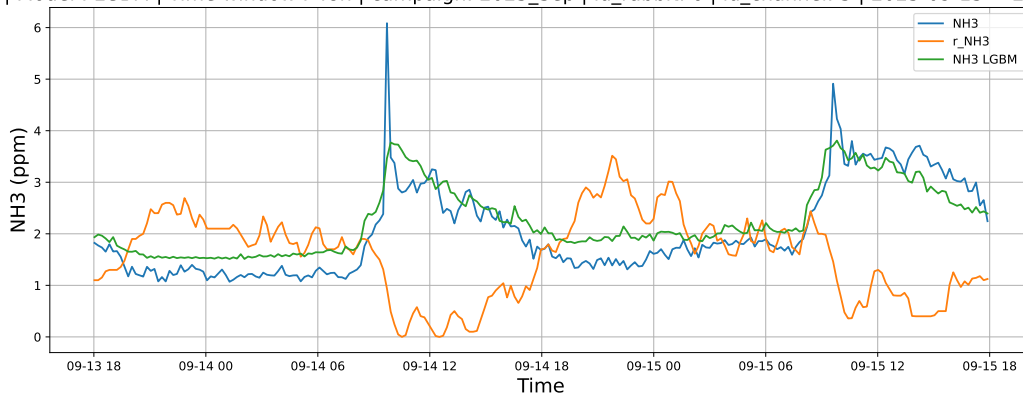


4.1.8.2 Time window : 48

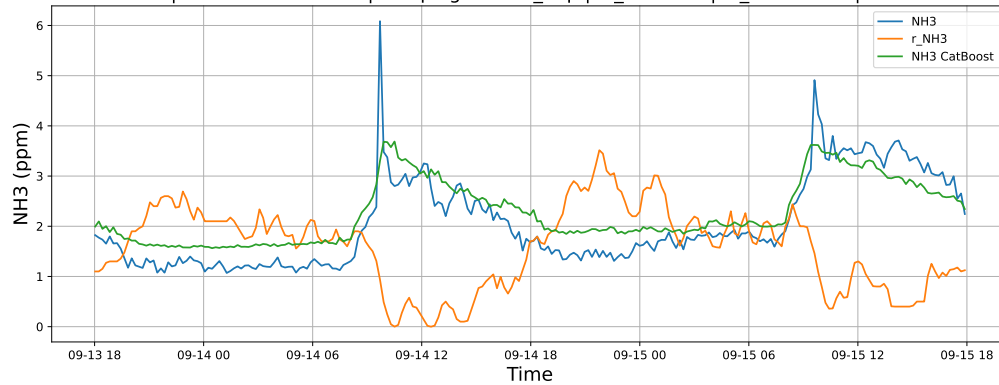
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

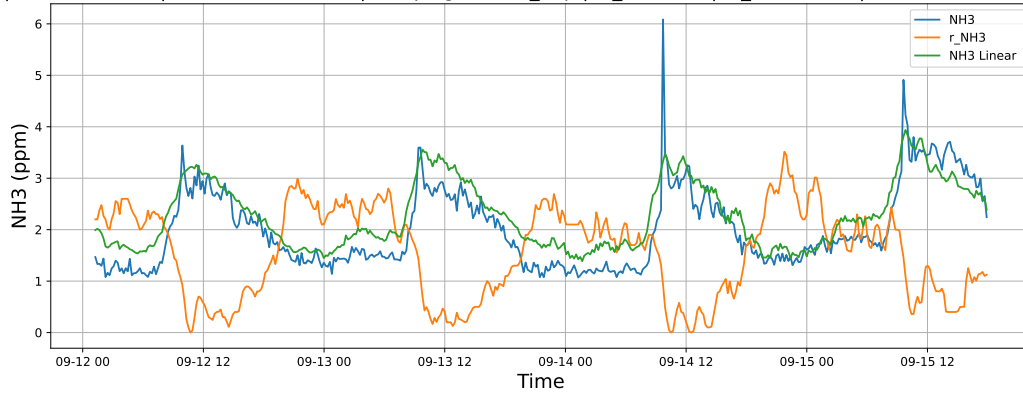


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

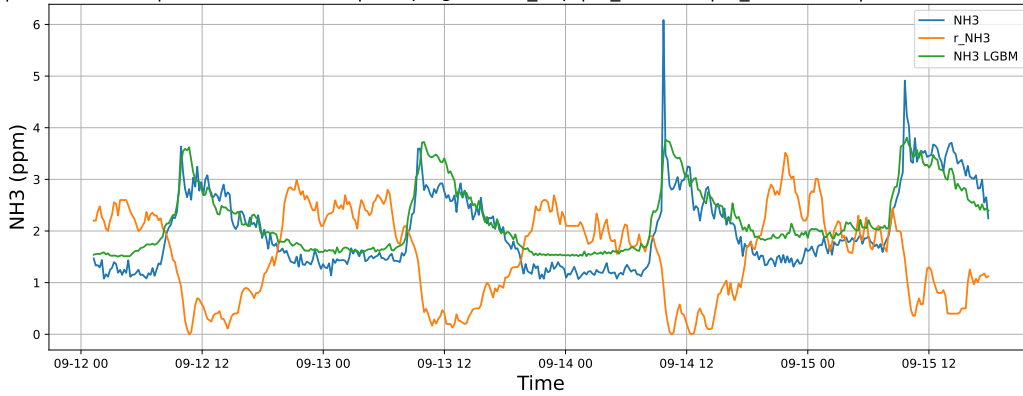


4.1.8.3 Time window : ALL

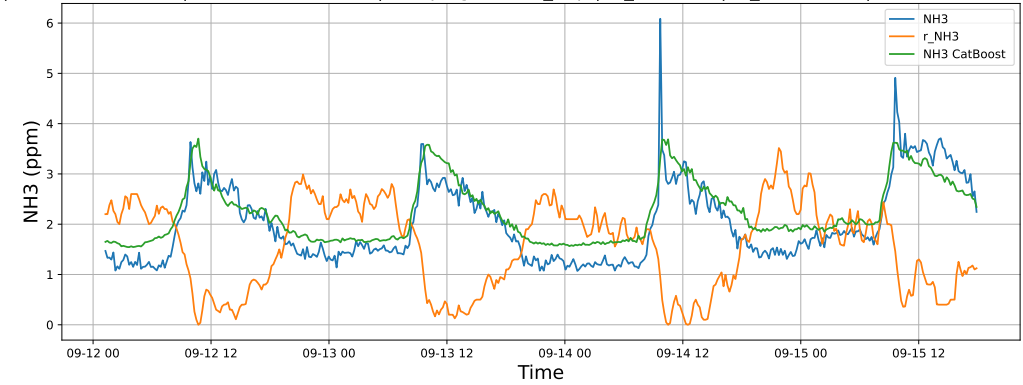
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-12 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-12 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-12 → 2025-09-15

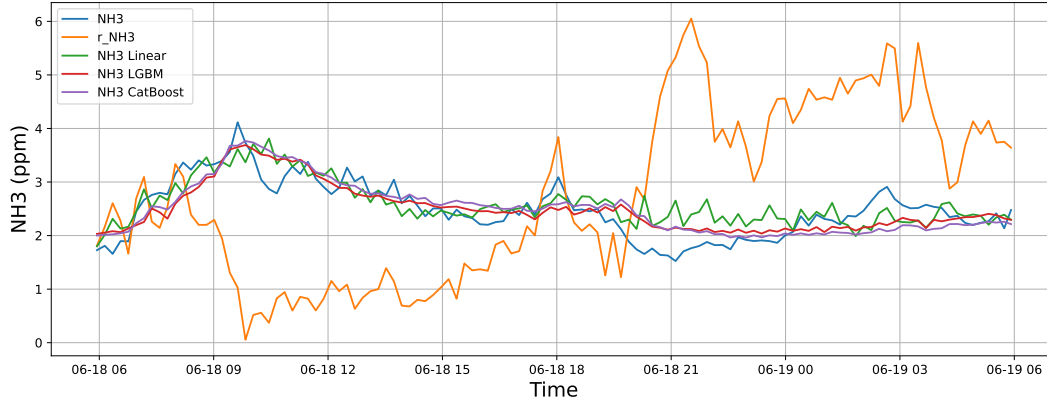


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

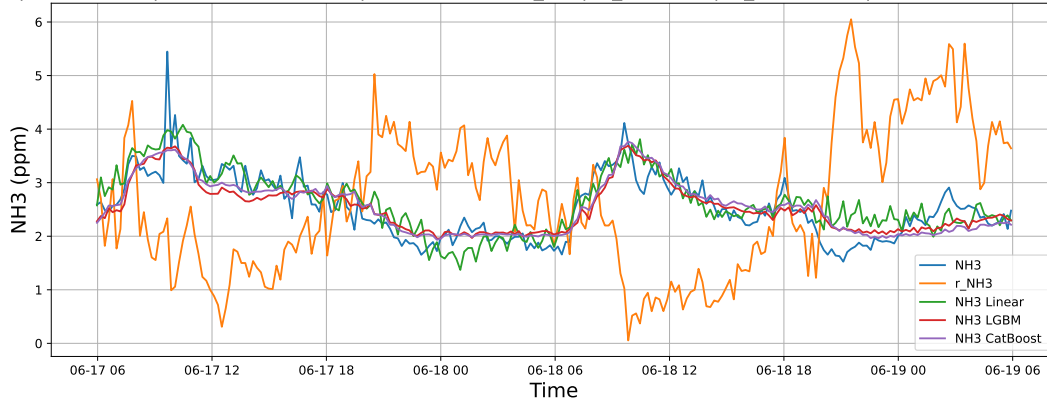
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



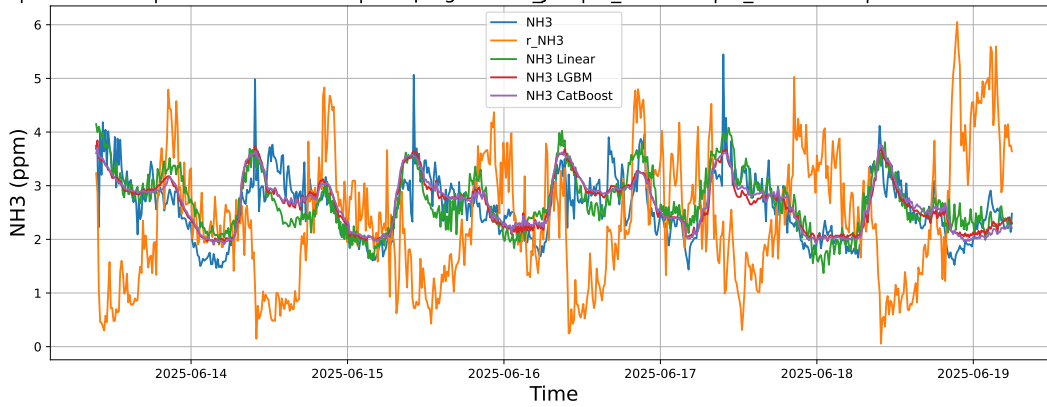
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

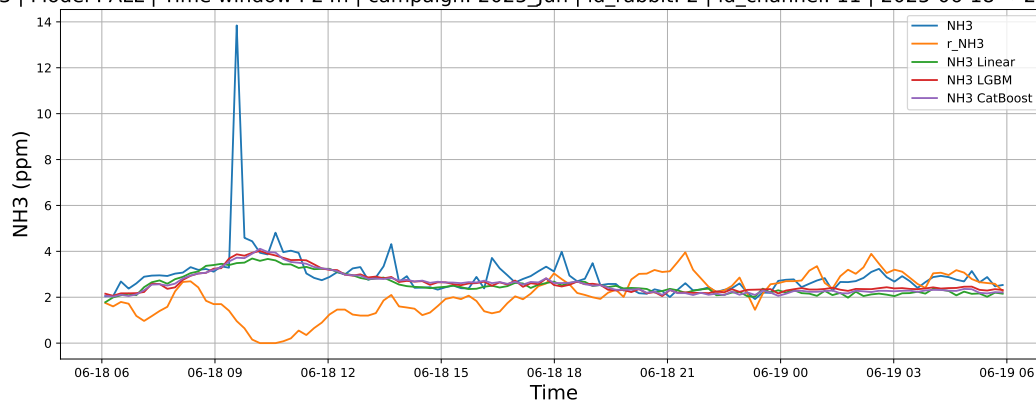
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-13 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

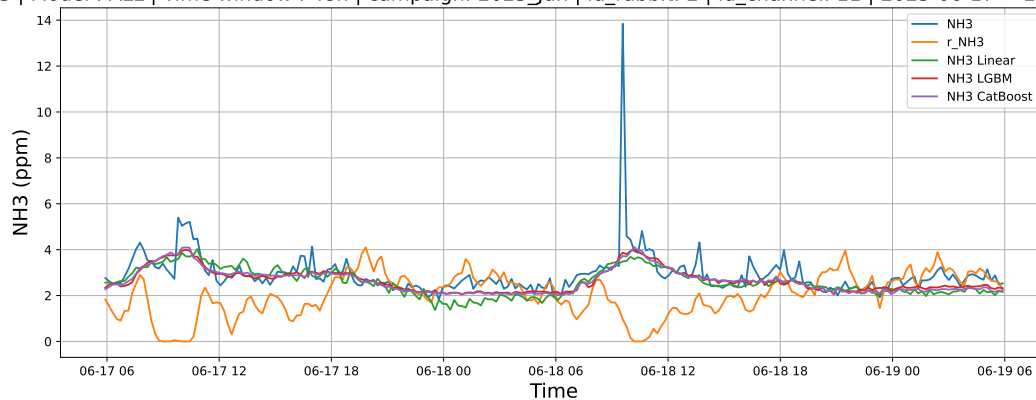
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



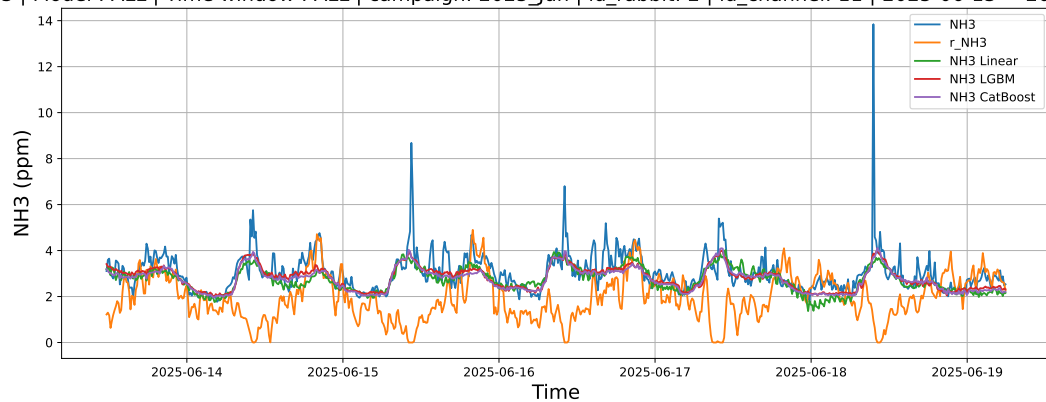
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

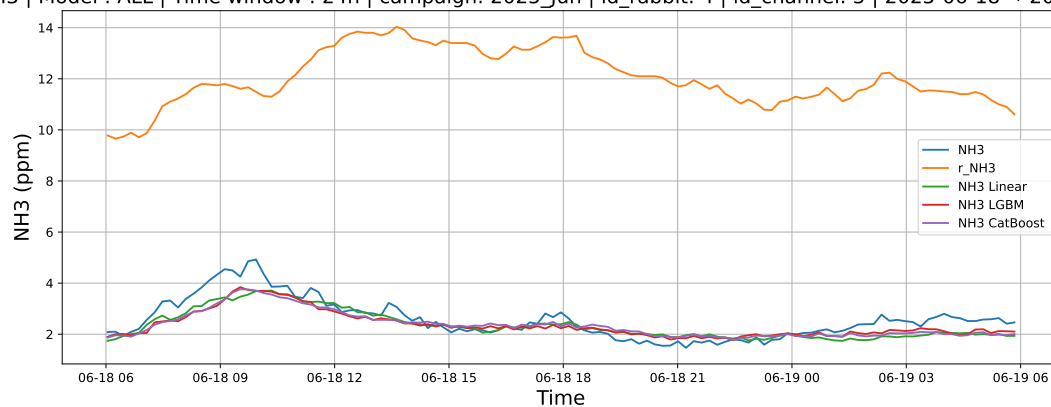
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-13 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

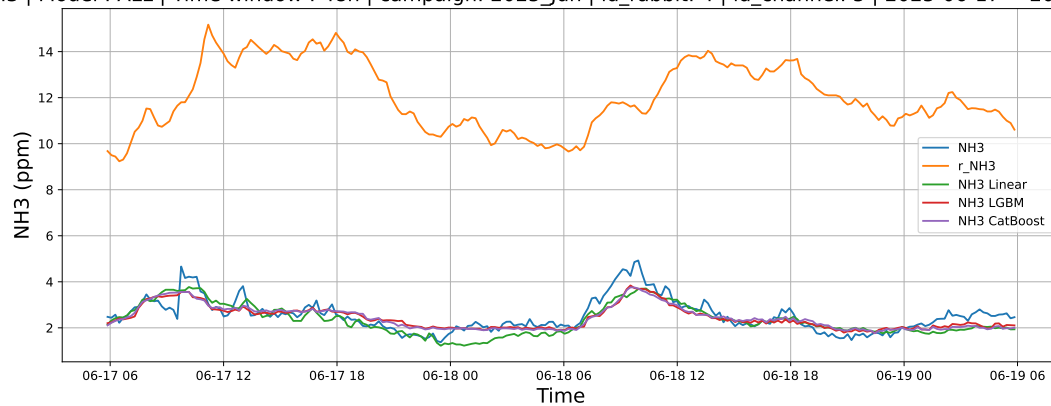
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



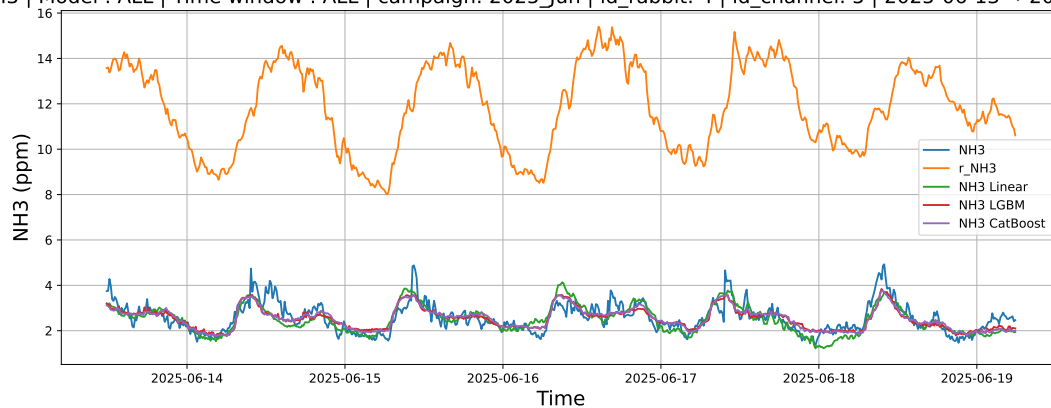
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

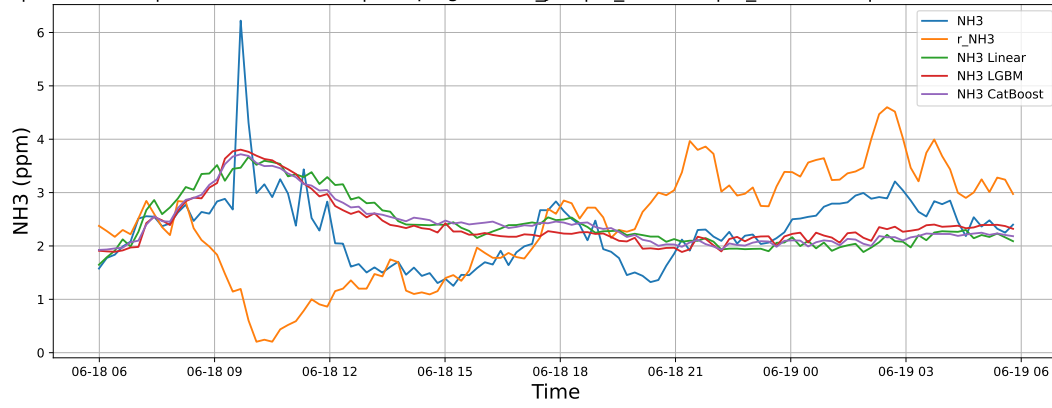
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-13 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

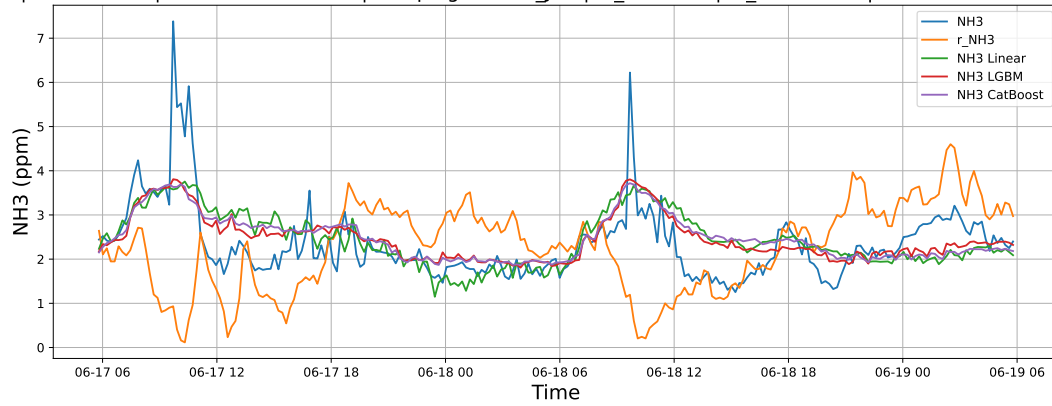
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



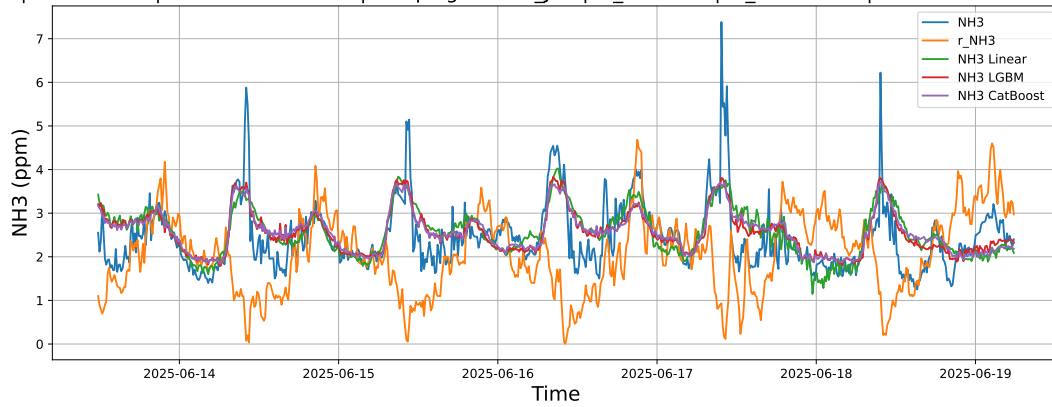
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

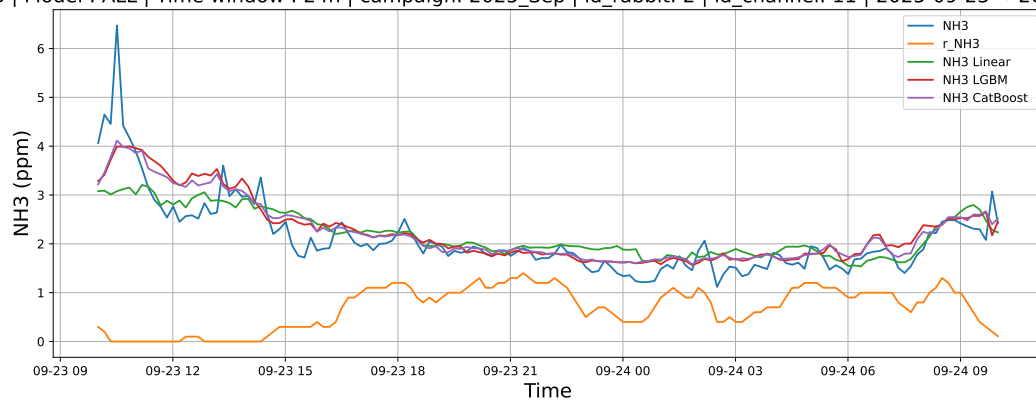
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-13 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

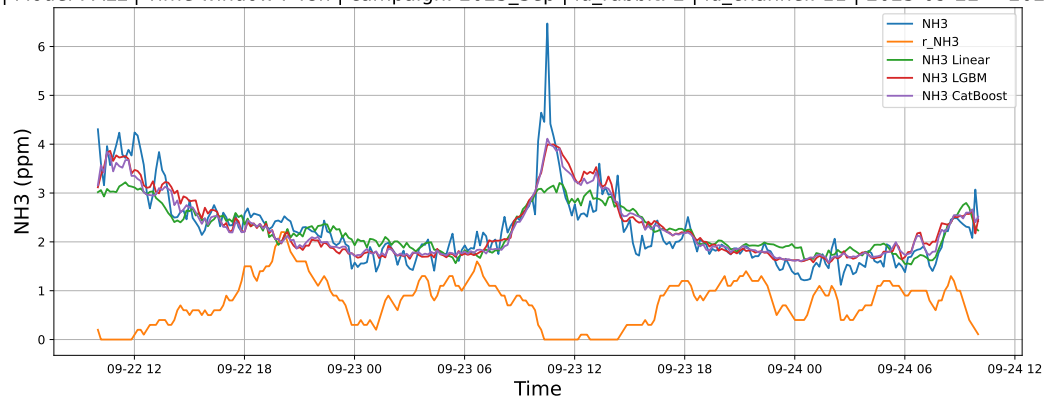
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



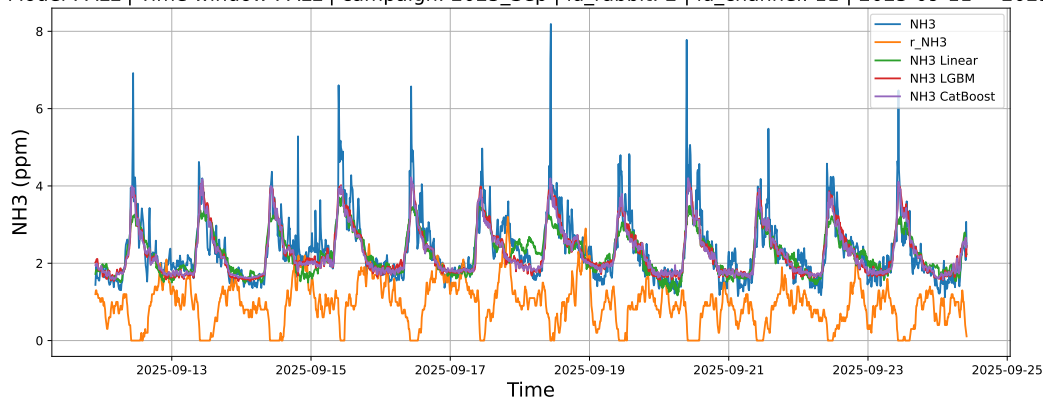
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

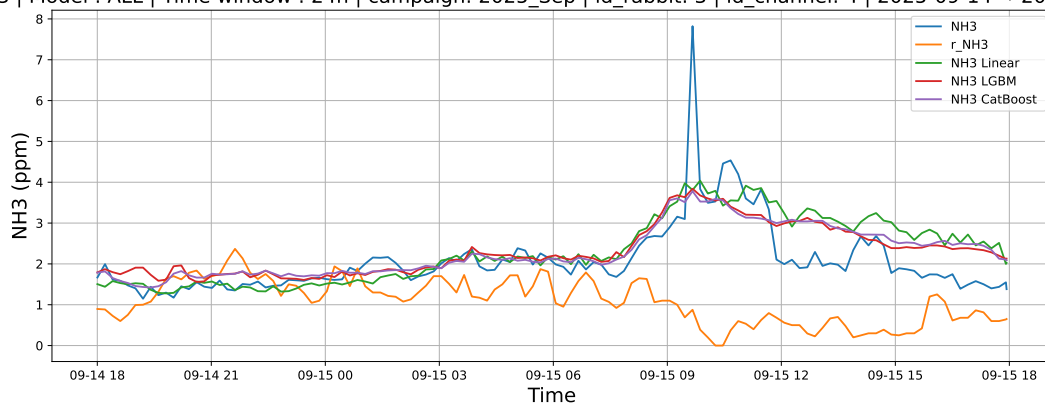
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep, id_rabbit: 3, id_channel: 4

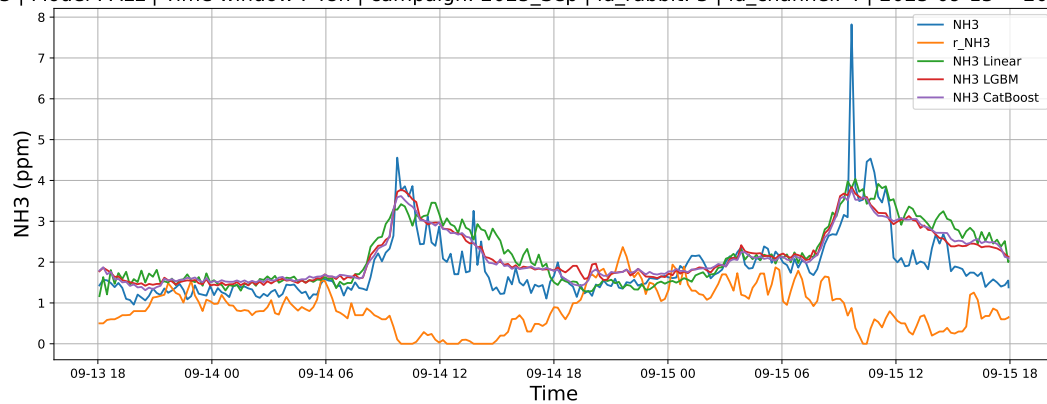
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



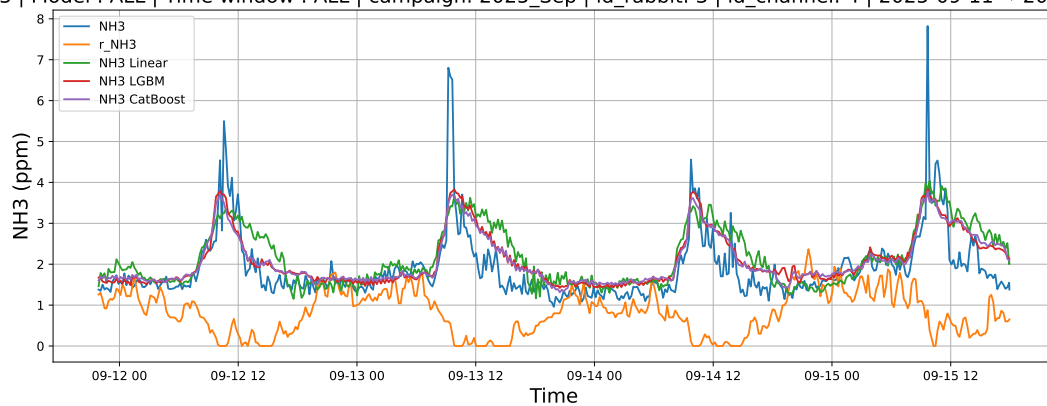
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

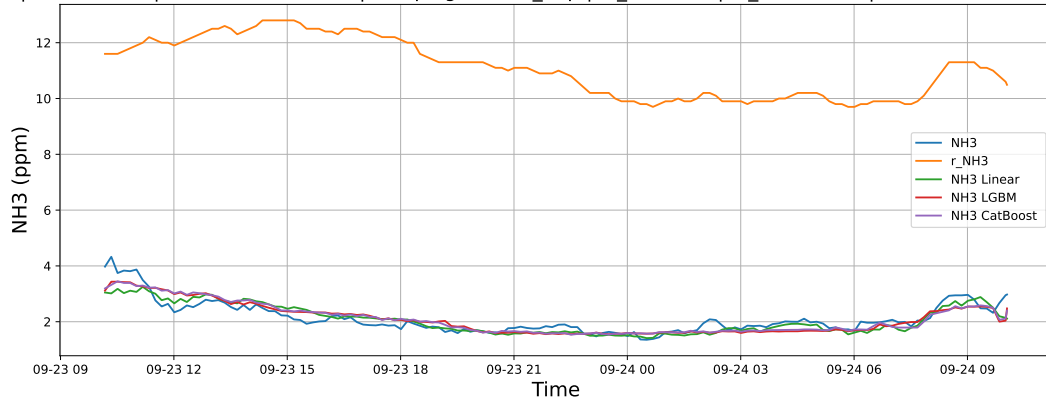
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

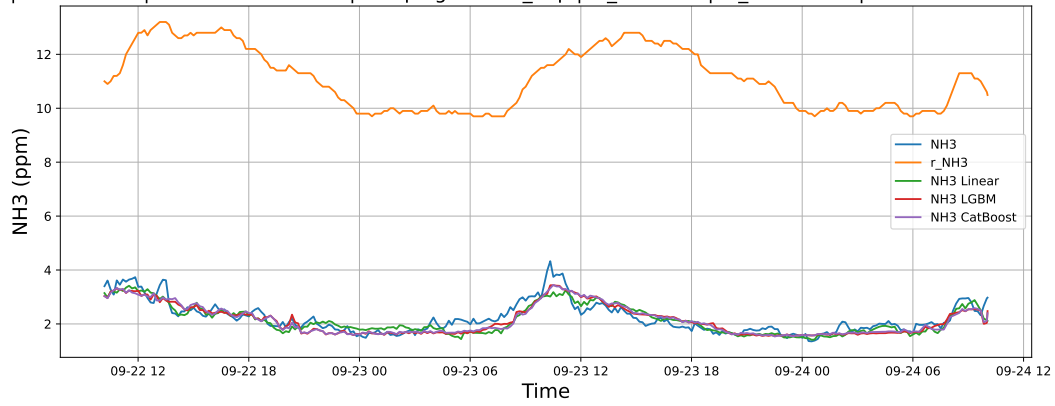
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



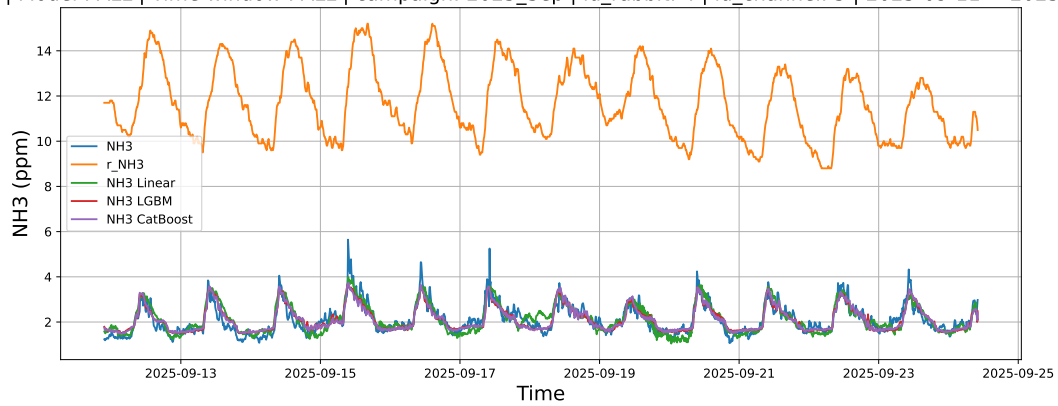
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

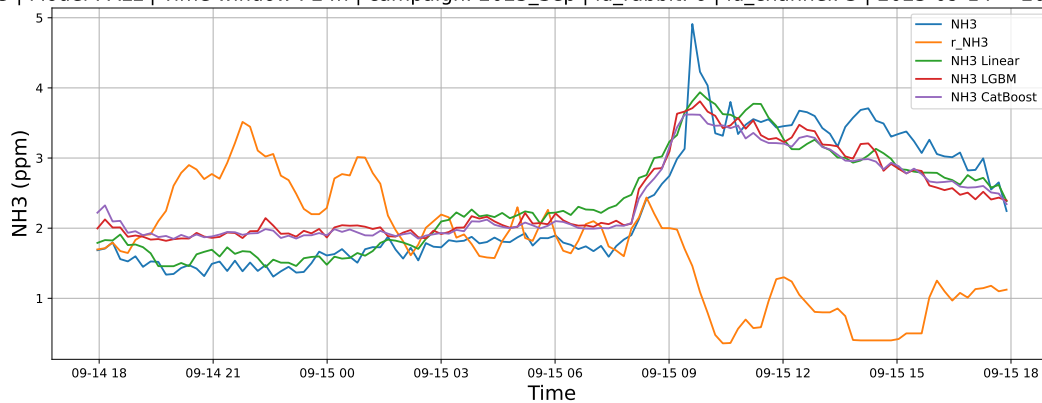
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

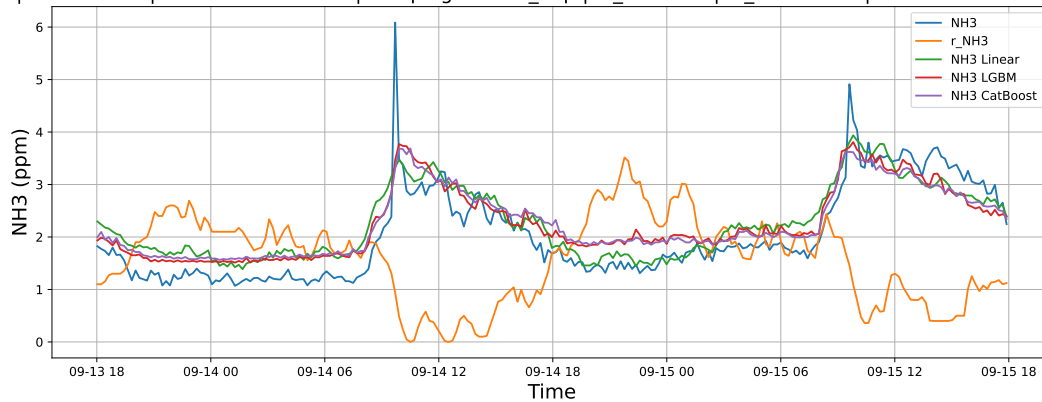
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-12 → 2025-09-15

